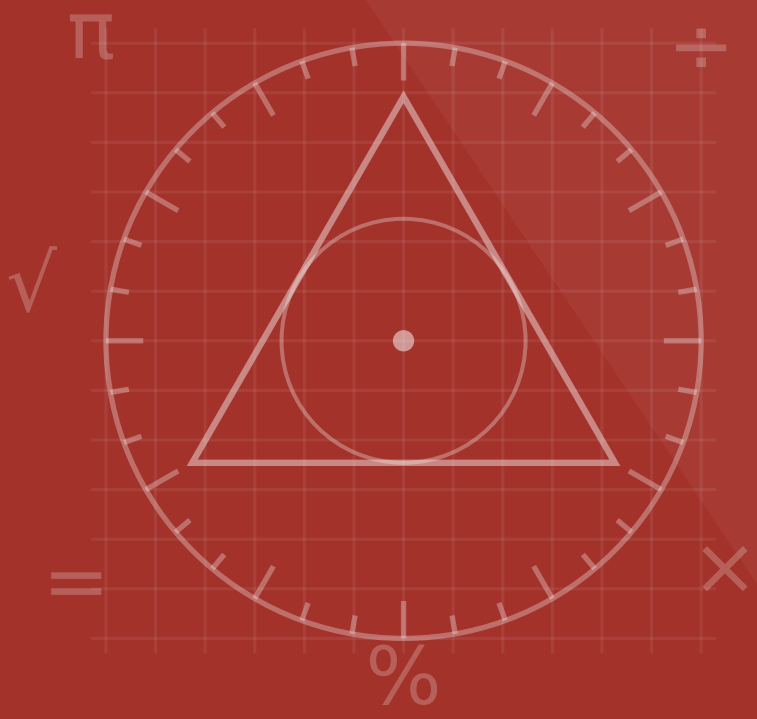
VENVLEARNING

STUDENT WORKBOOK



Mathematics

GRADE 7

Daily practice, weekly checks, and unit tests

9 units

A VenvLearning family homeschool curriculum

GRADE 7 · MATHEMATICS

VENVLEARNING

Grade 7
Mathematics

STUDENT WORKBOOK

Unit 1 Proportional Relationships
Unit 2 Percents
Unit 3 Adding and Subtracting Integers
Unit 4 Adding and Subtracting Rational Numbers
Unit 5 Multiplying and Dividing Rational Numbers
Unit 6 Expressions, Equations, and Inequalities
Unit 7 Statistics and Probability
Unit 8 Scale Drawings and Geometric Figures
Unit 9 Angles, Circles, Area, and Volume

This book belongs to _____

A VenvLearning family homeschool curriculum

How to use this workbook

Every lesson in your textbook has matching practice pages here. Show your thinking in the work space. When a problem has an answer line, write your final answer there.

- Warm-Up reviews something you already know.
- Practice uses today's lesson.
- Think Deeper problems ask you to explain or find a mistake.
- Exit Ticket shows your teacher what you learned today.

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UNIT 1

Proportional Relationships

Ratios, unit rates, the constant of proportionality, and proportional tables, graphs, and equations.

Your practice pages for Unit 1. Each lesson matches the same lesson number in your textbook.

UNIT VOCABULARY

ratio	A comparison of two quantities.
unit rate	Amount per one unit.
proportional relationship	$y \div x$ is always the same.
constant of proportionality	$k = y \div x$.

LESSON 1.1 · PRACTICE
Week 1 · Lesson 1 of 4

Unit 1: Proportional Relationships

Lesson 1.1: Ratios and Rates

Name _____ Date _____

TODAY'S GOAL

I can write ratios and rates and explain what they mean.

Standards: 7.RP.1 (review of 6.RP.1–3)

WHAT TO DO

- [ ] Read Textbook Lesson 1.1 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Write two equivalent ratios, Part D Complete the double number line, Part E Solve, Part F Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Answer each question.

1. Write the ratio of 4 dogs to 7 cats three ways.

2. Is 10:15 equivalent to 2:3?

3. A recipe uses 2 eggs per 3 cups of flour. How many eggs for 12 cups of flour?

PART B Warm-Up

Directions: Solve each problem. Show your work.

4. 12 × 4

5. Simplify $\frac{18}{24}$

PART C Write two equivalent ratios

6. 2:5

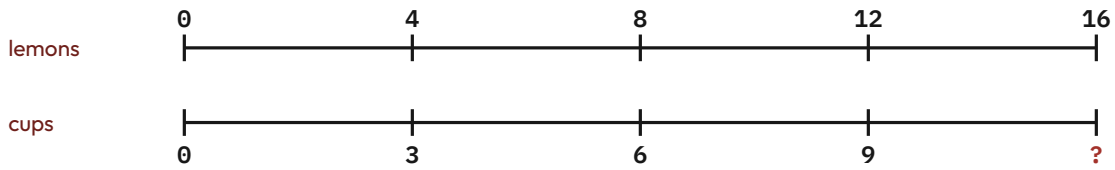
7. 7:3

8. 12:16

9. 1:4

PART D Complete the double number line

10. Lemonade: 4 lemons for every 3 cups of water. How much water for 16 lemons?



PART E Solve

11. A team won 12 games and lost 8. What is the ratio of wins to games played?

12. A printer prints 45 pages in 3 minutes. How many pages in 10 minutes?

PART F Exit Ticket

13. Explain why 6:8 and 9:12 are equivalent.

LESSON 1.2 · PRACTICE
Week 1 · Lesson 2 of 4

Unit 1: Proportional Relationships

Lesson 1.2: Unit Rates

Name _____ Date _____

TODAY'S GOAL

I can find a unit rate and use it to solve problems.

Standards: 7.RP.1

WHAT TO DO

- [ ] Read Textbook Lesson 1.2 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Find the unit rate, Part D Solve, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Answer each question.

1. $18 for 6 notebooks: unit price?

2. 390 miles on 15 gallons: miles per gallon?

3. Type 210 words in 5 minutes: words per minute?

PART B Warm-Up

Directions: Solve each problem. Show your work.

4. 4.20 ÷ 12

5. 54 ÷ 6

PART C Find the unit rate

6. $7.50 for 3 pounds

7. 120 km in 2.5 hours

8. 96 heartbeats in 1.5 minutes

9. $13.44 for 16 ounces

10. 8 laps in 20 minutes

11. $52 for 8 hours of work

PART D Solve

12. Lillie earns $11.50 per hour babysitting. How much for 6 hours?

13. A car uses 3 gallons per 84 miles. How far on 10 gallons?

PART E Exit Ticket

14. A 5-pound bag of rice costs $6.25. What is the price per pound? How much would 12 pounds cost?

LESSON 1.3 · PRACTICE
Week 1 · Lesson 3 of 4
Unit 1: Proportional Relationships

Lesson 1.3: Unit Rates with Fractions

Name _____ Date _____

TODAY'S GOAL

I can find unit rates when the quantities are fractions, like $\frac{1}{2}$ mile in $\frac{1}{4}$ hour.

Standards: 7.RP.1

WHAT TO DO

- [ ] Read Textbook Lesson 1.3 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Find the unit rate, Part D Solve, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Answer each question.

1. $\frac{3}{4}$ cup of sugar per $\frac{1}{2}$ batch. Cups per batch?

2. $\frac{1}{3}$ mile in $\frac{1}{12}$ hour. mph?

3. \$2.50 for $\frac{1}{2}$ pound. Price per pound?

PART B Warm-Up

Directions: Solve each problem. Show your work.

4. $\frac{1}{2} \div \frac{1}{4}$

5. $\frac{2}{3} \times 3$

PART C Find the unit rate

6. $\frac{1}{4}$ mile in $\frac{1}{8}$ hour

7. $\frac{2}{5}$ pound for $\frac{1}{5}$ dollar

8. $\frac{3}{4}$ page in $\frac{1}{4}$ minute

9. $\frac{5}{6}$ cup for $\frac{2}{3}$ batch

10. $1\frac{1}{2}$ miles in $\frac{1}{2}$ hour

11. $\frac{3}{8}$ gallon for $\frac{3}{4}$ acre

PART D Solve

12. A snail moves $\frac{1}{10}$ meter in $\frac{1}{5}$ minute. How far in 1 minute? In 30 minutes?

PART E Exit Ticket

13. Lillie mows $\frac{3}{5}$ of a lawn in $\frac{3}{4}$ hour. How much lawn per hour?

LESSON 1.4 · PRACTICE
Week 1 · Lesson 4 of 4
Unit 1: Proportional Relationships

Lesson 1.4: Comparing Unit Rates

Name _____ Date _____

TODAY'S GOAL

I can use unit rates to decide which deal or option is better.

Standards: 7.RP.1, 7.RP.3

WHAT TO DO

- [ ] Read Textbook Lesson 1.4 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Which is the better buy?, Part D Solve, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Answer each question.

1. 12 eggs for $3.00 or 18 eggs for $4.14?

2. Car A: 300 mi on 12 gal. Car B: 270 mi on 10 gal. Better mileage?

3. Who types faster: 150 words in 3 min or 200 words in 5 min?

PART B Warm-Up

Directions: Solve each problem. Show your work.

4. 3.84 ÷ 16

5. 5.28 ÷ 24

PART C Which is the better buy?

Directions: Find the unit price of each choice. Then name the better buy.

6. 8 oz for $2.40 or 12 oz for $3.24

7. 3 lb for $4.50 or 5 lb for $7.25

8. 6 pencils for $1.50 or 10 for $2.30

9. 2 L for $1.80 or 3 L for $2.85

PART D Solve

10. Dad's truck goes 342 miles on 18 gallons. Mom's car goes 406 miles on 14 gallons. Whose vehicle gets better gas mileage, and by how many mpg?

PART E Exit Ticket

11. Explain why a bigger package is not always the better buy.

WEEKLY CHECK
After Lesson 1.4

About 20 minutes

Weekly Check 1: Ratios and Unit Rates

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each question carefully. Show your thinking.
- [ ] When you finish, check that every question has an answer.

Show what you know

Directions: Do what each item asks. Show your work.

1. Write two ratios equivalent to 4:9.

2. $11.40 for 6 pounds: unit price

3. $\frac{2}{3}$ mile in $\frac{1}{6}$ hour: mph

4. Better buy: 10 oz for $2.50 or 16 oz for $3.84

5. A car goes 252 miles on 9 gallons. mpg?

6. At 28 mpg, how far on 13 gallons?

LESSON 1.5 · PRACTICE
Week 2 · Lesson 1 of 4

Unit 1: Proportional Relationships

Lesson 1.5: What Is a Proportional Relationship?

Name _____ Date _____

TODAY'S GOAL

I can tell when two quantities are in a proportional relationship.

Standards: 7.RP.2a

WHAT TO DO

- [ ] Read Textbook Lesson 1.5 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Proportional or not? Explain, Part D Real situations, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Answer each question.

1. Is distance proportional to time at a steady 60 mph?

2. Is a $20 gym fee plus $5 per visit proportional?

3. x: 2, 4, 6 → y: 7, 14, 21. Proportional?

PART B Warm-Up

Directions: Solve each problem. Show your work.

4. 21 ÷ 6

5. Is 4:10 equivalent to 6:15?

PART C Proportional or not? Explain.

6. x: 1, 2, 3, 4 → y: 5, 10, 15, 20

7. x: 1, 2, 3, 4 → y: 3, 5, 7, 9

8. x: 2, 5, 8 → y: 3, 7.5, 12

9. x: 3, 6, 9 → y: 1, 2, 4

PART D Real situations

Directions: Decide whether each situation is proportional. Explain why or why not.

10. Pay of $12 per hour

11. A taxi: $3 plus $2 per mile

12. Area of a square vs its side length

13. Cost of apples at $1.50 per pound

PART E Exit Ticket

14. Make a table that is NOT proportional. Explain why.

LESSON 1.6 · PRACTICE
Week 2 · Lesson 2 of 4

Unit 1: Proportional Relationships

Lesson 1.6: The Constant of Proportionality

Name _____ Date _____

TODAY'S GOAL

I can find the constant of proportionality k from a table.

Standards: 7.RP.2b

WHAT TO DO

- [ ] Read Textbook Lesson 1.6 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Find the constant of proportionality, Part D Complete the proportional table, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Find the constant of proportionality k ($k = y \div x$).

1. x: 2, 6, 10 → y: 9, 27, 45. Find k.

2. Hours 3 → $36 pay. k?

3. x: 4, 8 → y: 1, 2. k?

PART B Warm-Up

Directions: Solve each problem. Show your work.

4. 7.5 ÷ 6

5. 45 ÷ 10

PART C Find the constant of proportionality

6. x: 1, 3, 5 → y: 7, 21, 35

7. x: 2, 4, 10 → y: 3, 6, 15

8. x: 5, 10, 20 → y: 2, 4, 8

9. x: 6, 9, 12 → y: 4, 6, 8

PART D Complete the proportional table

Directions: Find k, then fill in each missing value.

10. x: 1, 2, 5, 8 → y: 6, ___, ___, ___

11. x: 4, 10, 15 → y: 10, ___, ___

PART E Exit Ticket

12. A car travels 165 miles in 3 hours at a steady speed. What is k? What does it mean?

LESSON 1.7 · PRACTICE
Week 2 · Lesson 3 of 4
Unit 1: Proportional Relationships

Lesson 1.7: Equations for Proportional Relationships

Name _____ Date _____

TODAY'S GOAL

I can write an equation y = kx for a proportional relationship.

Standards: 7.RP.2c

WHAT TO DO

- [ ] Read Textbook Lesson 1.7 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Write the equation, Part D Use the equation, Part E Solve, Part F Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Write the equation y = kx, or use it to find the missing value.

1. A plane flies 480 mph. Equation for distance d in t hours?

2. x: 3, 6 → y: 2, 4. Equation?

3. y = 12x. Find y when x = 7.5.

PART B Warm-Up

Directions: Solve each problem. Show your work.

4 16 ÷ 2

5 3.5 × 6

PART C Write the equation

Directions: Write an equation in the form y = kx.

6 k = 9

7 x: 4, 8 → y: 10, 20

8 $0.65 per ounce

9 15 pages per hour

PART D Use the equation

Directions: Substitute the given value and solve.

10 y = 6x; x = 12

11 y = 0.8x; x = 25

12 y = 4x; y = 52, find x

13 $y = \frac{3}{4}x$; x = 20

PART E Solve

14 Lillie earns $11.50 per hour. Write an equation. How much for 14 hours?

PART F Exit Ticket

15 Explain what k means in y = 45x if x is hours and y is miles.

LESSON 1.8 · PRACTICE
Week 2 · Lesson 4 of 4
Unit 1: Proportional Relationships

Lesson 1.8: Is It Proportional? Tables and Equations

Name _____ Date _____

TODAY'S GOAL

I can decide if a table or equation represents a proportional relationship.

Standards: 7.RP.2a

WHAT TO DO

- [ ] Read Textbook Lesson 1.8 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Proportional or not?, Part D Tables, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Write proportional or not proportional. Explain your answer.

1. $y = 7x$

2. $y = 2x - 1$

3. x: 0, 3, 6 → y: 0, 2, 4

PART B Warm-Up

Directions: Evaluate each expression for the given value.

4. $y = 7x$ when $x = 0$

5. $y = 2x - 1$ when $x = 0$

PART C Proportional or not?

Directions: Write proportional or not proportional for each equation.

6. $y = 9x$

7. $y = x + 9$

8. $y = {}^{1}/_{3}x$

9. $y = 3x + 0$

10. $y = 10 - x$

11. $y = 2.4x$

PART D Tables

Directions: Write proportional or not proportional for each table. Explain.

12. x: 1, 2, 3 → y: 4, 8, 13

13. x: 0, 5, 10 → y: 0, 3, 6

PART E Exit Ticket

14. A phone plan costs $25 plus $10 per GB. Is the cost proportional to GB? Explain.

WEEKLY CHECK
After Lesson 1.8

About 20 minutes

Weekly Check 2: Proportional Tables and Equations

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each question carefully. Show your thinking.
- [ ] When you finish, check that every question has an answer.

Show what you know

Directions: Do what each item asks. Show your work.

1. x: 2, 5, 9 → y: 7, 17.5, 31.5. Proportional? k?

2. x: 1, 2, 3 → y: 5, 9, 13. Proportional?

3. Write the equation for k = 0.6.

4. y = 15x; x = 8

5. y = 4x; y = 30, find x

6. Is y = 3x + 2 proportional? Why?

LESSON 1.9 · PRACTICE
Week 3 · Lesson 1 of 4
Unit 1: Proportional Relationships

Lesson 1.9: Graphing Proportional Relationships

Name _____ Date _____

TODAY'S GOAL

I can graph a proportional relationship on a coordinate plane.

Standards: 7.RP.2a

WHAT TO DO

- [ ] Read Textbook Lesson 1.9 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Make a table and graph, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Answer each question.

1. Plot y = 3x for x = 0, 1, 2, 3. Name the points.

2. Does the graph of y = 3x pass through (0, 0)?

3. Is (5, 15) on y = 3x?

PART B Warm-Up

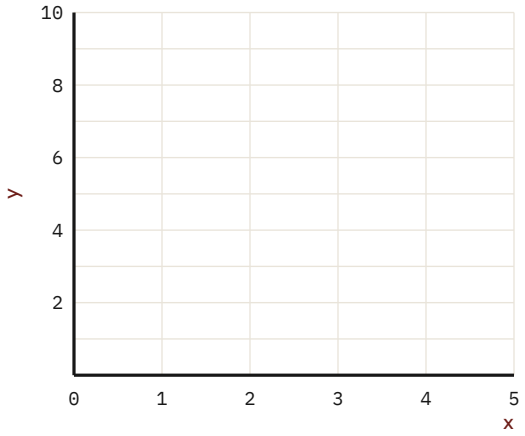
Directions: Evaluate, or name the point.

4 y = 5x when x = 6

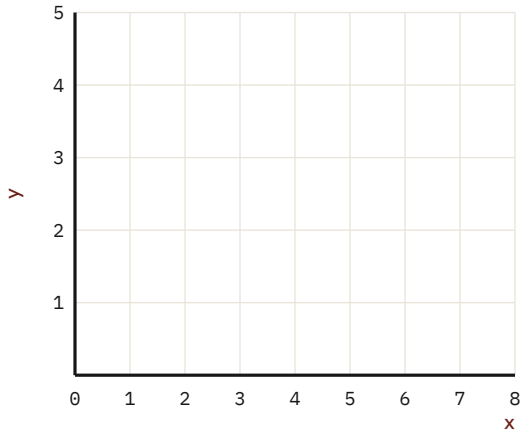
5 Name the point 4 right, 20 up

PART C Make a table and graph

6 y = 2x for x = 0 to 5



7 y = 0.5x for x = 0 to 8



PART D Exit Ticket

8 What do all proportional graphs have in common?

LESSON 1.10 · PRACTICE
Week 3 · Lesson 2 of 4

Unit 1: Proportional Relationships

Lesson 1.10: Recognizing Proportional Graphs

Name _____ Date _____

TODAY'S GOAL

I can tell from a graph whether a relationship is proportional.

Standards: 7.RP.2a

WHAT TO DO

- [ ] Read Textbook Lesson 1.10 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Proportional or not?, Part D Points on a line, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Answer each question.

1. A straight line through (0, 0) and (3, 12). Proportional? k?

2. A curved graph through (0, 0). Proportional?

3. A straight line through (0, 2) and (1, 5). Proportional?

PART B Warm-Up

Directions: Answer each item.

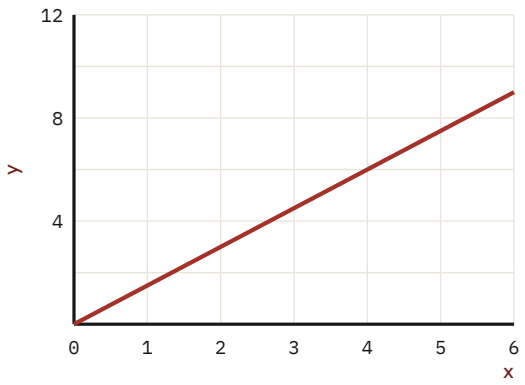
4 12 ÷ 3

5 Is (0, 0) the origin?

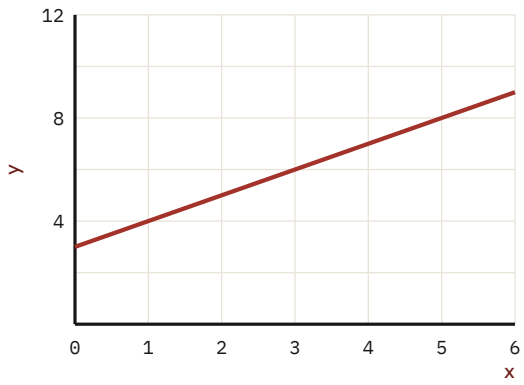
PART C Proportional or not?

Directions: Write proportional or not proportional for each graph. Explain.

6



7



PART D Points on a line

8 A proportional line goes through (4, 10). Name two more points on it.

9 Does a line through (2, 6) and (5, 15) show a proportional relationship?

PART E Exit Ticket

10 Sketch a graph that is a straight line but not proportional.

LESSON 1.11 · PRACTICE
Week 3 · Lesson 3 of 4

Unit 1: Proportional Relationships

Lesson 1.11: Interpreting Points on Proportional Graphs

Name _____ Date _____

TODAY'S GOAL

I can explain what a point (x, y) means on a proportional graph, including (0, 0) and (1, k).

Standards: 7.RP.2d

WHAT TO DO

- [ ] Read Textbook Lesson 1.11 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Interpret, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Answer each question about points on a proportional graph.

1. (1, 7.5) on a graph of hours and dollars means...

2. If (1, 12) is on a proportional graph, what is y when x = 5?

3. What does (0, 0) mean for pages read vs minutes?

PART B Warm-Up

Directions: Solve each problem. Show your work.

4 28 × 3

5 84 ÷ 3

PART C Interpret

6 A graph of cost vs pounds goes through (1, 4.5). What does it mean?

7 A graph of distance vs hours goes through (4, 260). Find k and explain.

8 A graph of cups of flour vs batches goes through (2, 5). What is the unit rate?

PART D Exit Ticket

9 Write a story for a proportional graph through (1, 15).

LESSON 1.12 · PRACTICE
Week 3 · Lesson 4 of 4

Unit 1: Proportional Relationships

Lesson 1.12: Comparing Proportional Relationships

Name _____ Date _____

TODAY'S GOAL

I can compare proportional relationships shown as tables, graphs, equations, and words.

Standards: 7.RP.2

WHAT TO DO

- [ ] Read Textbook Lesson 1.12 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Compare, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Answer each question.

1. Car A: y = 32x. Car B: 150 miles on 5 gallons. Better mileage?

2. Graph through (1, 2.75) vs y = 2.5x. Greater k?

3. Store A: $4 per pound. Store B: (5, 18). Cheaper?

PART B Warm-Up

Directions: Solve each problem. Show your work.

4 30 ÷ 3

5 18 ÷ 5

PART C Compare

6 Pump 1: 12 gallons per minute. Pump 2: y = 14x. Faster?

7 Runner A: table 3 mi in 24 min. Runner B: y = 7.5x (min per mile). Faster?

8 Recipe A: 2 cups sugar per 5 cups flour. Recipe B: y = 0.35x (sugar per flour). Sweeter?

9 Plan A: $45 for 3 months. Plan B: graph through (1, 16). Cheaper?

PART D Exit Ticket

10 Explain how to compare a table and a graph that both show proportional relationships.

WEEKLY CHECK
After Lesson 1.12

About 20 minutes

Weekly Check 3: Proportional Graphs

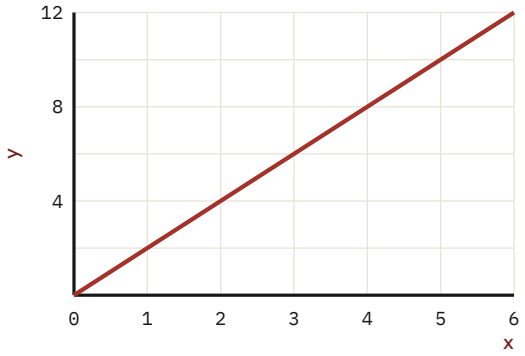
Name _____ Date _____

DIRECTIONS

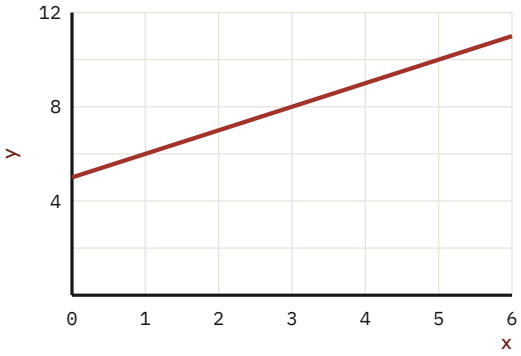
- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each question carefully. Show your thinking.
- [ ] When you finish, check that every question has an answer.

Show what you know

1 Proportional?



2 Proportional?



3 A proportional graph goes through (1, 6.5). What is k?

4 A proportional graph goes through (4, 18). What is y at x = 10?

5 Compare: y = 11x vs table 5 → 57.5. Greater rate?

6 What does (0, 0) mean on a graph of hours vs pay?

LESSON 1.13 · PRACTICE
Week 4 · Lesson 1 of 4

Unit 1: Proportional Relationships

Lesson 1.13: Solving Proportions

Name _____ Date _____

TODAY'S GOAL

I can solve problems with proportions using unit rates, scale factors, or equations.

Standards: 7.RP.3

WHAT TO DO

- [ ] Read Textbook Lesson 1.13 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Solve, Part D Word problems, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Answer each question.

1. 4 tickets cost $34. Cost of 10 tickets?

2. Solve $\frac{6}{15} = \frac{x}{25}$

3. A map: 2 cm = 15 km. 7 cm = ?

PART B Warm-Up

Directions: Solve each problem. Show your work.

4. $8.75 \div 5$

5. $96 \div 24$

PART C Solve

Directions: Solve each proportion for x.

6. $\frac{3}{8} = \frac{x}{40}$

7. $\frac{7}{x} = \frac{21}{33}$

8. $\frac{x}{12} = \frac{5}{4}$

9. $\frac{2.5}{4} = \frac{x}{10}$

PART D Word problems

10. 6 bags of mulch cover 72 sq ft. How many bags for 300 sq ft?

11. A 20-oz bottle has 240 calories. How many calories in 12 oz?

12. Ellie reads 45 pages in 1.5 hours. How long for 120 pages?

13. Paint: 2 gallons per 700 sq ft. Gallons for 1,750 sq ft?

PART E Exit Ticket

14. Solve two ways: 8 lemons make 3 quarts. How many lemons for 12 quarts?

LESSON 1.14 · PRACTICE
Week 4 · Lesson 2 of 4

Unit 1: Proportional Relationships

Lesson 1.14: Proportional Reasoning in the Real World

Name _____ Date _____

TODAY'S GOAL

I can solve real-world problems with proportional relationships, including recipes, maps, and speed.

Standards: 7.RP.3

WHAT TO DO

- [ ] Read Textbook Lesson 1.14 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Solve, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Answer each question.

1. A car uses 2 gallons per 54 miles. Gallons for 189 miles?

2. Map: 1 cm = 40 km. 6.5 cm = ?

3. $3/4$ cup sugar for 12 cookies. For 30 cookies?

PART B Warm-Up

Directions: Solve each problem. Show your work.

4 $3\frac{1}{2} \div 6$

5 3.6×25

PART C Solve

6 A recipe for 4 people uses 2 lb of chicken. How much for 14 people?

7 A cyclist rides 18 miles in 1.5 hours. At that rate, how far in 4 hours?

8 A model car is built at a scale of 1 in. : 18 in. The model is 9 in. long. How long is the real car in feet?

9 Lillie earns $57 for 6 hours of work. How many hours to earn $133?

PART D Exit Ticket

10 Write and solve a recipe-scaling problem for your family.

LESSON 1.15 · PRACTICE
Week 4 · Lesson 3 of 4
Unit 1: Proportional Relationships

Lesson 1.15: Multi-Step Ratio Problems

Name _____ Date _____

TODAY'S GOAL

I can solve multi-step problems involving ratios and rates.

Standards: 7.RP.3

WHAT TO DO

- [ ] Read Textbook Lesson 1.15 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Solve, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Answer each question.

1. Boys to girls 4:5; 36 students. How many girls?

2. Concrete mix 1:2:3 (cement:sand:gravel). 30 buckets total. Gravel?

3. Walk 1 hr at 3 mph, then jog 1 hr at 6 mph. Average speed?

PART B Warm-Up

Directions: Solve each problem. Show your work.

4 $20 \div 5 \times 3$

5 $100 + 195$

PART C Solve

6 A bag has red and green marbles in a ratio of 5:3. There are 48 marbles. How many are red?

7 A drink mix uses 2 scoops per 3 cups of water. How many scoops for a 9-quart cooler? (1 qt = 4 cups)

8 Lillie saves $3 for every $5 she earns. She earned $85. How much did she save? Spend?

9 A train travels 180 miles in 3 hours, then 120 miles in 1 hour. Average speed for the whole trip?

PART D Exit Ticket

10 A ratio of cats to dogs at a shelter is 2:7. There are 63 animals. How many more dogs than cats?

LESSON 1.16 · PRACTICE
Week 4 · Lesson 4 of 4
Unit 1: Proportional Relationships

Lesson 1.16: Proportional Relationships Problem Solving

Name _____ Date _____

TODAY'S GOAL

I can choose the best representation to solve proportional relationship problems.

Standards: 7.RP.1–3

WHAT TO DO

- [ ] Read Textbook Lesson 1.16 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Mixed practice, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Answer each question.

1. y = 18x: what is k?

2. Is y = x + 18 proportional?

3. Table: 4 → 14, 10 → 35. Equation?

PART B Warm-Up

Directions: Solve each problem. Show your work.

4 5.50 ÷ 50

5 0.12 × 100

PART C Mixed practice

6 $\frac{1}{2}$ mile in $\frac{1}{8}$ hour. mph?

7 Better buy: 15 oz for $3.30 or 20 oz for $4.20?

8 x: 2, 3, 7 → y: 5, 7.5, 17.5. Equation?

9 A graph passes through (3, 7). Proportional? Need more information?

10 Ratio 3:4, total 56. Larger part?

11 $42 for 3.5 hours. Hourly rate?

PART D Exit Ticket

12 Which representation do you find most useful for proportional relationships? Why?

UNIT REVIEW
Study before the unit test

About 40 minutes

Unit 1 Review

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each question carefully. Show your thinking.
- [ ] When you finish, check that every question has an answer.

PART A Rates and proportions

Directions: Find each unit rate, or solve the proportion.

1. $9.60 for 12 cans: unit price

2. $\frac{3}{4}$ pound for $\frac{1}{2}$ dollar: pounds per dollar

3. Solve $\frac{9}{12} = \frac{x}{20}$

4. Better buy: 6 for $4.50 or 10 for $7.20

PART B Proportional relationships

5. x: 3, 6, 12 → y: 2, 4, 8. Equation?

6. Is $y = 5x - 5$ proportional?

7. A proportional graph passes through (1, 3.25). Write the equation.

8. Ratio 2:5; total 63. Smaller part?

PART C Solve

9. A recipe for 8 people uses 3 cups of broth. How much for 20 people?

UNIT TEST
At the end of the unit
About 45 minutes

Unit 1 Test

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] This test has 20 points.
- [ ] Read each question carefully. Show your thinking.
- [ ] When you finish, check that every question has an answer.

PART A · Part A · Rates (1 point each)

Directions: Find each unit rate, or choose the better buy. Show your work.

1. $13.75 for 5 pounds: unit price

2. $1/2$ cup per $2/3$ batch: cups per batch

3. Better buy: 24 oz for $4.08 or 32 oz for $5.12

4. Write two ratios equivalent to 6:10.

PART B · Part B · Proportional relationships (1 point each)

Directions: Do what each item asks.

5. x: 4, 6, 10 → y: 10, 15, 25. Proportional? k?

6. x: 2, 4, 6 → y: 3, 7, 11. Proportional?

7. Write the equation: $8.25 per hour

8. y = 0.4x; x = 35

9. Is y = 7 + x proportional?

10. A proportional graph passes through (5, 40). k?

11 Interpret (1, 12) on a graph of hours vs miles.

PART C Part C · Problem solving (3 points each)

12 A car travels 196 miles on 7 gallons. Write an equation for miles m and gallons g. How far can it go on a full 15-gallon tank?

13 Sunscreen: 3 oz for $7.20 or 8 oz for $17.60. Which is the better buy? By how much per ounce?

14 Lemonade is 2 parts juice to 7 parts water. How much juice is in 45 cups of lemonade?

UNIT 2

Percents

Percent as a rate per 100, percent equations, discounts, tax, tips, percent change, interest, and percent error.

Your practice pages for Unit 2. Each lesson matches the same lesson number in your textbook.

UNIT VOCABULARY

percent	A rate per 100.
discount	Amount off a price.
markup	Amount added to cost.
percent change	Change ÷ original.
simple interest	$I = P \times r \times t$.
percent error	|estimate − actual| ÷ actual.

LESSON 2.1 · PRACTICE
Week 1 · Lesson 1 of 4
Unit 2: Percents

Lesson 2.1: What Is a Percent?

Name _____ Date _____

TODAY'S GOAL

I can explain a percent as a rate per 100.

Standards: 7.RP.3

WHAT TO DO

- [ ] Read Textbook Lesson 2.1 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Write as a percent, Part D Solve, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Write each amount as a percent.

1. 40 out of 100 = ___%

2. 7 out of 20 = ___%

3. What percent is 3 out of 4?

PART B Warm-Up

Directions: Multiply.

4. 25 × 4

5. 18 × 4

PART C Write as a percent

6. 13 out of 100

7. 9 out of 10

8. 12 out of 50

9. 3 out of 5

10. 7 out of 25

11. 1 out of 4

12. 20 out of 20

13. 11 out of 20

PART D Solve

14. A basketball player made 17 of 20 free throws. What percent did she make?

15. Lillie answered 46 of 50 questions correctly. Her percent score?

PART E Exit Ticket

16. Explain why 50% of something is half of it.

LESSON 2.2 · PRACTICE
Week 1 · Lesson 2 of 4
Unit 2: Percents

Lesson 2.2: Percents, Fractions, and Decimals

Name _____ Date _____

TODAY'S GOAL

I can write percents as fractions and decimals, and fractions and decimals as percents.

Standards: 7.RP.3, 7.NS.2d

WHAT TO DO

- [ ] Read Textbook Lesson 2.2 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Complete, Part D Order from least to greatest, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Rewrite each number in the form asked for.

1. 0.06 as a percent

2. 145% as a decimal

3. $\frac{7}{20}$ as a percent

PART B Warm-Up

Directions: Solve each problem. Show your work.

4. $3 \div 8$

5. 0.4×100

PART C Complete

Directions: Fill in each blank with the equivalent decimal or percent.

6. 62% = ___ (decimal)

7. 8% = ___ (decimal)

8. 0.9 = ___%

9. 1.25 = ___%

10. $\frac{2}{5}$ = ___%

11. $\frac{5}{8}$ = ___%

12. 30% = ___ (fraction, simplest)

13. 64% = ___ (fraction)

PART D Order from least to greatest

14. $\frac{2}{3}$, 0.65, 64%

PART E Exit Ticket

15. Explain why 0.05 is 5%, not 50%.

LESSON 2.3 · PRACTICE
Week 1 · Lesson 3 of 4
Unit 2: Percents

Lesson 2.3: Finding a Percent of a Number

Name _____ Date _____

TODAY'S GOAL

I can find a percent of a number using models and multiplication.

Standards: 7.RP.3

WHAT TO DO

- [ ] Read Textbook Lesson 2.3 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Find, Part D Solve, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Find the percent of each number. Show your work.

1. 20% of 85

2. 7% of 300

3. 120% of 50

PART B Warm-Up

Directions: Solve each problem. Show your work.

4. 10% of 60

5. 0.15 × 240

PART C Find

Directions: Find the percent of each number.

6. 10% of 90

7. 25% of 64

8. 40% of 35

9. 5% of 180

10. 75% of 48

11. 12% of 250

12. 150% of 30

13. 1% of 600

PART D Solve

14. A school has 450 students. 22% play an instrument. How many?

15. A $25 meal plus an 18% tip. How much is the tip?

PART E Exit Ticket

16. Find 35% of 80 two different ways.

LESSON 2.4 · PRACTICE
Week 1 · Lesson 4 of 4
Unit 2: Percents

Lesson 2.4: Finding What Percent

Name _____ Date _____

TODAY'S GOAL

I can find what percent one number is of another.

Standards: 7.RP.3

WHAT TO DO

- [ ] Read Textbook Lesson 2.4 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C What percent?, Part D Solve, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Answer each question.

1. What percent of 80 is 20?

2. What percent of 50 is 7?

3. What percent of 40 is 52?

PART B Warm-Up

Directions: Solve each problem. Show your work.

4. 42 ÷ 48

5. 0.875 = ___%

PART C What percent?

Directions: Write what percent the first number is of the second.

6. 9 of 36

7. 24 of 30

8. 13 of 52

9. 45 of 60

10. 3 of 12

11. 66 of 55

PART D Solve

12. A team won 27 of 36 games. What percent did it win?

13. A store sold 340 of its 400 jackets. What percent are left?

PART E Exit Ticket

14. A shirt costs $12. A sweater costs $30. The shirt's price is what percent of the sweater's?

WEEKLY CHECK
After Lesson 2.4

About 15 minutes

Weekly Check 4: Percent Basics

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each question carefully. Show your thinking.
- [ ] When you finish, check that every question has an answer.

Show what you know

Directions: Write each as a percent.

1. 9 out of 20 = ___%

2. $^{3}/_{8}$ = ___%

3. 0.07 = ___%

4. 30% of 70

5. What percent of 60 is 45?

6. A team won 14 of 16 games. Percent won?

LESSON 2.5 · PRACTICE
Week 2 · Lesson 1 of 4
Unit 2: Percents

Lesson 2.5: Finding the Whole

Name _____ Date _____

TODAY'S GOAL

I can find the whole when I know a part and its percent.

Standards: 7.RP.3

WHAT TO DO

- [ ] Read Textbook Lesson 2.5 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Find the whole, Part D Solve, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Answer each question.

1. 20% of what number is 14?

2. 75% of what number is 60?

3. 8 is 40% of what?

PART B Warm-Up

Directions: Solve each problem. Show your work.

4. 45 ÷ 0.30

5. If 10% is 7, 100% is ___

PART C Find the whole

Directions: Find the whole that makes each sentence true.

6. 10% of ___ is 12

7. 25% of ___ is 9

8. 50% of ___ is 37

9. 60% of ___ is 42

10. 5% of ___ is 3

11. 125% of ___ is 50

PART D Solve

12. A player scored 24 points, which was 40% of the team's points. How many points did the team score?

13. Ellie paid a $6 deposit, which is 15% of the price. What is the price?

PART E Exit Ticket

14. 18 students walk to school. That is 12% of the school. How many students attend?

LESSON 2.6 · PRACTICE
Week 2 · Lesson 2 of 4
Unit 2: Percents

Lesson 2.6: The Percent Equation

Name _____ Date _____

TODAY'S GOAL

I can use part = percent × whole to solve any percent problem.

Standards: 7.RP.3, 7.EE.4

WHAT TO DO

- [ ] Read Textbook Lesson 2.6 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Write an equation and solve, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Answer each question.

1. What is 64% of 75?

2. 27 is what percent of 90?

3. 33 is 55% of what?

PART B Warm-Up

Directions: Solve each problem. Show your work.

4. 36 ÷ 0.45

5. 0.64 × 75

PART C Write an equation and solve

6. What is 18% of 350?

7. 42 is what percent of 168?

8. 91 is 70% of what?

9. What is 2.5% of 400?

10. 60 is what percent of 48?

11. 15 is 6% of what?

PART D Exit Ticket

12. Write and solve: 14 is 35% of what number?

LESSON 2.7 · PRACTICE
Week 2 · Lesson 3 of 4
Unit 2: Percents

Lesson 2.7: Percents Greater Than 100% and Less Than 1%

Name _____ Date _____

TODAY'S GOAL

I can work with percents greater than 100% and less than 1%.

Standards: 7.RP.3

WHAT TO DO

- [ ] Read Textbook Lesson 2.7 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Find, Part D Solve, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Find each percent of the number, or rewrite the percent as a decimal.

1. 300% of 15

2. 0.2% of 5,000

3. 125% as a decimal

PART B Warm-Up

Directions: Solve each problem. Show your work.

4 2.5 × 80

5 0.005 × 2,000

PART C Find

Directions: Find the percent of each number.

6 200% of 34

7 110% of 60

8 0.1% of 9,000

9 0.75% of 400

10 175% of 20

11 0.4% as a decimal

PART D Solve

12 A puppy's weight is now 300% of its birth weight of 2.5 lb. How much does it weigh?

13 A savings account earns 0.6% per year on $1,500. How much interest in 1 year?

PART E Exit Ticket

14 Is 150% of a number more or less than the number? Explain.

LESSON 2.8 · PRACTICE
Week 2 · Lesson 4 of 4
Unit 2: Percents

Lesson 2.8: Estimating with Percents

Name _____ Date _____

TODAY'S GOAL

I can estimate percents using benchmarks like 10%, 25%, and 50%.

Standards: 7.RP.3, 7.EE.3

WHAT TO DO

- [ ] Read Textbook Lesson 2.8 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Estimate, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Answer each question.

1. Estimate 10% of $83.20

2. Estimate 15% of $59

3. Estimate 48% of 202

PART B Warm-Up

Directions: Solve each problem. Show your work.

4 10% of 47.80

5 Half of 4.80

PART C Estimate

Directions: Estimate each amount using friendly numbers. Show how you rounded.

6 20% of $31.75

7 25% of 397

8 5% of $198

9 33% of 61

10 Tip of 18% on $52

11 75% of 81

PART D Exit Ticket

12 Explain how to find 15% of a restaurant bill in your head.

WEEKLY CHECK
After Lesson 2.8

About 20 minutes

Weekly Check 5: Percent Equations

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each question carefully. Show your thinking.
- [ ] When you finish, check that every question has an answer.

Show what you know

Directions: Solve each problem. Show your work.

1. 40% of what is 26?

2. 18 is what percent of 24?

3. 250% of 18

4. 0.3% of 2,000

5. Estimate 15% of $79.

6. Write and solve: 63 is 35% of n.

LESSON 2.9 · PRACTICE
Week 3 · Lesson 1 of 4
Unit 2: Percents

Lesson 2.9: Discounts and Sale Prices

Name _____ Date _____

TODAY'S GOAL

I can find the sale price after a percent discount.

Standards: 7.RP.3

WHAT TO DO

- [ ] Read Textbook Lesson 2.9 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Find the sale price, Part D Solve, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Find the sale price, or the percent off. Show your work.

1. 40% off $45

2. 15% off $120

3. A $90 coat is on sale for $63. What percent off?

PART B Warm-Up

Directions: Solve each problem. Show your work.

4. 0.25×80

5. $100\% - 15\%$

PART C Find the sale price

6. 20% off $35.00

7. 30% off $50.00

8. 10% off $89.90

9. 35% off $60.00

10. 50% off $27.80

11. 12% off $250.00

PART D Solve

12. A video game costs $59.99. It is 30% off. What is the sale price?

13. A $150 bike is on sale for $105. What is the percent discount?

PART E Exit Ticket

14. Is 20% off and then another 20% off the same as 40% off? Try it with $100.

LESSON 2.10 · PRACTICE
Week 3 · Lesson 2 of 4
Unit 2: Percents

Lesson 2.10: Markups and Sales Tax

Name _____ Date _____

TODAY'S GOAL

I can find prices after a markup or sales tax.

Standards: 7.RP.3

WHAT TO DO

- [ ] Read Textbook Lesson 2.10 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Find the total, Part D Solve, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Find the tax, markup, or total. Show your work.

1. 7% tax on $15

2. Markup 40% on $30

3. Total with 8% tax on $62.50

PART B Warm-Up

Directions: Solve each problem. Show your work.

4 0.07 × 40

5 1.6 × 25

PART C Find the total

6 7% tax on $20.00

7 7% tax on $64.00

8 5% tax on $38.00

9 50% markup on $18.00

10 25% markup on $64.00

11 120% markup on $15.00

PART D Solve

12 A bookstore buys a book for $12 and sells it for $18. What is the percent markup?

13 Lillie buys $28.50 of supplies with 7% tax. What does she pay?

PART E Exit Ticket

14 A $50 item has a 20% markup, then a 20% discount. What is the final price?

LESSON 2.11 · PRACTICE
Week 3 · Lesson 3 of 4
Unit 2: Percents

Lesson 2.11: Percent Increase and Decrease

Name _____ Date _____

TODAY'S GOAL

I can find the percent of increase or decrease.

Standards: 7.RP.3

WHAT TO DO

- [ ] Read Textbook Lesson 2.11 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Find the percent change, Part D Solve, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Find the percent increase or decrease. Write increase or decrease.

1. $40 → $50

2. 80 → 60

3. 150 → 165

PART B Warm-Up

Directions: Solve each problem. Show your work.

4. $6 \div 8$

5. $300 \div 2{,}500$

PART C Find the percent change

Directions: Find the percent change. Write increase or decrease.

6. $20 \rightarrow 25$

7. $50 \rightarrow 35$

8. $12 \rightarrow 18$

9. $200 \rightarrow 170$

10. $64 \rightarrow 80$

11. $90 \rightarrow 63$

PART D Solve

12. A gallon of milk went from \$3.20 to \$3.60. Percent increase?

13. A plant was 15 cm tall and is now 24 cm. Percent increase?

PART E Exit Ticket

14. A price goes from \$50 to \$100, then back to \$50. Are the percent changes the same? Explain.

LESSON 2.12 · PRACTICE
Week 3 · Lesson 4 of 4
Unit 2: Percents

Lesson 2.12: Tips, Commissions, and Fees

Name _____ Date _____

TODAY'S GOAL

I can solve problems with tips, commissions, and fees.

Standards: 7.RP.3

WHAT TO DO

- [ ] Read Textbook Lesson 2.12 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Solve, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Find each tip, commission, or fee. Show your work.

1. 15% tip on $36

2. 4% commission on $8,500

3. 3% card fee on $120

PART B Warm-Up

Directions: Solve each problem. Show your work.

4. 0.06 × 1,250

5. 0.18 × 42

PART C Solve

Directions: Solve each problem. Show your work.

6. 20% tip on a $55 dinner. Total?

7. 5% commission on a $24,000 car

8. A $2 fee plus 10% on a $30 ticket. Total?

9. Lillie earns $8 per hour plus 10% of her sales. She works 5 hours and sells $300. Pay?

10. 15% tip on a $28.40 haircut

11. A real estate agent earns 3% on a $180,000 house

PART D Exit Ticket

12. Which earns more: 8% commission on $2,500 or a flat $175?

WEEKLY CHECK
After Lesson 2.12

About 20 minutes

Weekly Check 6: Discounts, Tax, and Change

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each question carefully. Show your thinking.
- [ ] When you finish, check that every question has an answer.

Show what you know

Directions: Solve each problem. Show your work.

1. 35% off $80

2. 7% tax on $45

3. $30 → $36 percent change

4. $250 → $200 percent change

5. 18% tip on $50

6. 6% commission on $3,500

LESSON 2.13 · PRACTICE
Week 4 · Lesson 1 of 4
Unit 2: Percents

Lesson 2.13: Simple Interest

Name _____ Date _____

TODAY'S GOAL

I can find simple interest and the total amount in an account or loan.

Standards: 7.RP.3, 7.EE.3

WHAT TO DO

- [ ] Read Textbook Lesson 2.13 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Find the interest and total, Part D Solve, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Use I = Prt to find the interest, and the total when asked.

1. $800 at 5% for 3 years

2. $2,000 at 4.5% for 2 years. Total?

3. $300 at 2% for 6 months (0.5 year)

PART B Warm-Up

Directions: Solve each problem. Show your work.

4. 500 × 0.03

5. 15 × 4

PART C Find the interest and total

6. $400.00 at 5% for 2 years

7. $1,500.00 at 3% for 4 years

8. $250.00 at 8% for 1 years

9. $6,000.00 at 2.5% for 3 years

PART D Solve

10. Which earns more: $1,000 at 4% for 3 years or $1,200 at 3% for 3 years?

PART E Exit Ticket

11. Explain why borrowing money costs more than the amount borrowed.

LESSON 2.14 · PRACTICE
Week 4 · Lesson 2 of 4
Unit 2: Percents

Lesson 2.14: Percent Error

Name _____ Date _____

TODAY'S GOAL

I can find the percent error of an estimate or measurement.

Standards: 7.RP.3

WHAT TO DO

- [ ] Read Textbook Lesson 2.14 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Find the percent error, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Find the percent error. Percent error = |estimate − actual| ÷ actual × 100.

1. Estimate 45, actual 50

2. Measured 8.4 cm, actual 8 cm

3. Guessed 300, actual 240

PART B Warm-Up

Directions: Solve each problem. Show your work.

4 $|120 - 150|$

5 $30 \div 150$

PART C Find the percent error

6 Estimate 90, actual 100

7 Estimate 55, actual 50

8 Estimate 18, actual 24

9 Estimate 420, actual 400

10 Estimate 3.6, actual 4

11 Estimate 130, actual 125

PART D Exit Ticket

Directions: Find the percent error. Show your work.

12 Lillie estimated a room was 12 ft long. It was 15 ft. Find the percent error.

LESSON 2.15 · PRACTICE
Week 4 · Lesson 3 of 4
Unit 2: Percents

Lesson 2.15: Multi-Step Percent Problems

Name _____ Date _____

TODAY'S GOAL

I can solve problems that combine discounts, tax, tips, and percent change.

Standards: 7.RP.3

WHAT TO DO

- [ ] Read Textbook Lesson 2.15 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Solve, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Solve each step in order. Show your work.

1. $40 shirt, 20% off, then 7% tax

2. $25 meal, 8% tax, then 20% tip on the pre-tax amount. Total?

3. $200 rises 25%, then falls 20%

PART B Warm-Up

Directions: Solve each problem. Show your work.

4 0.75 × 60

5 1.07 × 45

PART C Solve

6 A $120 jacket is 30% off. Tax is 7%. Total?

7 A family's dinner is $64. Tax is 7% and they tip 18% of $64. Total?

8 A stock worth $50 rises 20% and then falls 15%. What is it worth?

9 A store marks up a $30 item 50%, then puts it on sale for 20% off. Sale price?

PART D Exit Ticket

10 Does the order of a 10% discount and 7% tax matter for the final price? Test with $100.

LESSON 2.16 · PRACTICE
Week 4 · Lesson 4 of 4
Unit 2: Percents

Lesson 2.16: Percent Problem Solving

Name _____ Date _____

TODAY'S GOAL

I can choose and use the right percent strategy to solve real problems.

Standards: 7.RP.3

WHAT TO DO

- [ ] Read Textbook Lesson 2.16 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Mixed practice, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Solve each problem. Show your work.

1. What percent of 64 is 16?

2. 12% of what is 9?

3. $75 → $90 percent change

PART B Warm-Up

Directions: Solve each problem. Show your work.

4 0.8 × 35

5 1.07 × 64

PART C Mixed practice

Directions: Solve each problem. Show your work.

6 35% of 260

7 54 is what percent of 72?

8 27 is 45% of what?

9 Sale price: 15% off $86

10 I for $900 at 4% for 5 years

11 Percent error: estimate 38, actual 40

PART D Exit Ticket

12 Write a multi-step percent problem about something you want to buy. Solve it.

UNIT REVIEW
Study before the unit test

About 40 minutes

Unit 2 Review

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each question carefully. Show your thinking.
- [ ] When you finish, check that every question has an answer.

PART A Percent basics

Directions: Write each as a percent, or find the percent of the number.

1. 0.45 = ___%

2. $^{7}/_{8}$ = ___%

3. 24% of 150

4. What percent of 80 is 68?

5. 15 is 60% of what?

PART B Applications

Directions: Solve each problem. Show your work.

6. 25% off $64, then 7% tax

7. Percent change: 40 → 52

8. I = ? for $2,500 at 3% for 4 years

9. Percent error: guess 90, actual 75

UNIT TEST
At the end of the unit

About 45 minutes

Unit 2 Test

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] This test has 23 points.
- [ ] Read each question carefully. Show your thinking.
- [ ] When you finish, check that every question has an answer.

PART A Part A · Percent basics (1 point each)

Directions: Write each as a percent.

1. 13 out of 25 = ___%

2. 1.8 = ___%

3. $^{5}/_{8}$ = ___%

4. 0.4% as a decimal

PART B Part B · Percent equations (1 point each)

Directions: Solve each problem. Show your work.

5. 16% of 75

6. What percent of 120 is 42?

7. 51 is 85% of what?

8. 150% of 46

9. Estimate 18% of $41

PART C Part C · Applications (2 points each)

10 A $72 jacket is 35% off. Sale price?

11 A $36 meal with 7% tax and a 20% tip on $36. Total?

12 A town grew from 4,800 to 5,520 people. Percent increase?

13 $1,800 at 3.5% simple interest for 3 years. Interest?

PART D Part D · Problem solving (3 points each)

14 Store A sells a $90 bike at 30% off. Store B sells the same bike for $70 with a $5 coupon. Which is cheaper, and by how much?

15 Lillie guessed a pumpkin weighed 12 lb. It weighed 15 lb. Ellie guessed 17 lb. Who had the smaller percent error?

UNIT 3

Adding and Subtracting Integers

Opposites, absolute value, adding and subtracting integers with chips and number lines, and real-world applications.

Your practice pages for Unit 3. Each lesson matches the same lesson number in your textbook.

UNIT VOCABULARY

integer	Whole numbers and their opposites.
opposites	Same distance from 0, opposite sides.
absolute value	Distance from 0.
additive inverse	A number's opposite; sums to 0.

LESSON 3.1 · PRACTICE
Week 1 · Lesson 1 of 4

Unit 3: Adding and Subtracting Integers

Lesson 3.1: Integers, Opposites, and Absolute Value

Name _____ Date _____

TODAY'S GOAL

I can use integers to describe real situations and find opposites and absolute values.

Standards: 7.NS.1 (review of 6.NS.5–7)

WHAT TO DO

- [ ] Read Textbook Lesson 3.1 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Write an integer, Part D Find, Part E Order from least to greatest, Part F Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Write the integer, the opposite, or the absolute value.

1. Write an integer: a loss of 8 yards

2. Opposite of −15

3. |−23|

PART B Warm-Up

4 Which is colder: −3°F or −8°F?

5 Name a number between −2 and 0.

PART C Write an integer

Directions: Write an integer for each situation.

6 a deposit of $40

7 20 feet below sea level

8 a temperature drop of 6 degrees

9 a gain of 3 yards

PART D Find

Directions: Write the opposite or the absolute value.

10 opposite of 31

11 opposite of −4

12 |−18|

13 |0|

14 −|−6|

15 |14| − |−5|

PART E Order from least to greatest

16 3, −7, 0, −2, 5

PART F Exit Ticket

17 Can absolute value be negative? Explain.

LESSON 3.2 · PRACTICE
Week 1 · Lesson 2 of 4

Unit 3: Adding and Subtracting Integers

Lesson 3.2: Adding Integers with the Same Sign

Name _____ Date _____

TODAY'S GOAL

I can add two integers that have the same sign.

Standards: 7.NS.1b

WHAT TO DO

- [ ] Read Textbook Lesson 3.2 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Add, Part D Solve, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Add the integers.

1. $-5 + (-8)$

2. $12 + 9$

3. $-20 + (-15)$

PART B Warm-Up

Directions: Solve each problem. Show your work.

4. $|-9| + |-6|$

5. $4 + 3$

PART C Add

6. $-3 + (-4)$

7. $-11 + (-2)$

8. $6 + 8$

9. $-7 + (-7)$

10. $-15 + (-9)$

11. $-24 + (-36)$

12. $13 + 27$

13. $-50 + (-50)$

PART D Solve

14. A diver is at −12 m and dives 8 m deeper. Where is she now?

15. The temperature is −5°F and drops 9 more degrees. New temperature?

PART E Exit Ticket

16. Use chips to show $-4 + (-5)$. What is the sum?

LESSON 3.3 · PRACTICE
Week 1 · Lesson 3 of 4

Unit 3: Adding and Subtracting Integers

Lesson 3.3: Adding Integers with Different Signs

Name _____ Date _____

TODAY'S GOAL

I can add two integers with different signs.

Standards: 7.NS.1b

WHAT TO DO

- [ ] Read Textbook Lesson 3.3 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Add, Part D Solve, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Add the integers.

1. $7 + (-10)$

2. $-6 + 15$

3. $-25 + 18$

PART B Warm-Up

Directions: Solve each problem. Show your work.

4. $-4 + (-5)$

5. $14 - 9$

PART C Add

6. $5 + (-2)$

7. $-5 + 2$

8. $9 + (-13)$

9. $-12 + 20$

10. $30 + (-45)$

11. $-17 + 17$

12. $-8 + 3$

13. $21 + (-6)$

PART D Solve

14. Your account has $35. You buy a $52 game with an overdraft. Balance?

15. A football team gains 6 yards, then loses 9. Total?

PART E Exit Ticket

16. Explain why $-7 + 7 = 0$ using zero pairs.

LESSON 3.4 · PRACTICE
Week 1 · Lesson 4 of 4

Unit 3: Adding and Subtracting Integers

Lesson 3.4: Additive Inverses

Name _____ Date _____

TODAY'S GOAL

I can explain why a number plus its opposite equals 0.

Standards: 7.NS.1a, 7.NS.1b

WHAT TO DO

- [ ] Read Textbook Lesson 3.4 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Find the sum using inverses, Part D Missing number, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Add, using opposites that make zero, or find the missing number.

1. $-42 + 42$

2. $6 + (-9) + 9$

3. $___ + 17 = 0$

PART B Warm-Up

Directions: Add.

4. $-6 + 6$

5. $15 + (-15)$

PART C Find the sum using inverses

6. $-7 + 12 + 7$

7. $25 + (-18) + (-25)$

8. $-3 + 9 + (-9) + 3$

9. $40 + (-11) + (-40)$

PART D Missing number

Directions: Write the number that makes each sentence true.

10. $-8 + ___ = 0$

11. $___ + (-26) = 0$

12. $-5 + ___ = 2$

13. $14 + ___ = -1$

PART E Exit Ticket

14. Describe a real situation where two opposite quantities combine to make 0.

WEEKLY CHECK
After Lesson 3.4

About 15 minutes

Weekly Check 7: Adding Integers

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each question carefully. Show your thinking.
- [ ] When you finish, check that every question has an answer.

Show what you know

Directions: Add.

1. $-9 + (-14)$

2. $-9 + 14$

3. $21 + (-30)$

4. $|-17|$

5. $-13 + ___ = 0$

6. $-6 + 11 + 6$

LESSON 3.5 · PRACTICE
Week 2 · Lesson 1 of 4

Unit 3: Adding and Subtracting Integers

Lesson 3.5: Adding Integers on a Number Line

Name _____ Date _____

TODAY'S GOAL

I can use a number line to add integers.

Standards: 7.NS.1b

WHAT TO DO

- [ ] Read Textbook Lesson 3.5 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Use the number line, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Add. Use a number line to show each move.

1. $-4 + 6$ (use a number line)

2. $2 + (-5) + (-1)$

3. $-3 + (-3) + 8$

PART B Warm-Up

Directions: Add.

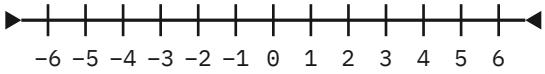
4 $-7 + 4$

5 $-2 + (-6)$

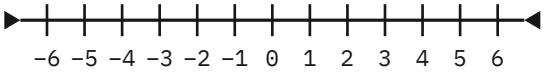
PART C Use the number line

Directions: Add. Draw the moves on a number line.

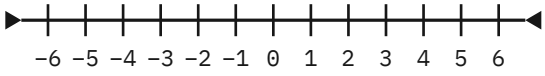
6 $-5 + 8$



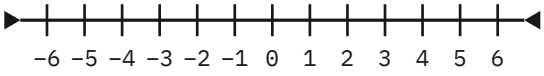
7 $4 + (-9)$



8 $-1 + (-4) + 3$



9 $6 + (-2) + (-7)$



PART D Exit Ticket

10 Draw a number line for $-6 + 10$ and give the sum.

LESSON 3.6 · PRACTICE
Week 2 · Lesson 2 of 4
Unit 3: Adding and Subtracting Integers

Lesson 3.6: Subtracting Integers

Name _____ Date _____

TODAY'S GOAL

I can subtract integers by adding the opposite.

Standards: 7.NS.1c

WHAT TO DO

- [ ] Read Textbook Lesson 3.6 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Subtract, Part D Rewrite as addition, then solve, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Subtract. Add the opposite.

1. $5 - 12$

2. $-8 - (-3)$

3. $-4 - 10$

PART B Warm-Up

Directions: Answer each item.

4 Opposite of −9

5 −6 + (−4)

PART C Subtract

6 7 − 10

7 −3 − 5

8 4 − (−6)

9 −9 − (−2)

10 −1 − (−8)

11 0 − (−15)

12 12 − (−12)

13 −20 − 13

PART D Rewrite as addition, then solve

14 15 − (−7)

15 −11 − 4

PART E Exit Ticket

16 Explain why 6 − (−2) is greater than 6.

LESSON 3.7 · PRACTICE
Week 2 · Lesson 3 of 4

Unit 3: Adding and Subtracting Integers

Lesson 3.7: Subtraction as Distance

Name _____ Date _____

TODAY'S GOAL

I can use a number line to show subtraction as the change from one number to another.

Standards: 7.NS.1c

WHAT TO DO

- [ ] Read Textbook Lesson 3.7 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Find the change (end − start), Part D Solve, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Answer each question.

1. From −3 to 9: change?

2. From 5 to −7: change?

3. From −10 to −2: change?

PART B Warm-Up

Directions: Subtract.

4. 11 − (−4)

5. −8 − 25

PART C Find the change (end − start)

6. from −2 to 6

7. from 8 to −1

8. from −15 to −5

9. from 0 to −9

10. from −7 to −20

11. from 14 to −3

PART D Solve

12. A submarine at −120 m rises to −45 m. Change?

13. A company's profit goes from $3 million to −$2 million. What is the change?

PART E Exit Ticket

14. Explain how a number line shows 4 − (−3) = 7.

LESSON 3.8 · PRACTICE
Week 2 · Lesson 4 of 4
Unit 3: Adding and Subtracting Integers

Lesson 3.8: Distance Between Two Numbers

Name _____ Date _____

TODAY'S GOAL

I can find the distance between two numbers using absolute value.

Standards: 7.NS.1c

WHAT TO DO

- [ ] Read Textbook Lesson 3.8 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Find the distance, Part D Solve, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Find the distance between the numbers.

1. Distance between −7 and 2

2. Distance between −12 and −4

3. $|-6 - 9|$

PART B Warm-Up

Directions: Solve each problem. Show your work.

4 $|-8|$

5 $3 - (-5)$

PART C Find the distance

6 −3 and 4

7 10 and −10

8 −25 and −9

9 0 and −14

10 −6 and 6

11 −100 and 35

PART D Solve

12 Highest Mississippi point: Woodall Mountain, 807 ft. Lowest: sea level on the Gulf, 0 ft. Difference?

13 Low of −11°F and high of 23°F. Temperature range?

PART E Exit Ticket

14 Why do we use absolute value for distance?

WEEKLY CHECK
After Lesson 3.8

About 15 minutes

Weekly Check 8: Subtracting Integers

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each question carefully. Show your thinking.
- [ ] When you finish, check that every question has an answer.

Show what you know

Directions: Subtract.

1. $4 - 11$

2. $-7 - (-3)$

3. $-5 - 8$

4. $10 - (-6)$

5. Distance between -8 and 5

6. From $-6°F$ to $9°F$: change?

LESSON 3.9 · PRACTICE
Week 3 · Lesson 1 of 4

Unit 3: Adding and Subtracting Integers

Lesson 3.9: Integers: Temperature and Elevation

Name _____ Date _____

TODAY'S GOAL

I can solve temperature and elevation problems with integers.

Standards: 7.NS.1, 7.NS.3

WHAT TO DO

- [ ] Read Textbook Lesson 3.9 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Solve, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Write an expression and solve. Include units.

1. A diver at −30 ft rises 12 ft.

2. 8°F drops 15°.

3. A balloon at 400 ft descends 650 ft into a canyon. New elevation?

PART B Warm-Up

Directions: Solve each problem. Show your work.

4 −6 + 23

5 9 − 5

PART C Solve

6 The temperature was −12°F and rose 19°. New temperature?

7 A plane at 2,500 ft descends 3,100 ft over the Dead Sea basin. New elevation?

8 A miner is 450 ft underground and goes up 120 ft. Elevation?

9 Morning −5°F, afternoon 14°F. How much warmer?

PART D Exit Ticket

10 Write a temperature story for −8 + 13 and solve it.

LESSON 3.10 · PRACTICE
Week 3 · Lesson 2 of 4

Unit 3: Adding and Subtracting Integers

Lesson 3.10: Integers and Money

Name _____ Date _____

TODAY'S GOAL

I can use integers to track money: deposits, withdrawals, profits, and debts.

Standards: 7.NS.1, 7.NS.3

WHAT TO DO

- [ ] Read Textbook Lesson 3.10 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Track the balance, Part D Solve, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Find each new balance or total.

1. $10 balance, buy a $14 item

2. −$20 balance, deposit $35

3. Profits: $50, −$65, $20. Total?

PART B Warm-Up

Directions: Add.

4 25 + (−40)

5 −15 + 30

PART C Track the balance

Directions: Find the balance after each step. Write the final balance.

6 Start $0. Earn $45, spend $28, spend $30, earn $20.

7 Start −$12. Deposit $50, withdraw $65.

PART D Solve

8 A business lost $1,200 in January and earned $2,050 in February. Total?

9 You owe $35 and pay back $22. How much do you owe now?

PART E Exit Ticket

10 Make a 4-transaction account record that ends at −$5.

LESSON 3.11 · PRACTICE
Week 3 · Lesson 3 of 4
Unit 3: Adding and Subtracting Integers

Lesson 3.11: Multi-Step Integer Problems

Name _____ Date _____

TODAY'S GOAL

I can solve multi-step problems with integer addition and subtraction.

Standards: 7.NS.3, 7.EE.3

WHAT TO DO

- [ ] Read Textbook Lesson 3.11 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Evaluate, Part D Solve, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Solve each problem. Show your work.

1. $-8 + 15 - 6 + 4$

2. A temperature of 3°F drops 10°, rises 4°, drops 2°.

3. Elevation −20 m: up 35, down 12, up 6

PART B Warm-Up

Directions: Add.

4. 12 + 25

5. −7 + (−3)

PART C Evaluate

Directions: Evaluate each expression from left to right.

6. −6 + 9 − 4

7. 15 − 22 + 3

8. −10 − 5 + 18

9. 7 − (−3) − 12

10. −4 + (−8) − (−6)

11. 20 − 35 + 10 − 5

PART D Solve

12. A hiker starts at −40 ft (in a valley), climbs 125 ft, descends 60 ft, climbs 15 ft. Elevation?

PART E Exit Ticket

13. Write a 3-step temperature story and solve it.

LESSON 3.12 · PRACTICE
Week 3 · Lesson 4 of 4
Unit 3: Adding and Subtracting Integers

Lesson 3.12: Properties and Mental Math with Integers

Name _____ Date _____

TODAY'S GOAL

I can use properties of addition to add integers mentally.

Standards: 7.NS.1d

WHAT TO DO

- [ ] Read Textbook Lesson 3.12 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Use properties to evaluate mentally, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Use properties (like grouping opposites) to evaluate each sum mentally.

1. $-25 + 38 + 25$

2. $14 + (-9) + (-14) + 9$

3. $-36 + 50 + (-14)$

PART B Warm-Up

Directions: Add.

4. $-17 + 17$

5. $45 + (-5)$

PART C Use properties to evaluate mentally

6. $-8 + 27 + 8$

7. $45 + (-19) + (-45)$

8. $-13 + (-7) + 50$

9. $64 - 30 - 34$

10. $-99 + 50 + 99 + (-50)$

11. $125 + (-75) + (-25)$

PART D Exit Ticket

12. Show two ways to group $-6 + 16 + (-4)$.

WEEKLY CHECK
After Lesson 3.12

About 15 minutes

Weekly Check 9: Integer Applications

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each question carefully. Show your thinking.
- [ ] When you finish, check that every question has an answer.

Show what you know

Directions: Solve each problem. Show your work.

1. −15 m diver rises 9 m

2. $20 balance, spend $32

3. −4 + 10 − 9 + 2

4. −29 + 46 + 29 (mentally)

5. Temp 6°F, drops 11°, rises 3°

6. Profits $40, −$55, $30: total

LESSON 3.13 · PRACTICE
Week 4 · Lesson 1 of 4

Unit 3: Adding and Subtracting Integers

Lesson 3.13: Integer Expressions

Name _____ Date _____

TODAY'S GOAL

I can evaluate expressions with integers and variables.

Standards: 7.NS.1, 7.EE.1

WHAT TO DO

- [ ] Read Textbook Lesson 3.13 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Evaluate when x = −5 and y = 3, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Evaluate each expression for the given values.

1. $x - 7$ when $x = -3$

2. $-y + 5$ when $y = -2$

3. $a + b - c$ when $a = 4$, $b = -9$, $c = -2$

PART B Warm-Up

Directions: Evaluate.

4 $-6 - (-11)$

5 $-(-4)$

PART C Evaluate when $x = -5$ and $y = 3$

6 $x + y$

7 $x - y$

8 $y - x$

9 $-x$

10 $x + x + y$

11 $|x| - y$

PART D Exit Ticket

12 Find a value of x so that $x - 8$ is negative and $x + 8$ is positive.

LESSON 3.14 · PRACTICE
Week 4 · Lesson 2 of 4

Unit 3: Adding and Subtracting Integers

Lesson 3.14: Explaining Integer Rules with Models

Name _____ Date _____

TODAY'S GOAL

I can explain integer addition and subtraction rules using chips and number lines.

Standards: 7.NS.1

WHAT TO DO

- [ ] Read Textbook Lesson 3.14 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Explain with a model and solve, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Answer each question.

1. Explain $-6 + 6 = 0$ with chips.

2. Explain $2 - 7 = -5$ on a number line.

3. Explain $-1 - (-4) = 3$ with chips.

PART B Warm-Up

Directions: Add or subtract.

4 $4 - (-2)$

5 $-3 + 5$

PART C Explain with a model and solve

Directions: Draw a model (chips or a number line) and solve.

6 $-5 + 2$

7 $3 - 8$

8 $-4 - (-6)$

9 $0 - (-3)$

PART D Exit Ticket

10 Which model do you prefer and why?

LESSON 3.15 · PRACTICE
Week 4 · Lesson 3 of 4
Unit 3: Adding and Subtracting Integers

Lesson 3.15: Finding and Fixing Integer Errors

Name _____ Date _____

TODAY'S GOAL

I can find and fix mistakes in integer calculations.

Standards: 7.NS.1

WHAT TO DO

- [ ] Read Textbook Lesson 3.15 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Find and fix the error, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Find the error in each problem. Write the correct answer.

1. Fix: $-7 + (-2) = -5$

2. Fix: $3 - 10 = 7$

3. Fix: $-6 - (-6) = -12$

PART B Warm-Up

Directions: Subtract.

4. $-8 - 3$

5. $5 - (-2)$

PART C Find and fix the error

Directions: Each answer is wrong. Write the correct answer and explain the mistake.

6. $-12 + 5 = -17$

7. $9 - (-4) = 5$

8. $-3 - 3 = 0$

9. $-10 + 15 = -5$

10. $0 - 8 = 8$

11. $|-2| + |-3| = -5$

PART D Exit Ticket

12. What is the most common integer mistake you make? How will you avoid it?

LESSON 3.16 · PRACTICE
Week 4 · Lesson 4 of 4

Unit 3: Adding and Subtracting Integers

Lesson 3.16: Integer Problem Solving

Name _____ Date _____

TODAY'S GOAL

I can solve problems with integer addition and subtraction from start to finish.

Standards: 7.NS.1, 7.NS.3

WHAT TO DO

- [ ] Read Textbook Lesson 3.16 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Mixed practice, Part D Solve, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Solve each problem. Show your work.

1. Range of −15 and 22

2. −9 + 14 − 20

3. Distance from −6 to −18

PART B Warm-Up

Directions: Add or subtract.

4. $3 - (-7)$

5. $5 + (-17)$

PART C Mixed practice

Directions: Add or subtract.

6. $-23 + 15$

7. $-23 - 15$

8. $15 - (-23)$

9. $|-9 - 4|$

10. $-6 + 13 - (-5) - 20$

11. $x - y$ when $x = -2$, $y = -9$

PART D Solve

12. A submarine at −250 ft rises 90 ft, dives 145 ft, then rises 60 ft. Final depth?

PART E Exit Ticket

13. Write and solve a problem that uses both addition and subtraction of negatives.

UNIT REVIEW
Study before the unit test

About 35 minutes

Unit 3 Review

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each question carefully. Show your thinking.
- [ ] When you finish, check that every question has an answer.

PART A Add and subtract

Directions: Add or subtract.

1. $-18 + (-7)$

2. $25 + (-40)$

3. $-12 - 9$

4. $6 - (-14)$

5. $-3 - (-10)$

6. $|-5 - 12|$

PART B Applications

Directions: Solve each problem. Show your work.

7. $-8°F$ rises $21°$

8. Elevation -45 ft to 30 ft: change

9. $x - y + 3$ when $x = -1$, $y = -6$

10. Fix: $-9 - (-4) = -13$

UNIT TEST
At the end of the unit

About 40 minutes

Unit 3 Test

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] This test has 21 points.
- [ ] Read each question carefully. Show your thinking.
- [ ] When you finish, check that every question has an answer.

PART A · Part A · Integers (1 point each)

Directions: Do what each item asks.

1. Opposite of −21

2. |−34|

3. Order: 4, −9, 0, −3

4. −17 + ___ = 0

PART B · Part B · Compute (1 point each)

Directions: Add or subtract.

5. −15 + (−26)

6. −38 + 22

7. 44 + (−19)

8. 9 − 25

9. −14 − (−30)

10. −6 − 18

11 $|-11 - 7|$

12 $-8 + 20 - (-5) - 13$

PART C Part C · Applications (2 points each)

13 At 5 a.m. it was −9°F. By 2 p.m. it was 16°F. How much did it rise?

14 A diver at −42 ft rises 15 ft, then dives 28 ft. Where is she?

15 Evaluate $a - b + c$ for $a = -3$, $b = -10$, $c = -8$.

PART D Part D · Explain (3 points)

16 Use a chip model or number line to explain why $2 - (-6) = 8$.

UNIT 4

Adding and Subtracting Rational Numbers

4

Signed fractions and decimals, converting fractions to decimals, and real-world problems.

Your practice pages for Unit 4. Each lesson matches the same lesson number in your textbook.

UNIT VOCABULARY

rational number	A number that can be written as a fraction of integers.
terminating decimal	A decimal that ends.
repeating decimal	A decimal with a repeating pattern.

LESSON 4.1 · PRACTICE
Week 1 · Lesson 1 of 4

Unit 4: Adding and Subtracting Rational Numbers

Lesson 4.1: Rational Numbers on the Number Line

Name _____ Date _____

TODAY'S GOAL

I can place positive and negative fractions and decimals on a number line.

Standards: 7.NS.1

WHAT TO DO

- [ ] Read Textbook Lesson 4.1 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Compare (write <, >, or =), Part D Order from least to greatest, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Answer each item.

1. Which is less: −1.2 or −1.02?

2. Opposite of $-3\frac{1}{2}$

3. $|-4.75|$

PART B Warm-Up

Directions: Answer each item.

4. Which is less: −5 or −2?

5. $\frac{3}{4}$ as a decimal

PART C Compare (write <, >, or =)

Directions: Write <, >, or = in each circle.

6. −0.6 ◯ −0.06

7. $-\frac{1}{2}$ ◯ −0.5

8. $-\frac{2}{5}$ ◯ $-\frac{1}{3}$

9. −3.4 ◯ −3.45

10. $\frac{5}{8}$ ◯ $-\frac{7}{8}$

11. −1.75 ◯ $-1\frac{3}{4}$

PART D Order from least to greatest

12. 0.5, $-\frac{3}{4}$, −0.7, $\frac{1}{4}$, −1

PART E Exit Ticket

13. Explain why −2.1 is less than −2.

LESSON 4.2 · PRACTICE
Week 1 · Lesson 2 of 4

Unit 4: Adding and Subtracting Rational Numbers

Lesson 4.2: Adding Signed Fractions

Name _____ Date _____

TODAY'S GOAL

I can add positive and negative fractions.

Standards: 7.NS.1b, 7.NS.1d

WHAT TO DO

- [ ] Read Textbook Lesson 4.2 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Add, Part D Solve, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Add. Simplify your answer.

1. $-\frac{2}{5} + \frac{4}{5}$

2. $\frac{3}{4} + (-\frac{5}{6})$

3. $-\frac{1}{6} + (-\frac{2}{3})$

PART B Warm-Up

Directions: Add.

4. $-5 + 3$

5. $\frac{1}{2} + \frac{1}{3}$

PART C Add

Directions: Add. Simplify your answer.

6. $-\frac{3}{7} + \frac{1}{7}$

7. $\frac{5}{9} + (-\frac{8}{9})$

8. $-\frac{1}{4} + (-\frac{3}{8})$

9. $\frac{2}{3} + (-\frac{1}{6})$

10. $-\frac{7}{10} + \frac{2}{5}$

11. $-\frac{5}{12} + (-\frac{1}{3})$

12. $\frac{4}{5} + (-\frac{1}{2})$

13. $-\frac{3}{4} + \frac{3}{4}$

PART D Solve

14. A water level was $-\frac{3}{4}$ ft (below normal). It rose $\frac{1}{2}$ ft. New level?

PART E Exit Ticket

Directions: Add. Simplify your answer.

15. $-\frac{5}{6} + \frac{3}{4}$

LESSON 4.3 · PRACTICE
Week 1 · Lesson 3 of 4

Unit 4: Adding and Subtracting Rational Numbers

Lesson 4.3: Subtracting Signed Fractions

Name _____ Date _____

TODAY'S GOAL

I can subtract positive and negative fractions by adding the opposite.

Standards: 7.NS.1c

WHAT TO DO

- [ ] Read Textbook Lesson 4.3 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Subtract, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Subtract. Simplify your answer.

1. $1/3 - 2/3$

2. $-3/4 - (-1/8)$

3. $-1/2 - 3/10$

PART B Warm-Up

Directions: Answer each item.

4. $2 - (-5)$

5. Opposite of $-\frac{3}{8}$

PART C Subtract

Directions: Subtract. Simplify your answer.

6. $\frac{1}{5} - \frac{4}{5}$

7. $-\frac{2}{9} - \frac{5}{9}$

8. $\frac{3}{8} - (-\frac{1}{4})$

9. $-\frac{5}{6} - (-\frac{1}{3})$

10. $-\frac{1}{2} - \frac{1}{4}$

11. $\frac{2}{3} - \frac{3}{4}$

12. $-\frac{7}{12} - (-\frac{3}{4})$

13. $0 - (-\frac{5}{8})$

PART D Exit Ticket

14. Explain why $\frac{1}{2} - (-\frac{1}{2}) = 1$.

LESSON 4.4 · PRACTICE
Week 1 · Lesson 4 of 4

Unit 4: Adding and Subtracting Rational Numbers

Lesson 4.4: Adding and Subtracting Signed Mixed Numbers

Name _____ Date _____

TODAY'S GOAL

I can add and subtract signed mixed numbers like $-2\frac{1}{3} + 1\frac{1}{2}$.

Standards: 7.NS.1

WHAT TO DO

- [ ] Read Textbook Lesson 4.4 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Compute, Part D Solve, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Add or subtract. Write the answer as a mixed number.

1. $-2\frac{1}{4} + 3\frac{1}{2}$

2. $1\frac{2}{3} - 3\frac{1}{4}$

3. $-2\frac{1}{5} + (-1\frac{1}{2})$

PART B Warm-Up

Directions: Rewrite each number as asked.

4. $2\frac{1}{3}$ as an improper fraction

5. $\frac{17}{6}$ as a mixed number

PART C Compute

Directions: Add or subtract. Write the answer as a mixed number.

6. $-2\frac{1}{2} + 4\frac{1}{4}$

7. $3\frac{1}{3} + (-3\frac{5}{6})$

8. $-1\frac{5}{8} + (-\frac{3}{4})$

9. $-3\frac{1}{2} - 1\frac{1}{4}$

10. $2\frac{1}{4} - (-3\frac{2}{3})$

11. $-3\frac{1}{5} - (-2\frac{1}{10})$

PART D Solve

12. A lake was $1\frac{1}{4}$ ft below normal. It dropped $2\frac{2}{3}$ ft more. New level?

PART E Exit Ticket

Directions: Subtract. Write the answer as a mixed number.

13. $-2\frac{1}{2} - (-5\frac{3}{4})$

WEEKLY CHECK
After Lesson 4.4

About 20 minutes

Weekly Check 10: Signed Fractions

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each question carefully. Show your thinking.
- [ ] When you finish, check that every question has an answer.

Show what you know

Directions: Add or subtract. Simplify your answer.

1. $-\frac{3}{5} + \frac{1}{5}$

2. $-\frac{1}{4} + \left(-\frac{2}{3}\right)$

3. $\frac{1}{6} - \frac{5}{6}$

4. $-\frac{3}{8} - \left(-\frac{1}{2}\right)$

5. $-3\frac{1}{2} + 2\frac{1}{4}$

6. Order: $-\frac{2}{3}$, -0.6, $\frac{1}{2}$

LESSON 4.5 · PRACTICE
Week 2 · Lesson 1 of 4

Unit 4: Adding and Subtracting Rational Numbers

Lesson 4.5: Fractions as Decimals

Name _____ Date _____

TODAY'S GOAL

I can write fractions as terminating or repeating decimals using long division.

Standards: 7.NS.2d

WHAT TO DO

- [ ] Read Textbook Lesson 4.5 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Write as a decimal. Terminating or repeating?, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Write each fraction as a decimal. Use a bar for repeating digits.

1. $\frac{5}{8}$ as a decimal

2. $\frac{4}{9}$ as a decimal

3. $-\frac{7}{4}$ as a decimal

PART B Warm-Up

Directions: Divide.

4. $3 \div 8$

5. $1 \div 3$

PART C Write as a decimal. Terminating or repeating?

6. $\frac{3}{5}$

7. $\frac{1}{6}$

8. $\frac{7}{8}$

9. $\frac{2}{9}$

10. $\frac{9}{20}$

11. $\frac{4}{11}$

PART D Exit Ticket

12. Write $-2\frac{1}{3}$ as a decimal.

LESSON 4.6 · PRACTICE
Week 2 · Lesson 2 of 4

Unit 4: Adding and Subtracting Rational Numbers

Lesson 4.6: Comparing and Ordering Rational Numbers

Name _____ Date _____

TODAY'S GOAL

I can compare and order signed fractions and decimals.

Standards: 7.NS.1, 7.NS.2d

WHAT TO DO

- [ ] Read Textbook Lesson 4.6 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Order from least to greatest, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Compare or order the numbers.

1. $-\frac{5}{8}$ ◯ -0.6

2. Order: -1.3, $-\frac{4}{3}$, -1.25

3. Greatest: -0.09, -0.9, -0.19

PART B Warm-Up

Directions: Write each fraction as a decimal.

4. $^{5}/_{8}$ as a decimal

5. $^{4}/_{3}$ as a decimal

PART C Order from least to greatest

6. $-0.4, -^{3}/_{8}, 0.3, -^{1}/_{2}$

7. $1^{1}/_{3}, -1.3, 1.3, -1^{1}/_{3}$

8. $-2.05, -2.5, -2^{1}/_{5}, -2$

PART D Exit Ticket

9. Which is greater, $-^{2}/_{3}$ or -0.67? Explain.

LESSON 4.7 · PRACTICE
Week 2 · Lesson 3 of 4

Unit 4: Adding and Subtracting Rational Numbers

Lesson 4.7: Adding Signed Decimals

Name _____ Date _____

TODAY'S GOAL

I can add positive and negative decimals.

Standards: 7.NS.1

WHAT TO DO

- [ ] Read Textbook Lesson 4.7 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Add, Part D Solve, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Add.

1. $-7.2 + 3.5$

2. $-0.45 + (-1.8)$

3. $12.06 + (-12.6)$

PART B Warm-Up

Directions: Add or subtract.

4. 40.75 − 12.50

5. 3.6 + 0.85

PART C Add

Directions: Add the decimals.

6. −4.3 + 2.1

7. 6.75 + (−9.5)

8. −0.8 + (−0.15)

9. −15.2 + 15.2

10. 3.05 + (−0.7)

11. −22.4 + (−7.65)

PART D Solve

12. A temperature of −3.8°C rises 5.6°C. New temperature?

13. A stock changed −1.35, +0.80, and −0.25 dollars. Total change?

PART E Exit Ticket

Directions: Add.

14. −8.06 + 5.9

LESSON 4.8 · PRACTICE
Week 2 · Lesson 4 of 4

Unit 4: Adding and Subtracting Rational Numbers

Lesson 4.8: Subtracting Signed Decimals

Name _____ Date _____

TODAY'S GOAL

I can subtract positive and negative decimals.

Standards: 7.NS.1c

WHAT TO DO

- [ ] Read Textbook Lesson 4.8 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Subtract, Part D Solve, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Subtract.

1. $4.2 - 6.5$

2. $-3.8 - (-1.25)$

3. $-0.9 - 0.45$

PART B Warm-Up

Directions: Add.

4. 163.1 + 2.4

5. −0.75 + (−1.6)

PART C Subtract

Directions: Subtract the decimals.

6. 2.5 − 7.1

7. −6.3 − 2.4

8. 1.75 − (−0.5)

9. −8.2 − (−8.2)

10. −0.05 − 0.5

11. 10 − (−3.75)

PART D Solve

12. From −12.5°F to 4.3°F: change?

13. A submarine at −85.4 m dives to −112.9 m. Change?

PART E Exit Ticket

Directions: Subtract.

14. −5.08 − (−7.3)

WEEKLY CHECK
After Lesson 4.8

About 20 minutes

Weekly Check 11: Signed Decimals

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each question carefully. Show your thinking.
- [ ] When you finish, check that every question has an answer.

Show what you know

Directions: Write each fraction as a decimal, or add.

1. $^{7}/_{8}$ as a decimal

2. $^{5}/_{9}$ as a decimal

3. $-6.4 + 2.75$

4. $-0.35 + (-1.2)$

5. $3.1 - 5.85$

6. $-2.5 - (-7.25)$

LESSON 4.9 · PRACTICE
Week 3 · Lesson 1 of 4

Unit 4: Adding and Subtracting Rational Numbers

Lesson 4.9: Rational Numbers and Money

Name _____ Date _____

TODAY'S GOAL

I can solve money problems with positive and negative decimals.

Standards: 7.NS.3, 7.EE.3

WHAT TO DO

- [ ] Read Textbook Lesson 4.9 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Solve, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Find each new balance or total.

1. $5.20 − $9.85

2. −$32.10 + $50

3. Profits: $12.40, −$20.15, $6.80

PART B Warm-Up

Directions: Subtract.

4 24.6 − 18.95

5 5.65 − 7.4

PART C Solve

Directions: Find the final amount. Show each step.

6 Start at −$8.25. Deposit $30, spend $14.60.

7 A lemonade stand: +$18.50, −$6.75 (supplies), +$12.25, −$4.30. Total?

8 A stock is at $45.20. It changes −2.85, +1.10, −0.65. New price?

9 Ellie owes $12.40 and pays $7.95. How much does she owe now?

PART D Exit Ticket

10 Create a 4-transaction account record that ends below $0.

LESSON 4.10 · PRACTICE
Week 3 · Lesson 2 of 4

Unit 4: Adding and Subtracting Rational Numbers

Lesson 4.10: Rational Numbers in Measurement

Name _____ Date _____

TODAY'S GOAL

I can solve elevation, temperature, and measurement problems with signed fractions and decimals.

Standards: 7.NS.3

WHAT TO DO

- [ ] Read Textbook Lesson 4.10 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Solve, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Write an expression and solve. Include units.

1. A plant $3\frac{1}{2}$ in. tall shrinks $\frac{3}{4}$ in. New height?

2. −1.8°F to 6.5°F: change

3. Water level $-\frac{2}{3}$ ft then rises $1\frac{1}{6}$ ft

PART B Warm-Up

Directions: Subtract.

4 $5\frac{1}{4} - 4\frac{3}{8}$

5 $7.2 - (-4.5)$

PART C Solve

6 A diver at $-10\frac{1}{2}$ ft rises $4\frac{3}{4}$ ft. Depth?

7 A runner's times: 12.45 s and 11.9 s. Improvement?

8 −6.25°F drops 3.8°F. New temperature?

9 Rope $12\frac{1}{3}$ m; cut $4\frac{5}{6}$ m. Left?

PART D Exit Ticket

10 Write a measurement story for $2\frac{1}{2} - 3\frac{3}{4}$. Solve.

LESSON 4.11 · PRACTICE
Week 3 · Lesson 3 of 4

Unit 4: Adding and Subtracting Rational Numbers

Lesson 4.11: Properties for Rational Number Sums

Name _____ Date _____

TODAY'S GOAL

I can use properties to add and subtract rational numbers efficiently.

Standards: 7.NS.1d

WHAT TO DO

- [ ] Read Textbook Lesson 4.11 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Compute mentally, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Use properties to evaluate each sum mentally.

1. $-3.7 + 8.25 + 3.7$

2. $2/5 - 1.5 - 2/5 + 0.5$

3. $-0.25 + 6 + (-0.75)$

PART B Warm-Up

Directions: Add or subtract.

4. $-\frac{3}{4} + \frac{3}{4}$

5. $2.6 - 0.6$

PART C Compute mentally

Directions: Group numbers that make zero or whole numbers first. Then evaluate.

6. $4.8 + (-2.3) + (-4.8)$

7. $-\frac{5}{6} + 3 + \frac{5}{6}$

8. $-1.25 + 7 + (-0.75)$

9. $2\frac{1}{2} - 4 + \frac{1}{2}$

10. $-9.9 + 5 + 9.9 + (-5)$

11. $0.75 + (-\frac{1}{4}) + (-2)$

PART D Exit Ticket

12. Show how grouping helps evaluate $-6.4 + 3.3 + 6.4 - 1.3$.

LESSON 4.12 · PRACTICE
Week 3 · Lesson 4 of 4

Unit 4: Adding and Subtracting Rational Numbers

Lesson 4.12: Multi-Step Rational Number Problems

Name _____ Date _____

TODAY'S GOAL

I can solve multi-step problems with signed fractions and decimals.

Standards: 7.EE.3

WHAT TO DO

- [ ] Read Textbook Lesson 4.12 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Solve, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Solve each problem. Show your work.

1. $-2.4 + 5.1 - 3.25$

2. $2/3 - 5/6 + (-1/2)$

3. A hiker at −15.5 m climbs 40.25 m then descends 8.75 m.

PART B Warm-Up

Directions: Evaluate.

4 $2 - 5 + 10 - 3$

5 $4\frac{4}{4}$

PART C Solve

6 Temperatures: −2.4°C, then +6.8°C, then −5.1°C. Final?

7 A stock: start $18.40. Changes: $-3\frac{3}{4}$, +$1.20, −$0.55. Final price? (Use 0.75 for $3\frac{3}{4}$.)

8 Lillie's garden bed is $8\frac{1}{2}$ ft. She uses $2\frac{3}{4}$ ft for tomatoes and $3\frac{1}{3}$ ft for peppers. Space left?

PART D Exit Ticket

9 Write and solve a three-step problem with negative decimals.

WEEKLY CHECK
After Lesson 4.12

About 20 minutes

Weekly Check 12: Rational Number Applications

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each question carefully. Show your thinking.
- [ ] When you finish, check that every question has an answer.

Show what you know

Directions: Solve each problem. Show your work.

1. Start −$6.40, deposit $25, spend $11.85

2. −3.2°C to 5.9°C: change

3. Board $6\frac{1}{4}$ ft, cut $2\frac{7}{8}$ ft. Left?

4. −4.5 + 2.75 + 4.5 (mentally)

5. −1.6 + 3.25 − 0.9

6. $\frac{1}{2} - \frac{3}{4} - \frac{1}{8}$

LESSON 4.13 · PRACTICE
Week 4 · Lesson 1 of 4

Unit 4: Adding and Subtracting Rational Numbers

Lesson 4.13: Distance Between Rational Numbers

Name _____ Date _____

TODAY'S GOAL

I can find the distance between two rational numbers using absolute value.

Standards: 7.NS.1c

WHAT TO DO

- [ ] Read Textbook Lesson 4.13 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Find the distance, Part D Solve, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Find the distance between the numbers.

1. Distance between -3.2 and 1.5

2. Distance between $-\frac{5}{6}$ and $-\frac{1}{3}$

3. $|-0.8 - 2.45|$

PART B Warm-Up

Directions: Evaluate.

4. $|-7.5|$

5. $1\frac{3}{4} + 2\frac{1}{2}$

PART C Find the distance

6. -4.6 and 2.9

7. $-\frac{2}{3}$ and $\frac{1}{6}$

8. -10.25 and -3.5

9. $3\frac{1}{2}$ and $-1\frac{1}{4}$

PART D Solve

10. Death Valley: -86 m. Mount McKinley (Denali): 6,190 m. Distance?

PART E Exit Ticket

11. Explain why the distance from -2.5 to 4 equals the distance from 4 to -2.5.

LESSON 4.14 · PRACTICE
Week 4 · Lesson 2 of 4

Unit 4: Adding and Subtracting Rational Numbers

Lesson 4.14: Expressions with Rational Numbers

Name _____ Date _____

TODAY'S GOAL

I can evaluate expressions with signed fractions and decimals.

Standards: 7.NS.1, 7.EE.1

WHAT TO DO

- [ ] Read Textbook Lesson 4.14 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Evaluate when p = −2.5 and q = 1.25, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Evaluate each expression for the given values.

1. a + b when a = −0.6, b = 2.35

2. $x - {}^{1}\!/_{2}$ when $x = -{}^{3}\!/_{4}$

3. −m + n when m = −4.2, n = −1.8

PART B Warm-Up

Directions: Evaluate.

4 $-1.75 + 3$

5 $-(-4.2)$

PART C Evaluate when $p = -2.5$ and $q = 1.25$

6 $p + q$

7 $p - q$

8 $q - p$

9 $-p - q$

10 $p + p + q$

11 $|p| - |q|$

PART D Exit Ticket

12 Evaluate $r - s$ when $r = {}^{2}/_{3}$ and $s = -{}^{5}/_{6}$.

LESSON 4.15 · PRACTICE
Week 4 · Lesson 3 of 4

Unit 4: Adding and Subtracting Rational Numbers

Lesson 4.15: Error Analysis with Rational Numbers

Name _____ Date _____

TODAY'S GOAL

I can find and fix errors in rational-number computations.

Standards: 7.NS.1

WHAT TO DO

- [ ] Read Textbook Lesson 4.15 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Find and fix, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Find the error in each problem. Write the correct answer.

1. Fix: $-3.2 - 1.5 = -1.7$

2. Fix: $\frac{1}{4} - (-\frac{3}{4}) = -\frac{1}{2}$

3. Fix: $-0.8 + 0.25 = -1.05$

PART B Warm-Up

Directions: Add or subtract.

4 $-\frac{1}{2} - \frac{1}{3}$

5 $-2.5 + 1.3$

PART C Find and fix

Directions: Each answer is wrong. Write the correct answer and explain the mistake.

6 $-6.1 + 2.4 = -8.5$

7 $\frac{2}{3} - \frac{5}{6} = \frac{1}{6}$

8 $-4 - (-1.25) = -5.25$

9 $0.3 - 0.45 = 0.15$

PART D Exit Ticket

10 Describe how estimating first can catch a sign error.

LESSON 4.16 · PRACTICE
Week 4 · Lesson 4 of 4

Unit 4: Adding and Subtracting Rational Numbers

Lesson 4.16: Rational Number Problem Solving

Name _____ Date _____

TODAY'S GOAL

I can solve problems with signed fractions and decimals.

Standards: 7.NS.1, 7.NS.3, 7.EE.3

WHAT TO DO

- [ ] Read Textbook Lesson 4.16 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Mixed practice, Part D Solve, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Solve each problem. Show your work.

1. $-\frac{3}{8} + \frac{5}{6}$

2. $-7.05 - (-2.6)$

3. Distance between $-\frac{3}{4}$ and 2.5

PART B Warm-Up

Directions: Add or subtract.

4. $-4 - 1.5$

5. $-7.75 + 0.75$

PART C Mixed practice

Directions: Add or subtract. Simplify your answer.

6. $-\frac{7}{8} + \frac{1}{4}$

7. $\frac{2}{5} - \left(-\frac{1}{2}\right)$

8. $-12.4 + 8.75$

9. $0.6 - 2.15$

10. Order: -0.3, $-\frac{1}{3}$, -0.33

11. $\frac{7}{12}$ as a decimal

PART D Solve

12. A business had monthly results of −$1,250.50, +$3,400.75, and −$875.25. Total?

PART E Exit Ticket

13. Which is harder for you: signed fractions or signed decimals? Make up and solve one of each.

UNIT REVIEW
Study before the unit test

About 35 minutes

Unit 4 Review

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each question carefully. Show your thinking.
- [ ] When you finish, check that every question has an answer.

PART A Compute

Directions: Add or subtract. Simplify your answer.

1. $-\frac{5}{6} + \frac{1}{3}$

2. $-\frac{3}{4} - \frac{5}{8}$

3. $-2\frac{3}{4} + (-1\frac{1}{2})$

4. $-9.6 + 4.35$

5. $-1.2 - (-3.05)$

6. $\frac{3}{11}$ as a decimal

PART B Apply

Directions: Do what each item asks.

7. Order: -1.5, $-1\frac{2}{3}$, -1.6

8. Distance between -2.75 and 4.1

9. $x - y$ when $x = -0.4$, $y = 1.35$

10. A balance of $-\$15.60$ gets a $\$40$ deposit and a $\$9.25$ charge.

UNIT TEST
At the end of the unit

About 45 minutes

Unit 4 Test

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] This test has 21 points.
- [ ] Read each question carefully. Show your thinking.
- [ ] When you finish, check that every question has an answer.

PART A Part A · Rational numbers (1 point each)

Directions: Write each as a decimal, or put the numbers in order.

1. $^{9}/_{16}$ as a decimal

2. $^{2}/_{3}$ as a decimal

3. Order: -0.8, $-^{3}/_{4}$, $-^{5}/_{6}$

4. $|-6.05|$

PART B Part B · Compute (1 point each)

Directions: Add or subtract. Simplify your answer.

5. $-^{4}/_{9} + (-^{2}/_{3})$

6. $^{7}/_{10} + (-1\,^{1}/_{5})$

7. $^{3}/_{8} - (-^{5}/_{12})$

8. $-4\,^{1}/_{4} - 1\,^{5}/_{6}$

9. $-14.25 + 6.8$

10. $0.45 - 2.6$

11 $-8.3 - (-12.05)$

12 $-3.75 + 9.2 + 3.75$ (mentally)

PART C Part C · Applications (2 points each)

Directions: Solve each problem. Show your work.

13 Temperatures change −2.6°, +4.15°, −0.8° from −1.5°C. Final temperature?

14 A diver at $-12\,\frac{1}{2}$ ft rises $7\,\frac{3}{4}$ ft. Depth now?

15 Distance between −18.4 m and 22.75 m

PART D Part D · Explain (3 points)

Directions: Explain the mistake and solve the problem correctly.

16 Ellie says $-2\,\frac{1}{4} + 1\,\frac{3}{4} = -3\,\frac{2}{4}$. Explain her mistake and give the correct answer.

UNIT 5

Multiplying and Dividing Rational Numbers

Sign rules and why they work, signed fractions and decimals, order of operations, rates, and averages.

Your practice pages for Unit 5. Each lesson matches the same lesson number in your textbook.

UNIT VOCABULARY

product	Result of multiplication.
quotient	Result of division.
reciprocal	Number whose product with another is 1.
mean	Sum ÷ count.

LESSON 5.1 · PRACTICE
Week 1 · Lesson 1 of 4

Unit 5: Multiplying and Dividing Rational Numbers

Lesson 5.1: Multiplying Integers: Patterns

Name _____ Date _____

TODAY'S GOAL

I can use patterns to discover how to multiply positive and negative integers.

Standards: 7.NS.2a

WHAT TO DO

- [ ] Read Textbook Lesson 5.1 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Multiply, Part D Complete the pattern, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Multiply.

1. $5 \times (-6)$

2. $(-7) \times (-3)$

3. $(-9) \times 4$

PART B Warm-Up

Directions: Add or multiply.

4. $-3 + (-3) + (-3) + (-3)$

5. 6×7

PART C Multiply

6. $3 \times (-8)$

7. $(-5) \times 7$

8. $(-6) \times (-6)$

9. $(-11) \times 2$

10. $(-4) \times (-9)$

11. $12 \times (-3)$

12. $(-1) \times (-15)$

13. $0 \times (-23)$

PART D Complete the pattern

Directions: Fill in the blanks to continue the pattern.

14. $3 \times (-5) = -15$, $2 \times (-5) =$ ___, $1 \times (-5) =$ ___, $0 \times (-5) =$ ___, $-1 \times (-5) =$ ___

PART E Exit Ticket

15. Use a pattern to explain why $-3 \times (-4)$ is positive.

LESSON 5.2 · PRACTICE
Week 1 · Lesson 2 of 4

Unit 5: Multiplying and Dividing Rational Numbers

Lesson 5.2: Multiplying Integers: Rules and Reasons

Name _____ Date _____

TODAY'S GOAL

I can multiply integers, including more than two factors, and explain the sign rules.

Standards: 7.NS.2a, 7.NS.2c

WHAT TO DO

- [ ] Read Textbook Lesson 5.2 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Multiply, Part D Sign only, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Multiply. Count the negative signs to find the sign of the answer.

1. $(-4)(5)(-2)$

2. $(-1)(-1)(-1)(-1)(-1)$

3. $(-3)^2$

PART B Warm-Up

Directions: Multiply.

4. $(-4) \times 5$

5. $(-20) \times (-2)$

PART C Multiply

6. $(-2)(6)(-3)$

7. $(-5)(-2)(-4)$

8. $(7)(-1)(3)$

9. $(-3)(-3)(-3)$

10. $(-10)(-10)$

11. $(-2)(-2)(-2)(-2)$

PART D Sign only

12. Is the product of 9 negative numbers positive or negative?

13. Is $(-8)(-7)(-6)(-5)$ positive or negative?

PART E Exit Ticket

14. Explain why $(-4)^2$ and -4^2 are different.

LESSON 5.3 · PRACTICE
Week 1 · Lesson 3 of 4

Unit 5: Multiplying and Dividing Rational Numbers

Lesson 5.3: Dividing Integers

Name _____ Date _____

TODAY'S GOAL

I can divide integers using the related multiplication fact.

Standards: 7.NS.2b

WHAT TO DO

- [ ] Read Textbook Lesson 5.3 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Divide, Part D Solve, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Divide.

1. $-42 \div 6$

2. $56 \div (-8)$

3. $-81 \div (-9)$

PART B Warm-Up

Directions: Multiply.

4. $5 \times (-12)$

5. $(-5) \times 12$

PART C Divide

6. $-36 \div 4$

7. $45 \div (-9)$

8. $-72 \div (-8)$

9. $100 \div (-25)$

10. $-90 \div 10$

11. $-64 \div (-16)$

12. $0 \div (-7)$

13. $-150 \div 30$

PART D Solve

14. The temperature dropped 18°F in 6 hours at a steady rate. Change per hour?

15. A stock lost $48 over 4 days, the same each day. Daily change?

PART E Exit Ticket

16. Why is $-5 \div 0$ undefined?

LESSON 5.4 · PRACTICE
Week 1 · Lesson 4 of 4

Unit 5: Multiplying and Dividing Rational Numbers

Lesson 5.4: Negative Fractions and Signs of Quotients

Name _____ Date _____

TODAY'S GOAL

I can explain why $-(\frac{p}{q}) = \frac{-p}{q} = \frac{p}{-q}$.

Standards: 7.NS.2b

WHAT TO DO

- [ ] Read Textbook Lesson 5.4 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Simplify, Part D Same or different?, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Answer each question.

1. Simplify $\frac{-18}{6}$

2. Simplify $\frac{20}{-5}$

3. Simplify $\frac{-7}{-14}$

PART B Warm-Up

Directions: Divide.

4 $-18 \div 6$

5 $20 \div (-5)$

PART C Simplify

Directions: Simplify each fraction to an integer.

6 $\frac{-24}{8}$

7 $\frac{35}{-7}$

8 $\frac{-9}{-3}$

9 $\frac{-5}{10}$

10 $\frac{6}{-8}$

11 $\frac{-15}{-20}$

PART D Same or different?

Directions: Write same or different for each pair. Explain. Circle the letter of the best answer.

12 $-\frac{2}{5}$ and $\frac{-2}{5}$

A same

B different

13 $\frac{-4}{-9}$ and $-\frac{4}{9}$

A same

B different

PART E Exit Ticket

14 Write $-\frac{5}{6}$ two other ways.

WEEKLY CHECK
After Lesson 5.4

About 15 minutes

Weekly Check 13: Multiplying and Dividing Integers

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each question carefully. Show your thinking.
- [ ] When you finish, check that every question has an answer.

Show what you know

Directions: Multiply.

1. $(-7) \times 8$

2. $(-6) \times (-9)$

3. $(-2)(-3)(-4)$

4. $-63 \div 7$

5. $-48 \div (-6)$

6. Simplify $\frac{14}{-21}$

LESSON 5.5 · PRACTICE
Week 2 · Lesson 1 of 4

Unit 5: Multiplying and Dividing Rational Numbers

Lesson 5.5: Multiplying Signed Fractions

Name _____ Date _____

TODAY'S GOAL

I can multiply positive and negative fractions and mixed numbers.

Standards: 7.NS.2a, 7.NS.2c

WHAT TO DO

- [ ] Read Textbook Lesson 5.5 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Multiply, Part D Solve, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Multiply. Simplify your answer.

1. $(-\frac{2}{5}) \times \frac{3}{4}$

2. $(-\frac{5}{6}) \times (-\frac{3}{10})$

3. $(-2\frac{1}{4}) \times \frac{2}{3}$

PART B Warm-Up

Directions: Multiply.

4. $\frac{2}{3} \times \frac{3}{4}$

5. $(-2) \times 3$

PART C Multiply

6. $(-\frac{1}{3}) \times \frac{6}{7}$

7. $\frac{4}{5} \times (-\frac{5}{8})$

8. $(-\frac{3}{4}) \times (-\frac{2}{9})$

9. $(-3\frac{1}{2}) \times \frac{4}{7}$

10. $(-2\frac{1}{2}) \times (-2\frac{1}{2})$

11. $3\frac{1}{3} \times (-\frac{9}{20})$

PART D Solve

12. A diver descends at $2\frac{1}{4}$ ft per second for $\frac{4}{3}$ seconds. Change in depth?

PART E Exit Ticket

Directions: Multiply. Simplify your answer.

13. $(-3\frac{3}{4}) \times (-1\frac{3}{5})$

LESSON 5.6 · PRACTICE
Week 2 · Lesson 2 of 4

Unit 5: Multiplying and Dividing Rational Numbers

Lesson 5.6: Dividing Signed Fractions

Name _____ Date _____

TODAY'S GOAL

I can divide positive and negative fractions.

Standards: 7.NS.2b, 7.NS.2c

WHAT TO DO

- [ ] Read Textbook Lesson 5.6 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Divide, Part D Solve, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Divide. Multiply by the reciprocal and simplify.

1. $(-\frac{2}{3}) \div \frac{4}{9}$

2. $\frac{7}{8} \div (-\frac{7}{16})$

3. $(-2\frac{1}{2}) \div (-\frac{3}{4})$

PART B Warm-Up

Directions: Answer each item.

4 Reciprocal of $-\frac{3}{8}$

5 $\frac{3}{4} \times \frac{8}{3}$

PART C Divide

Directions: Divide. Multiply by the reciprocal and simplify.

6 $(-\frac{1}{2}) \div \frac{1}{4}$

7 $\frac{3}{5} \div (-\frac{9}{10})$

8 $(-\frac{4}{7}) \div (-\frac{2}{7})$

9 $(-6) \div \frac{3}{4}$

10 $1\frac{1}{4} \div (-2\frac{1}{2})$

11 $(-4\frac{1}{2}) \div (-\frac{3}{8})$

PART D Solve

12 A hiker descends $4\frac{1}{2}$ miles in $\frac{3}{4}$ hour. Change in elevation per hour (in miles)?

PART E Exit Ticket

Directions: Divide. Simplify your answer.

13 $(-3\frac{1}{3}) \div \frac{5}{6}$

LESSON 5.7 · PRACTICE
Week 2 · Lesson 3 of 4

Unit 5: Multiplying and Dividing Rational Numbers

Lesson 5.7: Multiplying and Dividing Signed Decimals

Name _____ Date _____

TODAY'S GOAL

I can multiply and divide positive and negative decimals.

Standards: 7.NS.2

WHAT TO DO

- [ ] Read Textbook Lesson 5.7 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Compute, Part D Solve, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Multiply or divide.

1. $(-0.6) \times 4.5$

2. $(-7.2) \div (-0.9)$

3. $(-2.5) \times (-0.4)$

PART B Warm-Up

Directions: Multiply or divide.

4 8×1.25

5 $10 \div 4$

PART C Compute

6 $(-3.2) \times 5$

7 $(-0.25) \times (-12)$

8 $1.5 \times (-0.6)$

9 $(-8.4) \div 4$

10 $6.3 \div (-0.7)$

11 $(-0.48) \div (-0.06)$

PART D Solve

12 A submarine changes depth by −12.5 m per minute. Change in 3.2 minutes?

13 A temperature fell 7.2°F in 1.5 hours. Change per hour?

PART E Exit Ticket

Directions: Divide.

14 $(-13.5) \div 0.45$

LESSON 5.8 · PRACTICE
Week 2 · Lesson 4 of 4

Unit 5: Multiplying and Dividing Rational Numbers

Lesson 5.8: Rational Number Operations Practice

Name _____ Date _____

TODAY'S GOAL

I can multiply and divide any rational numbers fluently.

Standards: 7.NS.2

WHAT TO DO

- [ ] Read Textbook Lesson 5.8 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Compute, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Multiply or divide. Simplify your answer.

1. $(-0.5) \times {}^{4}\!/_{5}$

2. ${}^{9}\!/_{10} \div (-0.3)$

3. $(-1\,{}^{1}\!/_{4})(-0.8)$

PART B Warm-Up

Directions: Multiply or divide.

4. 0.75×0.8

5. $1.2 \div 1.5$

PART C Compute

Directions: Multiply or divide. Simplify your answer.

6. $(-0.25)(-\frac{4}{5})$

7. $-\frac{3}{8} \div 0.75$

8. $(-2.4)(\frac{5}{6})$

9. $-2\frac{1}{2} \div (-0.5)$

10. $(-1.5)(-1.5)$

11. $\frac{1}{3} \div (-\frac{1}{9})$

PART D Exit Ticket

12. Evaluate $(-\frac{2}{5})(1.5) \div (-0.3)$.

WEEKLY CHECK
After Lesson 5.8

About 20 minutes

Weekly Check 14: Rational Products and Quotients

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each question carefully. Show your thinking.
- [ ] When you finish, check that every question has an answer.

Show what you know

Directions: Multiply or divide. Simplify your answer.

1. $(-\frac{3}{8}) \times \frac{4}{9}$

2. $(-2\frac{1}{2}) \times (-1\frac{1}{5})$

3. $(-\frac{4}{5}) \div \frac{2}{15}$

4. $2\frac{1}{3} \div (-1\frac{5}{9})$

5. $(-1.6) \times 0.5$

6. $(-2.7) \div (-0.09)$

LESSON 5.9 · PRACTICE
Week 3 · Lesson 1 of 4

Unit 5: Multiplying and Dividing Rational Numbers

Lesson 5.9: Order of Operations with Rational Numbers

Name _____ Date _____

TODAY'S GOAL

I can use the order of operations with positive and negative numbers.

Standards: 7.NS.3, 7.EE.3

WHAT TO DO

- [ ] Read Textbook Lesson 5.9 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Evaluate, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Evaluate. Follow the order of operations.

1. $-8 + 3 \times (-2)$

2. $(-4)^2 - 10$

3. $12 \div (-3) \times (-2)$

PART B Warm-Up

Directions: Evaluate.

4 $(-2)^2$

5 $-3 + 2$

PART C Evaluate

Directions: Evaluate. Follow the order of operations.

6 $5 - 2 \times (-4)$

7 $-18 \div 3 + (-2)$

8 $(-3)^2 - (-3)$

9 $-2(5 - 9)$

10 $(1 - 7)(-2) \div 4$

11 $-4^2 + (-4)^2$

PART D Exit Ticket

12 Evaluate $-0.5(6 - 10) + 1.5^2$.

LESSON 5.10 · PRACTICE
Week 3 · Lesson 2 of 4

Unit 5: Multiplying and Dividing Rational Numbers

Lesson 5.10: The Distributive Property with Rational Numbers

Name _____ Date _____

TODAY'S GOAL

I can use the distributive property with negative numbers.

Standards: 7.NS.2a, 7.EE.1

WHAT TO DO

- [ ] Read Textbook Lesson 5.10 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Use the distributive property, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Use the distributive property to rewrite and evaluate or expand.

1. $-5(2 + 0.4)$

2. $-2(y - 7)$

3. $-\frac{1}{2}(8 - 6)$

PART B Warm-Up

Directions: Multiply.

4 $(-4) \times 3$

5 $(-4)(-0.25)$

PART C Use the distributive property

6 -6×1.5 (think $6 \times (1 + 0.5)$)

7 $-8(10 - 0.5)$

8 $-3(4x + 2)$

9 $-(a - 9)$

10 $\frac{2}{3}(9 - 12)$

11 $-0.5(4n - 10)$

PART D Exit Ticket

12 Show two ways to evaluate $-7(5 - 2)$.

LESSON 5.11 · PRACTICE
Week 3 · Lesson 3 of 4

Unit 5: Multiplying and Dividing Rational Numbers

Lesson 5.11: Repeated Change: Rate × Time

Name _____ Date _____

TODAY'S GOAL

I can model repeated change with multiplication of signed numbers.

Standards: 7.NS.2a, 7.NS.3

WHAT TO DO

- [ ] Read Textbook Lesson 5.11 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Solve, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Multiply the rate by the time. Write the change with its sign and units.

1. Rate −2°F/hour; 5 hours later

2. Rate −2°F/hour; 5 hours ago

3. Rate +3 cm/day; 4 days ago

PART B Warm-Up

Directions: Multiply.

4 $(-4.5) \times 3$

5 $(-4.5) \times (-3)$

PART C Solve

6 A tank drains 1.5 gallons per minute. Change after 8 minutes?

7 Same tank: change compared to 8 minutes ago?

8 Temperature rises 1.2° per hour. Compared to now, what was it 6 hours ago?

9 A diver ascends 0.75 m/s. Change in 20 s?

PART D Exit Ticket

10 Explain $(-3)(-4) = 12$ using a temperature that drops 3° per hour.

LESSON 5.12 · PRACTICE
Week 3 · Lesson 4 of 4

Unit 5: Multiplying and Dividing Rational Numbers

Lesson 5.12: Choosing the Operation

Name _____ Date _____

TODAY'S GOAL

I can decide which operation fits a situation with rational numbers.

Standards: 7.NS.3

WHAT TO DO

- [ ] Read Textbook Lesson 5.12 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Which operation? Then solve, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Answer each question.

1. Lost 3.5 lb a week for 6 weeks: total change?

2. From −8°F to 5°F: change?

3. −$84 over 7 days: average daily change?

PART B Warm-Up

Directions: Multiply or divide.

4 $-1{,}240 \div 4$

5 $(-3.5) \times 6$

PART C Which operation? Then solve.

6 A balance of −$45 gets a $60 deposit.

7 A plane descends 300 ft/min for 12 min.

8 Temperatures −6° and 9°: how far apart?

9 A stock drops $36.60 over 6 days. Average?

PART D Exit Ticket

10 Write a situation for each operation using negative numbers.

WEEKLY CHECK
After Lesson 5.12

About 20 minutes

Weekly Check 15: Operations and Models

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each question carefully. Show your thinking.
- [ ] When you finish, check that every question has an answer.

Show what you know

Directions: Evaluate, or expand the expression.

1. $-4 + 6 \times (-2)$

2. $(-5)^2 - 3(-4)$

3. $-3(2x - 7)$

4. Rate $-1.5°$/hr; 4 hours ago?

5. A \$45 loss over 5 days: average daily change

6. From 30 to −10 in 8 hours: rate

LESSON 5.13 · PRACTICE
Week 4 · Lesson 1 of 4

Unit 5: Multiplying and Dividing Rational Numbers

Lesson 5.13: Average Change

Name _____ Date _____

TODAY'S GOAL

I can find the mean of positive and negative numbers.

Standards: 7.NS.3

WHAT TO DO

- [ ] Read Textbook Lesson 5.13 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Find the mean, Part D Solve, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Find the mean: add the numbers, then divide by how many there are.

1. Mean of −3, 5, −8, 2

2. Mean of −1.5, −2.5, 1

3. Golf scores −2, 0, −3, +1: mean

PART B Warm-Up

Directions: Evaluate.

4 $-4 + 2 - 7 + 1 - 2$

5 $-10 \div 5$

PART C Find the mean

Directions: Find the mean of each set.

6 $-6, 4, -2, 0$

7 $-10, -15, -5$

8 $3, -9, 6, -12, 2$

9 $-1, -1, -1, 7$

PART D Solve

10 A stock changed −\$1.20, +\$0.60, −\$0.90, +\$0.30 over 4 days. Mean daily change?

PART E Exit Ticket

11 Find 4 numbers with a mean of −3.

LESSON 5.14 · PRACTICE
Week 4 · Lesson 2 of 4

Unit 5: Multiplying and Dividing Rational Numbers

Lesson 5.14: Rate Problems with Negatives

Name _____ Date _____

TODAY'S GOAL

I can solve rate problems that involve decreases.

Standards: 7.NS.3, 7.RP.1

WHAT TO DO

- [ ] Read Textbook Lesson 5.14 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Solve, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Answer each question.

1. A pool level drops from 60 in. to 48 in. in 4 days. Rate?

2. A balance goes from $50 to −$25 in 5 weeks. Rate?

3. Elevation 2,400 ft to 600 ft in 3 hours. Rate?

PART B Warm-Up

Directions: Subtract or divide.

4 −6 − 18

5 −24 ÷ 8

PART C Solve

6 A submarine goes from −40 m to −220 m in 12 minutes. Rate?

7 At that rate, how long from −220 m to −400 m?

8 A plant's height changes from 14.5 cm to 11.3 cm in 4 days (drought). Rate?

9 −$2.40 per day. After how many days is the change −$16.80?

PART D Exit Ticket

10 Temperature drops 2.5° per hour from 10°. What is it after 6 hours?

LESSON 5.15 · PRACTICE
Week 4 · Lesson 3 of 4

Unit 5: Multiplying and Dividing Rational Numbers

Lesson 5.15: Multi-Step Rational Number Problems

Name _____ Date _____

TODAY'S GOAL

I can solve multi-step problems using all four operations with rational numbers.

Standards: 7.EE.3, 7.NS.3

WHAT TO DO

- [ ] Read Textbook Lesson 5.15 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Solve, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Evaluate. Follow the order of operations.

1. $4(-2.5) + 3(1.5)$

2. $(-\frac{3}{4} + \frac{1}{2}) \times 8$

3. $(-12 - 6) \div (-3) + 2$

PART B Warm-Up

Directions: Multiply.

4 $3 \times (-1{,}250)$

5 $2 \times 2{,}400$

PART C Solve

6 A diver descends 2.4 m every 10 seconds for 50 seconds, then rises 3 m. Final change?

7 Ellie's savings change by −$12.50 a week for 4 weeks, then +$20 a week for 3 weeks. Net change?

8 The temperature was 4°C. It fell 1.5°C per hour for 6 hours, then rose 2°C. Final temperature?

PART D Exit Ticket

9 Write a problem whose answer is −8 using all four operations.

LESSON 5.16 · PRACTICE
Week 4 · Lesson 4 of 4

Unit 5: Multiplying and Dividing Rational Numbers

Lesson 5.16: Rational Number Operations Problem Solving

Name _____ Date _____

TODAY'S GOAL

I can use all four operations with rational numbers to solve problems.

Standards: 7.NS.1–3

WHAT TO DO

- [ ] Read Textbook Lesson 5.16 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Mixed practice, Part D Solve, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Solve each problem. Show your work.

1. $(-2.5)(-4) \div (-5)$

2. $-\frac{2}{3} \div \frac{4}{9}$

3. Mean of $-4.2, 1.8, -3$

PART B Warm-Up

Directions: Multiply or add.

4. $4 \times (-3)$

5. $5 + (-12)$

PART C Mixed practice

Directions: Multiply or divide. Simplify your answer.

6. $(-8) \times 7$

7. $-96 \div (-12)$

8. $(-\frac{5}{6}) \times \frac{9}{10}$

9. $(-1\frac{3}{4}) \div 3\frac{1}{2}$

10. $(-0.4) \times (-2.75)$

11. $-3 + (-2)^3$

PART D Solve

12. A stock's price changes −$0.85 per day for 6 days, then +$1.20 per day for 4 days. Net change? Average per day?

PART E Exit Ticket

13. Explain how you decide the sign of a product of five numbers.

UNIT REVIEW
Study before the unit test

About 35 minutes

Unit 5 Review

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each question carefully. Show your thinking.
- [ ] When you finish, check that every question has an answer.

PART A Compute

Directions: Multiply or divide. Simplify your answer.

1. $(-12) \times (-5)$

2. $84 \div (-7)$

3. $(-\frac{2}{3}) \times (-\frac{9}{14})$

4. $(-\frac{5}{8}) \div \frac{15}{16}$

5. $3.5 \times (-0.6)$

6. $(-4.5) \div 1.5$

7. $(-1)^6$

8. $10 - 3(-2)^2$

PART B Apply

Directions: Solve each problem. Show your work.

9. Mean of −8, 3, −5, −2

10. Temp falls from 12° to −9° in 7 hours. Rate?

11. $-0.5(8x - 6)$

12. A pool drains 2.5 in./hour. Change after 3.5 hours?

UNIT TEST
At the end of the unit

About 45 minutes

Unit 5 Test

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] This test has 21 points.
- [ ] Read each question carefully. Show your thinking.
- [ ] When you finish, check that every question has an answer.

PART A · Part A · Integers (1 point each)

Directions: Multiply.

1. $(-9) \times 6$

2. $(-11) \times (-4)$

3. $(-1)(-2)(-3)(-4)$

4. $-108 \div 9$

5. $-75 \div (-15)$

6. Simplify $\frac{-16}{24}$

PART B · Part B · Rational numbers (1 point each)

Directions: Multiply or divide. Simplify your answer.

7. $(-\frac{4}{7}) \times 2\frac{5}{8}$

8. $(-\frac{3}{5}) \div (-\frac{9}{20})$

9. $(-2.4) \times 1.25$

10. $6.4 \div (-0.8)$

11 $-8 + 12 \div (-4)$

12 $-4(3 - 2.5)$

PART C Part C · Applications (2 points each)

Directions: Solve each problem. Show your work.

13 The temperature fell from 15°F to −9°F in 6 hours. Find the rate of change.

14 Mean of −3.5, 2, −4.5, 1, −5

15 A submarine changes depth by −8.5 m per minute. Where was it, relative to now, 4 minutes ago?

PART D Part D · Explain (3 points)

16 Use a pattern or the distributive property to explain why $(-2)(-5) = 10$.

UNIT 6

Expressions, Equations, and Inequalities

Writing and simplifying expressions, solving two-step equations, and solving and graphing inequalities.

Your practice pages for Unit 6. Each lesson matches the same lesson number in your textbook.

UNIT VOCABULARY

variable	A letter for a number.
expression	No equals sign.
equation	Two expressions are equal.
inequality	Comparison with <, >, ≤, ≥.
like terms	Same variable part.

LESSON 6.1 · PRACTICE
Week 1 · Lesson 1 of 4

Unit 6: Expressions, Equations, and Inequalities

Lesson 6.1: Writing Expressions

Name _____ Date _____

TODAY'S GOAL

I can write algebraic expressions from words and real situations.

Standards: 7.EE.2, 7.EE.4

WHAT TO DO

- [ ] Read Textbook Lesson 6.1 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Write an expression, Part D Write an expression for the situation, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Write an algebraic expression for each phrase.

1. 7 less than a number m

2. the quotient of a number and −3

3. A $25 bill split among k friends

PART B Warm-Up

Directions: Evaluate, or name the coefficient.

4. $9 \times 4 + 6$

5. Name the coefficient in $-5y$

PART C Write an expression

Directions: Write an algebraic expression for each phrase.

6. a number increased by 12

7. 8 times a number, minus 4

8. the difference of a number and −2

9. one-third of a number

10. 6 less than 5 times a number

11. twice the sum of a number and 3

PART D Write an expression for the situation

12. A phone plan costs $20 plus $0.10 per text t.

13. Lillie has d dollars and spends $7.50.

PART E Exit Ticket

14. Write a real situation for the expression $12h + 15$.

LESSON 6.2 · PRACTICE
Week 1 · Lesson 2 of 4

Unit 6: Expressions, Equations, and Inequalities

Lesson 6.2: Evaluating Expressions

Name _____ Date _____

TODAY'S GOAL

I can evaluate expressions for given values, including negative and fractional values.

Standards: 7.EE.3

WHAT TO DO

- [ ] Read Textbook Lesson 6.2 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Evaluate when x = −3, y = 4, z = 0.5, Part D Solve, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Evaluate each expression for the given values.

1. $4x - 7$ when $x = -2$

2. $\frac{1}{2}n + 3$ when $n = -8$

3. $a^2 - b$ when $a = -3$, $b = 4$

PART B Warm-Up

Directions: Evaluate.

4. $1.8 \times (-10)$

5. $-18 + 32$

PART C Evaluate when $x = -3$, $y = 4$, $z = 0.5$

6. $2x + y$

7. $y - 5x$

8. $10z - x$

9. xy

10. $x^2 + y^2$

11. $(y - x) \div z$

PART D Solve

12. A taxi costs $2.5m + 3$ dollars for m miles. Cost for 12 miles?

13. Convert 25°C to Fahrenheit with $1.8C + 32$.

PART E Exit Ticket

14. Evaluate $-2(x - 6)$ when $x = -1$.

LESSON 6.3 · PRACTICE
Week 1 · Lesson 3 of 4

Unit 6: Expressions, Equations, and Inequalities

Lesson 6.3: Combining Like Terms

Name _____ Date _____

TODAY'S GOAL

I can simplify expressions by combining like terms, including rational coefficients.

Standards: 7.EE.1

WHAT TO DO

- [ ] Read Textbook Lesson 6.3 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Simplify, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Simplify by combining like terms.

1. $5y - 8y + 3$

2. $-2m + 6 + 7m - 10$

3. $1.5x - 2 + 0.5x$

PART B Warm-Up

Directions: Subtract.

4. $4 - 6$

5. $0.5 - 0.75$

PART C Simplify

Directions: Simplify by combining like terms.

6. $9x + 4x$

7. $3a - 8 + 5a$

8. $-6n + 2 - n + 9$

9. $2.4p - 1.4p + 6$

10. $\frac{2}{3}k + \frac{1}{3}k - 5$

11. $4x + 3y - x - 5y$

PART D Exit Ticket

12. Is $5x + 3$ equal to $8x$? Explain.

LESSON 6.4 · PRACTICE
Week 1 · Lesson 4 of 4

Unit 6: Expressions, Equations, and Inequalities

Lesson 6.4: Expressions in Context

Name _____ Date _____

TODAY'S GOAL

I can rewrite an expression to show what it means in a situation.

Standards: 7.EE.2

WHAT TO DO

- [ ] Read Textbook Lesson 6.4 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Rewrite and explain, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Write an expression for each situation. Write it two ways when asked.

1. A 15% tip on bill b, two ways

2. A 30% markdown on price m

3. Equal sides: s + s + s

PART B Warm-Up

Directions: Evaluate.

4 $1 + 0.05$

5 $1 - 0.2$

PART C Rewrite and explain

Directions: Rewrite each expression in a simpler form. Explain what it means.

6 $p - 0.35p$

7 $c + 0.07c$

8 $2w + 2(w + 4)$

9 $x + x + 0.5x$

PART D Exit Ticket

10 Write two expressions for the price p after a 10% increase. Explain each.

WEEKLY CHECK
After Lesson 6.4

About 15 minutes

Weekly Check 16: Expressions

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each question carefully. Show your thinking.
- [ ] When you finish, check that every question has an answer.

Show what you know

1. Write: 4 less than 3 times a number

2. Evaluate $5x - 2y$ when $x = -2$, $y = 3$

3. Simplify $7a - 3 - 9a + 8$

4. Simplify $0.6k + 1.4k - 2$

5. Rewrite $p - 0.25p$

6. Evaluate $x^2 - x$ when $x = -4$

LESSON 6.5 · PRACTICE
Week 2 · Lesson 1 of 4

Unit 6: Expressions, Equations, and Inequalities

Lesson 6.5: The Distributive Property: Expanding

Name _____ Date _____

TODAY'S GOAL

I can expand expressions using the distributive property.

Standards: 7.EE.1

WHAT TO DO

- [ ] Read Textbook Lesson 6.5 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Expand, Part D Expand and simplify, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Expand. Use the distributive property.

1. $5(2a - 3)$

2. $-(4m + 1)$

3. $-\frac{1}{2}(6n - 10)$

PART B Warm-Up

Directions: Multiply.

4 $-2 \cdot (-5)$

5 $\frac{3}{4} \cdot 12$

PART C Expand

Directions: Use the distributive property to expand.

6 $3(x + 7)$

7 $-4(2y - 1)$

8 $0.5(6k + 8)$

9 $-(-3 + p)$

10 $\frac{2}{3}(9x - 3)$

11 $-1.5(4 - 2w)$

PART D Expand and simplify

12 $2(x - 4) + 3x$

13 $6 - 2(n + 5)$

14 $3(a + 1) - (a - 4)$

15 $-5(2 - y) + 4y$

PART E Exit Ticket

Directions: Expand and simplify.

16 Expand and simplify: $-3(2x + 4) + 7x$

LESSON 6.6 · PRACTICE
Week 2 · Lesson 2 of 4

Unit 6: Expressions, Equations, and Inequalities

Lesson 6.6: Factoring Expressions

Name _____ Date _____

TODAY'S GOAL

I can factor an expression by finding a common factor.

Standards: 7.EE.1

WHAT TO DO

- [ ] Read Textbook Lesson 6.6 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Factor, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Factor out the greatest common factor.

1. $10a + 15$

2. $8m - 12$

3. $-7n - 21$

PART B Warm-Up

Directions: Answer each item.

4 GCF of 6 and 9

5 Expand 3(2x + 3)

PART C Factor

Directions: Factor out the greatest common factor.

6 4x + 20

7 18y − 24

8 9k + 9

9 −5p + 30

10 12 − 16w

11 2.5x + 5

PART D Exit Ticket

12 A rectangle's area is 8x + 32 and one side is 8. Find the other side.

LESSON 6.7 · PRACTICE
Week 2 · Lesson 3 of 4

Unit 6: Expressions, Equations, and Inequalities

Lesson 6.7: Simplifying with Rational Coefficients

Name _____ Date _____

TODAY'S GOAL

I can simplify expressions with fraction and decimal coefficients.

Standards: 7.EE.1

WHAT TO DO

- [ ] Read Textbook Lesson 6.7 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Simplify, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Expand, then combine like terms.

1. $\frac{1}{4}(8a + 4) + a$

2. $0.2(10y - 5) - y$

3. $\frac{2}{5}(10 - 5m) + 3m$

PART B Warm-Up

Directions: Multiply.

4. $\frac{1}{2} \cdot 4x$

5. $0.4 \cdot 5n$

PART C Simplify

Directions: Expand, then combine like terms.

6. $\frac{1}{3}(6x + 3) - x$

7. $0.5(4k - 8) + 3k$

8. $-\frac{3}{4}(4p - 8) + p$

9. $1.2(5 + 10w) - 2w$

10. $\frac{1}{2}x + \frac{1}{4}x - \frac{3}{4}x$

11. $\frac{2}{3}(3n - 6) - \frac{1}{2}(2n + 4)$

PART D Exit Ticket

12. Simplify $\frac{3}{5}(10x - 5) - 0.5(2x + 4)$

LESSON 6.8 · PRACTICE
Week 2 · Lesson 4 of 4

Unit 6: Expressions, Equations, and Inequalities

Lesson 6.8: Equivalent Expressions

Name _____ Date _____

TODAY'S GOAL

I can tell whether two expressions are equivalent and justify my answer.

Standards: 7.EE.1, 7.EE.2

WHAT TO DO

- [ ] Read Textbook Lesson 6.8 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Equivalent? Justify, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Write equivalent or not equivalent for each pair. Show why.

1. $4(x - 1)$ and $4x - 4$

2. $5 - 2x$ and $2x - 5$

3. $x + x + x + 2$ and $3(x + 1) - 1$

PART B Warm-Up

Directions: Expand or evaluate.

4 Expand $2(x + 3)$

5 $3(1) + 2$

PART C Equivalent? Justify.

Directions: Write equivalent or not equivalent. Show your work to justify it.

6 $6x - 9$ and $3(2x - 3)$

7 $-(x - 4)$ and $-x - 4$

8 $0.5(4n + 2)$ and $2n + 1$

9 $3(a + 2) - a$ and $2a + 2$

PART D Exit Ticket

10 Write three expressions equivalent to $4x + 8$.

WEEKLY CHECK
After Lesson 6.8

About 15 minutes

Weekly Check 17: Equivalent Expressions

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each question carefully. Show your thinking.
- [ ] When you finish, check that every question has an answer.

Show what you know

Directions: Expand, factor, or simplify as each item asks.

1. Expand $-3(4x - 2)$

2. Factor $12y + 18$

3. Simplify $\frac{1}{2}(6n + 4) - n$

4. Simplify $5(x - 2) - 2(x + 1)$

5. Equivalent? $2(x + 5)$ and $2x + 5$

6. Factor $-8m + 24$

LESSON 6.9 · PRACTICE
Week 3 · Lesson 1 of 4

Unit 6: Expressions, Equations, and Inequalities

Lesson 6.9: One-Step Equations

Name _____ Date _____

TODAY'S GOAL

I can solve one-step equations with rational numbers.

Standards: 7.EE.4a

WHAT TO DO

- [ ] Read Textbook Lesson 6.9 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Solve, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Solve each equation. Check your answer.

1. $x + 7 = -3$

2. $-5m = -35$

3. $\frac{2}{3}k = 8$

PART B Warm-Up

Directions: Evaluate.

4 $-2 + 4.5$

5 $18 \div (-3)$

PART C Solve

Directions: Solve each equation. Check your answer.

6 $a + 12 = 5$

7 $b - 3.2 = -1.8$

8 $6c = -45$

9 $-d = 9$

10 $p \div (-2) = 11$

11 $\frac{3}{4}w = -9$

PART D Exit Ticket

12 Solve and check: $-\frac{1}{2}x = 7$.

LESSON 6.10 · PRACTICE
Week 3 · Lesson 2 of 4

Unit 6: Expressions, Equations, and Inequalities

Lesson 6.10: Two-Step Equations

Name _____ Date _____

TODAY'S GOAL

I can solve equations of the form $px + q = r$.

Standards: 7.EE.4a

WHAT TO DO

- [ ] Read Textbook Lesson 6.10 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Solve, Part D Fraction and decimal coefficients, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Solve each equation. Check your answer.

1. $3x + 5 = 20$

2. $-2x + 9 = -7$

3. $5x - 12 = 13$

PART B Warm-Up

Directions: Evaluate.

4 115 − 25

5 90 ÷ 15

PART C Solve

Directions: Solve each equation. Check your answer.

6 $2x + 7 = 19$

7 $6x - 4 = 32$

8 $-3x + 10 = 1$

9 $4x + 15 = 3$

10 $-7x - 2 = 26$

11 $8x + 5 = 9$

PART D Fraction and decimal coefficients

Directions: Solve each equation. Check your answer.

12 $\frac{1}{2}x + 4 = 10$

13 $0.25x - 3 = 2$

14 $\frac{2}{3}x - 5 = -1$

15 $1.5x + 2.5 = -5.5$

PART E Exit Ticket

16 Write and solve: a taxi charges $3 plus $2.50 per mile. The fare was $20.50. How many miles?

LESSON 6.11 · PRACTICE
Week 3 · Lesson 3 of 4

Unit 6: Expressions, Equations, and Inequalities

Lesson 6.11: Equations with Parentheses

Name _____ Date _____

TODAY'S GOAL

I can solve equations of the form $p(x + q) = r$.

Standards: 7.EE.4a

WHAT TO DO

- [ ] Read Textbook Lesson 6.11 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Solve, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Solve each equation. Check your answer.

1. $2(x - 4) = 10$

2. $-5(y + 1) = 20$

3. $\frac{1}{2}(n + 6) = -3$

PART B Warm-Up

Directions: Evaluate.

4 $33 \div 3$

5 $11 - 6$

PART C Solve

Directions: Solve each equation. Check your answer.

6 $4(x + 3) = 28$

7 $-2(y - 7) = 6$

8 $6(k - 1.5) = 21$

9 $-3(m + 4) = -3$

10 $\frac{2}{3}(p - 9) = 4$

11 $0.5(w + 8) = -1$

PART D Exit Ticket

12 The perimeter of a rectangle is $2(L + 7) = 46$. Find L.

LESSON 6.12 · PRACTICE
Week 3 · Lesson 4 of 4

Unit 6: Expressions, Equations, and Inequalities

Lesson 6.12: Equation Word Problems

Name _____ Date _____

TODAY'S GOAL

I can write and solve equations for word problems and compare to an arithmetic solution.

Standards: 7.EE.4a

WHAT TO DO

- [ ] Read Textbook Lesson 6.12 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Write an equation and solve, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Answer each question.

1. A plumber charges $60 plus $45 per hour. The bill is $240. Hours?

2. Lillie has $18 and saves $7.50 a week. When will she have $63?

3. The sum of three consecutive integers is 57. The smallest?

PART B Warm-Up

Directions: Solve or evaluate.

4 $2x + 12 = 54$

5 $54 - 12$

PART C Write an equation and solve

6 A family pays $28 for 4 movie tickets and a $4 snack. Price per ticket?

7 A water tank has 120 gallons and drains 7.5 gallons per minute. When will it have 30 gallons left?

8 Ellie's age times 3, minus 4, is 32. How old is she?

9 A temperature of −8°F rises 2.5° per hour. When will it reach 7°F?

PART D Exit Ticket

10 Write your own equation word problem with answer $x = -3$.

WEEKLY CHECK
After Lesson 6.12

About 20 minutes

Weekly Check 18: Equations

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each question carefully. Show your thinking.
- [ ] When you finish, check that every question has an answer.

Show what you know

Directions: Solve each equation.

1. $x - 6.5 = -2$

2. $-7y = 56$

3. $3x - 8 = 13$

4. $-5x + 4 = 29$

5. $4(n - 2) = -20$

6. A $15 fee plus $9 per hour; total $78. Hours?

LESSON 6.13 · PRACTICE
Week 4 · Lesson 1 of 4

Unit 6: Expressions, Equations, and Inequalities

Lesson 6.13: Writing and Graphing Inequalities

Name _____ Date _____

TODAY'S GOAL

I can write inequalities for situations and graph them on a number line.

Standards: 7.EE.4b

WHAT TO DO

- [ ] Read Textbook Lesson 6.13 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Write an inequality, Part D Write the inequality shown, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Write an inequality, or answer the question about its graph.

1. No more than 30 students: write it

2. Temperature below −5

3. Graph $x \geq -2$: open or closed circle?

PART B Warm-Up

4 Is $48 \geq 48$ true?

5 Is $48 > 48$ true?

PART C Write an inequality

Directions: Write an inequality for each situation. Use a variable.

6 a speed limit of 65 mph

7 more than 12 players

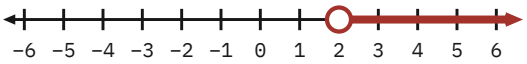
8 at least $20

9 colder than −10°

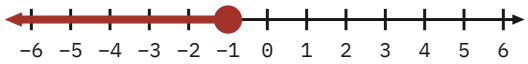
PART D Write the inequality shown

Directions: Write the inequality that each number line shows.

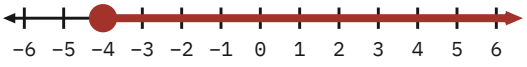
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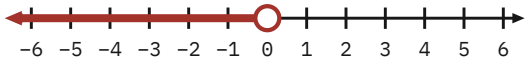
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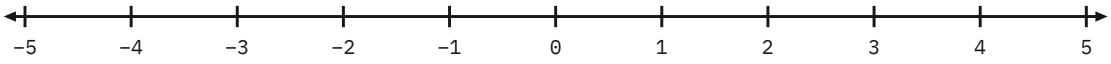


13



PART E Exit Ticket

14 Graph $t \leq 3$ on a number line.



LESSON 6.14 · PRACTICE
Week 4 · Lesson 2 of 4

Unit 6: Expressions, Equations, and Inequalities

Lesson 6.14: Solving One-Step Inequalities

Name _____ Date _____

TODAY'S GOAL

I can solve one-step inequalities, including multiplying or dividing by a negative.

Standards: 7.EE.4b

WHAT TO DO

- [ ] Read Textbook Lesson 6.14 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Solve, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Solve each inequality. Flip the sign if you multiply or divide by a negative.

1. $x + 6 \geq 2$

2. $-5y \leq 30$

3. $\frac{n}{-2} > 3$

PART B Warm-Up

Directions: Answer each item.

4 Is $-2 > -5$?

5 $12 \div (-3)$

PART C Solve

Directions: Solve each inequality. Then graph the solution on a number line.

6 $x - 4 < 7$

7 $4k \geq -20$

8 $-6m > 18$

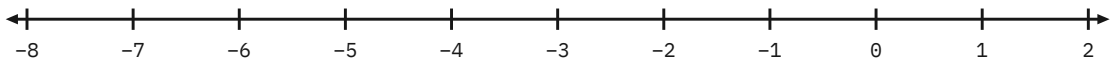
9 $-p \leq 2.5$

10 $\frac{1}{3}w < -2$

11 $-0.5z \geq -4$

PART D Exit Ticket

12 Solve $-2x \geq 10$ and graph it.



LESSON 6.15 · PRACTICE
Week 4 · Lesson 3 of 4

Unit 6: Expressions, Equations, and Inequalities

Lesson 6.15: Two-Step Inequalities

Name _____ Date _____

TODAY'S GOAL

I can solve inequalities of the form $px + q > r$ and graph the solutions.

Standards: 7.EE.4b

WHAT TO DO

- [ ] Read Textbook Lesson 6.15 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Solve, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Solve each inequality. Flip the sign if you multiply or divide by a negative.

1. $3x - 4 \leq 11$

2. $-4y + 2 > 14$

3. $\frac{1}{2}n + 3 \geq -1$

PART B Warm-Up

Directions: Evaluate.

4. $100 - 20$

5. $80 \div 8$

PART C Solve

Directions: Solve each inequality. Then graph the solution on a number line.

6. $2x + 7 > 15$

7. $5y - 3 \leq -18$

8. $-3k + 1 \geq 10$

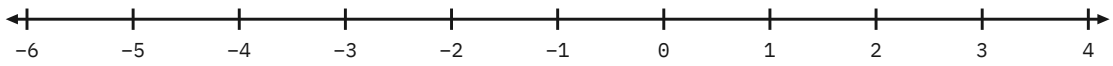
9. $9 - 2m < 1$

10. $0.5p + 4 > 1$

11. $-x - 6 \leq -2$

PART D Exit Ticket

12. Solve $6 - 3x \leq 12$ and graph it.



LESSON 6.16 · PRACTICE
Week 4 · Lesson 4 of 4

Unit 6: Expressions, Equations, and Inequalities

Lesson 6.16: Inequality Word Problems

Name _____ Date _____

TODAY'S GOAL

I can write, solve, and interpret inequalities for real situations.

Standards: 7.EE.4b

WHAT TO DO

- [ ] Read Textbook Lesson 6.16 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Write, solve, and interpret, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Answer each question.

1. A phone plan costs $30 + $5 per GB. Budget $60. Max GB?

2. Ellie needs at least 85 points. She has 62. Each task is worth 4 points. How many tasks?

3. An elevator holds at most 2,000 lb. A 180-lb operator rides with boxes of 65 lb. Max boxes?

PART B Warm-Up

Directions: Evaluate.

4 150 − 10

5 140 ÷ 12.5

PART C Write, solve, and interpret

6 A taxi charges $3 plus $2 per mile. You have $25. How far can you ride?

7 To keep a B, Lillie needs an average of at least 80 on 4 tests. Her first three scores are 76, 84, 78. What score does she need on test 4?

8 A freezer is at 5°F and cools 3° per hour. When will it be below −10°F?

PART D Exit Ticket

9 Write an inequality problem where the answer must be a whole number. Solve it.

UNIT REVIEW
Study before the unit test

About 40 minutes

Unit 6 Review

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each question carefully. Show your thinking.
- [ ] When you finish, check that every question has an answer.

PART A Expressions

Directions: Do what each item asks.

1. Evaluate $-2a + b^2$ when $a = 3$, $b = -4$

2. Simplify $3(2x - 1) - (x + 5)$

3. Factor $15k - 25$

4. Rewrite $1.08c$ in words

PART B Equations and inequalities

Directions: Solve each equation or inequality.

5. $4x + 9 = -11$

6. $\frac{2}{5}(x + 10) = 6$

7. $-3x + 4 > 19$

8. $7 + 2y \leq 1$

9. A pool has 900 gal and drains 25 gal/min. When is it below 400 gal?

UNIT TEST
At the end of the unit

About 45 minutes

Unit 6 Test

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] This test has 21 points.
- [ ] Read each question carefully. Show your thinking.
- [ ] When you finish, check that every question has an answer.

PART A · Part A · Expressions (1 point each)

Directions: Do what each item asks.

1. Write: twice the difference of a number and 5

2. Evaluate $3x - y$ when $x = -2$, $y = -7$

3. Simplify $-4m + 9 + 6m - 12$

4. Expand $-5(2 - 3k)$

5. Factor $21a + 14$

6. Simplify $\frac{3}{4}(8x - 4) - 2x$

PART B · Part B · Equations (1 point each)

Directions: Solve each equation.

7. $y + 8.25 = 3$

8. $-6x + 0 = 39$

9. $2x + 11 = -5$

10. $-3(p + 4) = 9$

11 $\frac{1}{3}n - 2 = 5$

PART C Part C · Inequalities (1 point each)

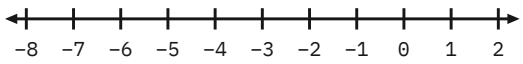
Directions: Solve each inequality.

12 $x - 3 \geq -8$

13 $-4y < 28$

14 $5 - 2k \leq 17$

15 Graph the answer to the previous problem.



PART D Part D · Problem solving (3 points each)

Directions: Write an equation or inequality, solve it, and answer the question in a sentence.

16 A plumber charges $55 plus $40 per hour. The bill was $235. Write and solve an equation to find the hours.

17 Ellie has $35 and wants to buy shirts at $8.50 each with a $3 shipping fee. Write and solve an inequality for the number of shirts. Interpret.

UNIT 7

Statistics and Probability

Sampling, comparing populations, probability models, and compound events.

Your practice pages for Unit 7. Each lesson matches the same lesson number in your textbook.

UNIT VOCABULARY

population	The whole group.
sample	Part of the population.
MAD	Average distance from the mean.
probability	A number 0–1 for likelihood.
sample space	All possible outcomes.

LESSON 7.1 · PRACTICE
Week 1 · Lesson 1 of 4

Unit 7: Statistics and Probability

Lesson 7.1: Populations and Samples

Name _____ Date _____

TODAY'S GOAL

I can tell the difference between a population and a sample.

Standards: 7.SP.1

WHAT TO DO

- [ ] Read Textbook Lesson 7.1 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Name the population and the sample, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Answer each question.

1. A chef tastes one spoonful of soup. Population? Sample?

2. A factory tests 50 of 10,000 light bulbs. Population? Sample?

3. When is a census a better choice than a sample?

PART B Warm-Up

4 What fraction of 2,000 is 80?

5 What percent of 2,000 is 80?

PART C Name the population and the sample

6 A city surveys 400 residents about a new park.

7 A coach times 6 of her 30 swimmers.

8 A scientist tags 25 frogs in a pond.

9 A teacher checks every student's homework.

PART D Exit Ticket

10 Describe a question you could answer with a sample. Name the population and the sample.

LESSON 7.2 · PRACTICE
Week 1 · Lesson 2 of 4

Unit 7: Statistics and Probability

Lesson 7.2: Random Samples and Bias

Name _____ Date _____

TODAY'S GOAL

I can tell whether a sample is random and representative, or biased.

Standards: 7.SP.1

WHAT TO DO

- [ ] Read Textbook Lesson 7.2 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Random or biased? Explain, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Write random or biased for each sampling method. Explain.

1. Survey the basketball team about favorite sports.

2. Roll a die to pick one student from each table group of 6.

3. Ask your friends whom they will vote for.

PART B Warm-Up

4 Name a population for “favorite sport in town.”

5 Name a sample of that population.

PART C Random or biased? Explain.

6 A school picks 40 student ID numbers by computer to survey about lunch.

7 A pet store asks its customers whether people in the city prefer cats or dogs.

8 A magazine asks readers to mail in opinions about the magazine.

9 A farmer tests corn from 10 spots chosen at random across the field.

PART D Exit Ticket

10 Describe a random way to choose 10 students from a school of 300.

LESSON 7.3 · PRACTICE
Week 1 · Lesson 3 of 4
Unit 7: Statistics and Probability

Lesson 7.3: Making Inferences from Samples

Name _____ Date _____

TODAY'S GOAL

I can use data from a random sample to estimate something about a population.

Standards: 7.SP.2

WHAT TO DO

- [ ] Read Textbook Lesson 7.3 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Estimate for the population, Part D Solve, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Use the sample to estimate for the whole population. Show your proportion.

1. 12 of 40 sampled students walk to school. School: 500. Estimate walkers.

2. 3 of 60 sampled apples are bruised. Shipment: 2,400. Estimate bruised.

3. 45 of 75 sampled voters favor a plan. Percent?

PART B Warm-Up

Directions: Solve each problem. Show your work.

4 18 ÷ 50 as a percent

5 0.36 × 600

PART C Estimate for the population

Directions: Use the sample to estimate for the whole population. Show your proportion.

6 9 of 30 in the sample; population 450

7 14 of 70 in the sample; population 1,500

8 6 of 25 in the sample; population 800

9 33 of 120 in the sample; population 2,400

PART D Solve

10 A random sample of 80 households found 52 own a pet. The town has 6,000 households. Estimate the number with pets.

11 A survey of 25 random students found 7 play an instrument. Estimate for 350 students.

PART E Exit Ticket

12 Why should the sample be random for this estimate to be trusted?

LESSON 7.4 · PRACTICE
Week 1 · Lesson 4 of 4
Unit 7: Statistics and Probability

Lesson 7.4: Variability in Samples

Name _____ Date _____

TODAY'S GOAL

I can use several samples to judge how much an estimate might vary.

Standards: 7.SP.2

WHAT TO DO

- [ ] Read Textbook Lesson 7.4 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Use the samples, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Answer each question.

1. What is the range of the five percents above?

2. Which gives a more reliable estimate: one sample of 20 or five samples of 20?

3. A sixth sample has 7 red. Does it change the estimate much?

PART B Warm-Up

Directions: Solve each problem. Show your work.

4 Mean of 6, 8, 5, 7, 9

5 7 out of 20 as a percent

PART C Use the samples

Directions: Answer each question using the samples.

6 Four random samples of 50 voters show 28, 31, 25, and 30 supporting a plan. Find the mean percent.

7 Use your answer to estimate supporters among 8,000 voters.

8 Lillie's sample of 10 says 70% like a book. Ellie's sample of 100 says 52%. Whose estimate is more reliable? Why?

PART D Exit Ticket

9 Why don't all random samples give the same result?

WEEKLY CHECK
After Lesson 7.4

About 15 minutes

Weekly Check 19: Sampling

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each question carefully. Show your thinking.
- [ ] When you finish, check that every question has an answer.

Show what you know

Directions: Answer each question.

1. A school surveys 60 of its 900 students. Name the population.

2. Surveying only the chess club about favorite games: random or biased?

3. 15 of 60 sampled students bike. Estimate bikers in 900.

4. Samples show 40%, 46%, 43%, 39%. Mean percent?

5. Use that mean to estimate for 2,000 people.

6. Which is more reliable: a sample of 30 or of 300?

LESSON 7.5 · PRACTICE
Week 2 · Lesson 1 of 4

Unit 7: Statistics and Probability

Lesson 7.5: Measures of Center and Variability

Name _____ Date _____

TODAY'S GOAL

I can find the mean, median, mean absolute deviation, range, and interquartile range of a data set.

Standards: 7.SP.4

WHAT TO DO

- [ ] Read Textbook Lesson 7.5 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Find the mean and MAD, Part D Find the median and IQR, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Find each measure. Show your work.

1. Mean and MAD of 12, 15, 15, 18, 20

2. Median of 7, 3, 9, 4, 10, 6

3. IQR of 1, 3, 4, 6, 7, 9, 12, 15

PART B Warm-Up

Directions: Answer each item.

4 Order: 9, 2, 7, 4, 5

5 Median of 2, 4, 5, 7, 9

PART C Find the mean and MAD

Directions: Find the mean, then the mean absolute deviation (MAD).

6 4, 6, 8, 10, 12

7 5, 5, 7, 9, 9

8 10, 12, 14, 16, 18, 20

9 1, 4, 4, 5, 6

PART D Find the median and IQR

Directions: Find the median, then the interquartile range (IQR).

10 2, 3, 5, 6, 8, 9, 11

11 10, 12, 13, 15, 18, 20, 21, 25

PART E Exit Ticket

12 Which set has more variability: 2, 4, 6, 8, 10 or 12, 15, 15, 18, 20? Use MAD.

LESSON 7.6 · PRACTICE
Week 2 · Lesson 2 of 4

Unit 7: Statistics and Probability

Lesson 7.6: Comparing Dot Plots

Name _____ Date _____

TODAY'S GOAL

I can compare two data sets using their centers and variability.

Standards: 7.SP.3

WHAT TO DO

- [ ] Read Textbook Lesson 7.6 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Compare the data sets, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Answer each question.

1. Do the dot plots overlap?

2. Which class has more variability?

3. Is a typical B student doing more homework?

PART B Warm-Up

Directions: Find the mean of each set.

4 Mean of 3, 4, 4, 5, 5, 5, 6, 6, 7

5 Mean of 6, 7, 7, 8, 8, 8, 9, 9, 10

PART C Compare the data sets

6 Team X jumps (ft): 10, 11, 12, 12, 13, 14. Team Y: 12, 13, 14, 14, 15, 16. Difference of means?

7 Find the MAD of either team.

8 How many MADs apart are the means?

9 Is the difference large or small? Explain.

PART D Exit Ticket

10 Two data sets have means 20 and 22 and MADs of 5. Are they very different? Why?

LESSON 7.7 · PRACTICE
Week 2 · Lesson 3 of 4

Unit 7: Statistics and Probability

Lesson 7.7: Comparing Box Plots

Name _____ Date _____

TODAY'S GOAL

I can compare two data sets displayed in box plots.

Standards: 7.SP.3, 7.SP.4

WHAT TO DO

- [ ] Read Textbook Lesson 7.7 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Compare, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Use the box plots in Lesson 7.7 to answer each question.

1. Range of Class A

2. Range of Class B

3. About what percent of Class B is taller than Class A's median?

PART B Warm-Up

4 What does the line inside a box show?

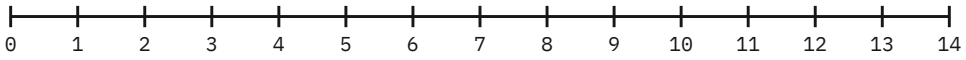
5 What does the box's length show?

PART C Compare

6 Data: 3, 5, 6, 8, 9, 11, 14. Five-number summary?

7 Data: 20, 22, 25, 26, 28, 30, 31, 35. Five-number summary?

8 Plot Set A {2, 4, 5, 7, 9} and Set B {5, 7, 8, 10, 12}. Which has the greater median?



PART D Exit Ticket

9 When might you compare medians instead of means?

LESSON 7.8 · PRACTICE
Week 2 · Lesson 4 of 4

Unit 7: Statistics and Probability

Lesson 7.8: Comparing Populations

Name _____ Date _____

TODAY'S GOAL

I can use random samples to compare two populations.

Standards: 7.SP.3, 7.SP.4

WHAT TO DO

- [ ] Read Textbook Lesson 7.8 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Compare the populations, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Answer each question.

1. If the difference were 0.5 MAD, what would you conclude?

2. Why do the samples need to be random?

3. Name one other way to compare these groups.

PART B Warm-Up

Directions: Find the mean of each set.

4 Mean of 4, 5, 6, 6, 7, 8

5 Mean of 2, 3, 3, 4, 4, 5

PART C Compare the populations

6 Random samples of tomato plants: Fertilizer A heights (cm) have mean 42, MAD 3. Fertilizer B: mean 36, MAD 3. How many MADs apart?

7 Is there a meaningful difference? Explain.

8 Samples of two dog breeds: mean weights 61 lb and 58 lb, each MAD 6 lb. Meaningful difference?

PART D Exit Ticket

9 Write a question comparing two populations you could answer with random samples.

WEEKLY CHECK
After Lesson 7.8

About 15 minutes

Weekly Check 20: Comparing Data

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each question carefully. Show your thinking.
- [ ] When you finish, check that every question has an answer.

Show what you know

Directions: Find each measure.

1. Mean of 6, 8, 10, 12, 14

2. MAD of 6, 8, 10, 12, 14

3. Median of 3, 7, 8, 12, 15, 20

4. IQR of 2, 4, 5, 7, 8, 10, 13, 14

5. Means 30 and 36, MAD 3. How many MADs apart?

6. Is that a meaningful difference?

LESSON 7.9 · PRACTICE
Week 3 · Lesson 1 of 4

Unit 7: Statistics and Probability

Lesson 7.9: Probability and Likelihood

Name _____ Date _____

TODAY'S GOAL

I can describe the likelihood of an event with a number from 0 to 1.

Standards: 7.SP.5

WHAT TO DO

- [ ] Read Textbook Lesson 7.9 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Describe the likelihood, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Describe each event as impossible, unlikely, as likely as not, likely, or certain.

1. Rolling a 7 on a standard number cube

2. A coin landing heads

3. Drawing a red marble from a bag of 9 red, 1 blue

PART B Warm-Up

4 Is $\frac{3}{4}$ closer to 0 or 1?

5 Write 0.2 as a fraction

PART C Describe the likelihood

Directions: Describe each probability as impossible, unlikely, as likely as not, likely, or certain.

6 Probability 0.95

7 Probability $\frac{1}{8}$

8 Probability 0

9 Probability 0.5

10 Snow in July in Texas

11 Choosing a vowel from A, E, I

PART D Exit Ticket

12 Describe an event that is likely but not certain. Estimate its probability.

LESSON 7.10 · PRACTICE
Week 3 · Lesson 2 of 4

Unit 7: Statistics and Probability

Lesson 7.10: Theoretical Probability

Name _____ Date _____

TODAY'S GOAL

I can find the theoretical probability of an event.

Standards: 7.SP.7a

WHAT TO DO

- [ ] Read Textbook Lesson 7.10 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C A spinner has 8 equal parts numbered 1–8. Find each probability, Part D Solve, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Use the spinner or number cube in Lesson 7.10. Write each probability as a fraction.

1. P(red)

2. P(even) on a number cube

3. P(not 6) on a number cube

PART B Warm-Up

Directions: Simplify, or subtract.

4 Simplify $^{5}/_{10}$

5 $1 - 0.3$

PART C A spinner has 8 equal parts numbered 1–8. Find each probability.

Directions: Write each probability as a fraction in simplest form.

6 P(3)

7 P(even)

8 P(less than 4)

9 P(9)

10 P(not 8)

11 P(multiple of 3)

PART D Solve

12 A class has 12 boys and 16 girls. A name is drawn at random. P(girl)?

13 The probability of rain is 0.35. What is the probability of no rain?

PART E Exit Ticket

14 Letters of MISSISSIPPI are on cards. P(S)?

LESSON 7.11 · PRACTICE
Week 3 · Lesson 3 of 4
Unit 7: Statistics and Probability

Lesson 7.11: Experimental Probability

Name _____ Date _____

TODAY'S GOAL

I can find experimental probability and compare it to theoretical probability.

Standards: 7.SP.6, 7.SP.7b

WHAT TO DO

- [ ] Read Textbook Lesson 7.11 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C A coin is flipped 50 times: 29 heads, Part D Predict, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Use the results in Lesson 7.11 to find each probability, or make the prediction.

1. Experimental P(3)

2. Experimental P(odd)

3. Predict how many 6s in 600 rolls of a fair cube

PART B Warm-Up

Directions: Solve each problem. Show your work.

4 15 ÷ 60 as a percent

5 $\frac{1}{6}$ of 600

PART C A coin is flipped 50 times: 29 heads.

Directions: Find each probability. Write it as a fraction or decimal.

6 Experimental P(heads)

7 Theoretical P(heads)

8 Experimental P(tails)

9 Predict heads in 1,000 flips of a fair coin

PART D Predict

10 A player made 18 of 24 free throws. Predict makes in 40 attempts.

11 A spinner landed on blue 12 times in 80 spins. Predict blues in 200 spins.

PART E Exit Ticket

12 Why might experimental and theoretical probability differ?

LESSON 7.12 · PRACTICE
Week 3 · Lesson 4 of 4

Unit 7: Statistics and Probability

Lesson 7.12: Probability Models

Name _____ Date _____

TODAY'S GOAL

I can build uniform and non-uniform probability models.

Standards: 7.SP.7a, 7.SP.7b

WHAT TO DO

- [ ] Read Textbook Lesson 7.12 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Build or check the model, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Answer each item about probability models.

1. Uniform model for a coin

2. Spinner P(not red)

3. A model lists 0.3, 0.5, and ? Find ?

PART B Warm-Up

Directions: Add or subtract.

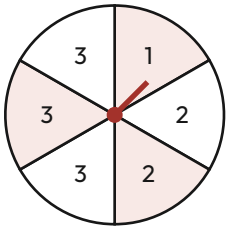
4 $\frac{1}{2} + \frac{1}{4} + \frac{1}{4}$

5 $1 - 0.3 - 0.5$

PART C Build or check the model

Directions: Do what each item asks. Remember that a model's probabilities must add to 1.

6 Spinner with sections 1, 2, 2, 3, 3, 3. Give P(1), P(2), P(3).



7 Is the model P(A) = 0.4, P(B) = 0.4, P(C) = 0.3 valid? Why?

8 Of 50 cars, 20 were white, 15 black, 10 silver, and 5 red. Write a model.

PART D Exit Ticket

Directions: Draw and label the spinner.

9 Design a spinner where P(green) = 0.5 and P(red) = 0.25.

WEEKLY CHECK
After Lesson 7.12

About 15 minutes

Weekly Check 21: Probability

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each question carefully. Show your thinking.
- [ ] When you finish, check that every question has an answer.

Show what you know

Directions: Find each probability, or describe the likelihood.

1. Bag: 4 red, 6 blue. P(red)

2. P(not 2) on a number cube

3. Probability 0.9 is...

4. 32 heads in 50 flips: experimental P(heads)

5. Model: 0.25, 0.45, ? Find ?

6. Coin landed heads 12 of 40 times. Predict heads in 200 flips with that rate.

LESSON 7.13 · PRACTICE
Week 4 · Lesson 1 of 4

Unit 7: Statistics and Probability

Lesson 7.13: Sample Spaces for Compound Events

Name _____ Date _____

TODAY'S GOAL

I can list the sample space of a compound event using lists and tables.

Standards: 7.SP.8b

WHAT TO DO

- [ ] Read Textbook Lesson 7.13 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Use the sum table, Part D List the sample space, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Use the table of sums in Lesson 7.13. Write each probability as a fraction.

1. P(sum is 2)

2. P(sum is 10)

3. P(doubles)

PART B Warm-Up

Directions: Multiply or simplify.

4. 6×6

5. Simplify $\frac{6}{36}$

PART C Use the sum table

Directions: Use the table of sums for two number cubes. Write each probability as a fraction.

6. P(sum is 8)

7. P(sum greater than 9)

8. P(sum is even)

9. P(sum less than 5)

10. P(sum is 12)

11. P(sum is 1)

PART D List the sample space

Directions: List every possible outcome.

12. Flip a coin and spin a 3-color spinner (R, B, Y). List all outcomes.

PART E Exit Ticket

13. Which sum of two number cubes is most likely? Why?

LESSON 7.14 · PRACTICE
Week 4 · Lesson 2 of 4

Unit 7: Statistics and Probability

Lesson 7.14: Tree Diagrams

Name _____ Date _____

TODAY'S GOAL

I can use a tree diagram to find the sample space of a compound event.

Standards: 7.SP.8b

WHAT TO DO

- [ ] Read Textbook Lesson 7.14 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Tree diagrams, Part D Count, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Use the tree diagram in Lesson 7.14 to answer each question.

1. P(outfit with khakis)

2. Add a yellow shirt. How many outfits?

3. Coin flipped 3 times: how many outcomes?

PART B Warm-Up

Directions: Multiply.

4 3×2

5 $2 \times 2 \times 2$

PART C Tree diagrams

Directions: Draw the tree diagram. Then answer each question.

6 Draw a tree: flip a coin twice. List outcomes.

7 P(exactly one head)

8 P(at least one tail)

PART D Count

Directions: Find how many combinations are possible. Show your multiplication.

9 4 sandwiches, 3 drinks: combos

10 2 sizes, 5 flavors: cones

11 Coin and number cube: outcomes

12 3 coins: P(all heads)

PART E Exit Ticket

13 A café offers 3 breads, 4 fillings, and 2 sauces. How many sandwiches?

LESSON 7.15 · PRACTICE
Week 4 · Lesson 3 of 4

Unit 7: Statistics and Probability

Lesson 7.15: Probability of Compound Events

Name _____ Date _____

TODAY'S GOAL

I can find the probability of a compound event using the sample space.

Standards: 7.SP.8a, 7.SP.8c

WHAT TO DO

- [ ] Read Textbook Lesson 7.15 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Find each probability, Part D Solve, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: A coin is flipped and a number cube is rolled. Find each probability as a fraction.

1. P(tails and 5)

2. P(heads and greater than 2)

3. P(tails or 6)

PART B Warm-Up

Directions: Multiply.

4 $\frac{1}{2} \times \frac{1}{3}$

5 2×6

PART C Find each probability

Directions: Find each probability as a fraction in simplest form.

6 Two coins: P(HH)

7 Two number cubes: P(both 6)

8 Two cubes: P(both even)

9 Coin + cube: P(T and 1 or 2)

10 Spinner 1–4 twice: P(sum 5)

11 Three coins: P(exactly 2 heads)

PART D Solve

12 Ellie guesses on 2 true/false questions. P(both correct)?

13 A family has 3 children. Assuming boy and girl are equally likely, P(at least one girl)?

PART E Exit Ticket

14 Roll two number cubes. P(sum is 4)?

LESSON 7.16 · PRACTICE
Week 4 · Lesson 4 of 4

Unit 7: Statistics and Probability

Lesson 7.16: Simulations

Name _____ Date _____

TODAY'S GOAL

I can design and use a simulation to estimate the probability of a compound event.

Standards: 7.SP.8c

WHAT TO DO

- [ ] Read Textbook Lesson 7.16 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Design a simulation, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Use the simulation in Lesson 7.16 to answer each question.

1. Experimental P(needing more than 6 boxes) from the table

2. Which tool simulates a 50% event?

3. Which tool simulates a 1-in-6 event?

PART B Warm-Up

Directions: Answer each item.

4 Mean of 6, 5, 7, 5, 7

5 What tool has 4 equal outcomes?

PART C Design a simulation

Directions: Describe the tool you would use, what each outcome means, and how you would run the trials.

6 A player makes 50% of shots. Design a simulation for P(making 3 in a row).

7 Rain is forecast at 30% each day. Design a simulation for P(rain on at least 1 of 2 days).

8 In 20 trials of the shot simulation, 3 had three heads. Experimental probability?

9 What is the theoretical probability of three heads?

PART D Exit Ticket

10 Why do more trials give a better estimate?

UNIT REVIEW
Study before the unit test

About 40 minutes

Unit 7 Review

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each question carefully. Show your thinking.
- [ ] When you finish, check that every question has an answer.

PART A Statistics

Directions: Answer each item.

1. Random or biased: surveying shoppers at a bookstore about reading habits

2. 24 of 80 sampled like a show. Estimate for 5,000.

3. Mean and MAD of 5, 7, 9, 11, 13

4. Five-number summary of 1, 4, 6, 7, 9, 12, 14

PART B Probability

Directions: Find each probability as a fraction.

5. Spinner 1–10: P(prime)

6. Two coins: P(no heads)

7. Two cubes: P(sum is 6)

8. 3 shirts, 4 pants, 2 shoes: outfits

9. Design a simulation for a 25% event.

UNIT TEST
At the end of the unit

About 45 minutes

Unit 7 Test

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] This test has 20 points.
- [ ] Read each question carefully. Show your thinking.
- [ ] When you finish, check that every question has an answer.

PART A Part A · Sampling (1 point each)

Directions: Answer each item.

1. A factory tests 100 of 20,000 parts. Name the sample.

2. Asking the first 20 people in line for a roller coaster about favorite rides: random or biased?

3. 9 of 45 sampled students have a summer job. Estimate for 1,200 students.

4. Samples: 12, 15, 9, 14 successes out of 30 each. Mean count?

PART B Part B · Comparing data (1 point each)

Directions: Find each measure.

5. Mean of 8, 10, 11, 13, 18

6. MAD of 8, 10, 11, 13, 18

7. IQR of 5, 6, 9, 10, 12, 15, 16, 20

8. Means 52 and 60, MADs 4. MADs apart?

PART C Part C · Probability (1 point each)

Directions: Find each probability as a fraction.

9 Bag: 2 red, 3 green, 5 yellow. P(green)

10 P(not yellow)

11 Coin + cube: P(H and 3)

12 Two cubes: P(sum is 9)

13 Three coins: P(at least one head)

14 18 of 30 spins landed on red. Experimental P(red)?

PART D Part D · Problem solving (3 points each)

Directions: Answer each problem fully. Show your work and explain your reasoning.

15 Random samples of two orchards: Orchard A apples mean 150 g, MAD 10 g; Orchard B mean 175 g, MAD 10 g. Compare the orchards. Is the difference meaningful? Explain.

16 Design a simulation to estimate the probability that a family with 4 children has at least 3 girls. Describe the tool, one trial, and what you record.

UNIT 8

Scale Drawings and Geometric Figures

Scaled copies, scale drawings and maps, drawing triangles from conditions, and cross sections.

Your practice pages for Unit 8. Each lesson matches the same lesson number in your textbook.

UNIT VOCABULARY

scale factor	Multiplier for every length.
scale	Drawing length : actual length.
triangle inequality	Two sides sum to more than the third.
cross section	The shape of a slice.

LESSON 8.1 · PRACTICE
Week 1 · Lesson 1 of 4

Unit 8: Scale Drawings and Geometric Figures

Lesson 8.1: Scaled Copies

Name _____ Date _____

TODAY'S GOAL

I can tell whether one figure is a scaled copy of another.

Standards: 7.G.1

WHAT TO DO

- [ ] Read Textbook Lesson 8.1 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Scaled copy? Write yes or no, Part D Draw a scaled copy of the L-shape with every length × 3, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Answer each question.

1. Is a 4 × 10 rectangle a scaled copy of 2 × 5?

2. Is a 5 × 7 rectangle a scaled copy of 3 × 5?

3. Is any square a scaled copy of any other square?

PART B Warm-Up

Directions: Divide.

4. $6 \div 3$

5. $8 \div 4$

PART C Scaled copy? Write yes or no.

6. 2×3 and 6×9

7. 4×6 and 6×8

8. 5×10 and 1×2

9. 3×7 and 9×14

10. 8×12 and 2×3

11. 1×4 and 3×7

PART D Draw a scaled copy of the L-shape with every length × 3.

Directions: Draw the scaled copy on the grid. Multiply every length by 3.

12.

PART E Exit Ticket

13. Lillie adds 2 to every side of a triangle. Is the new triangle a scaled copy? Explain.

LESSON 8.2 · PRACTICE
Week 1 · Lesson 2 of 4

Unit 8: Scale Drawings and Geometric Figures

Lesson 8.2: Scale Factors

Name _____ Date _____

TODAY'S GOAL

I can find and use a scale factor.

Standards: 7.G.1

WHAT TO DO

- [ ] Read Textbook Lesson 8.2 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Find the scale factor, Part D Find the missing length, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Answer each question.

1. Original 5 cm, copy 15 cm. Scale factor?

2. Original 12 ft, copy 3 ft. Scale factor?

3. Side 8 m, scale factor 0.75. Copy length?

PART B Warm-Up

Directions: Solve each problem. Show your work.

4. $10 \div 4$

5. 6×2.5

PART C Find the scale factor

6. $3 \to 12$

7. $10 \to 4$

8. $6 \to 9$

9. $20 \to 5$

10. $8 \to 8$

11. $4 \to 14$

PART D Find the missing length

12. Original 6 cm, scale factor 3

13. Original 12 cm, scale factor $\frac{1}{3}$

14. Original 9 cm, scale factor $1\frac{2}{3}$

15. Original 14 cm, scale factor 0.5

16. Original 10 cm, scale factor 1.5

17. Original 16 cm, scale factor 0.75

PART E Exit Ticket

18. A triangle has sides 6, 8, 10. A copy's shortest side is 9. Find the other two sides.

LESSON 8.3 · PRACTICE
Week 1 · Lesson 3 of 4

Unit 8: Scale Drawings and Geometric Figures

Lesson 8.3: Scale Factors and Area

Name _____ Date _____

TODAY'S GOAL

I can find how area changes when a figure is scaled.

Standards: 7.G.1

WHAT TO DO

- [ ] Read Textbook Lesson 8.3 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Find the new area, Part D Solve, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Answer each question.

1. Area 5, scale factor 4. New area?

2. Area 36, scale factor $\frac{1}{3}$

3. Area changed from 7 to 63. Scale factor?

PART B Warm-Up

Directions: Evaluate.

4. 2^2

5. $(\frac{1}{2})^2$

PART C Find the new area

6. Area 8, scale factor 2

7. Area 6, scale factor 5

8. Area 50, scale factor 0.2

9. Area 12, scale factor 1.5

10. Area 20, scale factor 0.5

11. Area 3, scale factor 10

PART D Solve

12. A garden 4 m by 5 m is redesigned with every length × 1.5. Find the new area.

13. A poster's area is 9 times a flyer's area. What is the scale factor?

PART E Exit Ticket

Directions: Explain the mistake in complete sentences.

14. Lillie says scaling by 3 triples the area. Explain her mistake.

LESSON 8.4 · PRACTICE
Week 1 · Lesson 4 of 4

Unit 8: Scale Drawings and Geometric Figures

Lesson 8.4: Scale Drawings

Name _____ Date _____

TODAY'S GOAL

I can use a scale to find actual lengths from a scale drawing.

Standards: 7.G.1

WHAT TO DO

- [ ] Read Textbook Lesson 8.4 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Find the actual length, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Answer each question.

1. Scale 1 cm : 5 m. Drawing 7 cm. Actual?

2. Scale 1 in. : 20 mi. Drawing 3.5 in. Actual?

3. Scale 2 cm : 1 m. Drawing 9 cm. Actual?

PART B Warm-Up

Directions: Multiply.

4. 1.5 × 8

5. 1.75 × 8

PART C Find the actual length

6. Scale 1 in. : 6 ft; drawing 4 in.

7. Scale 1 cm : 3 m; drawing 8.5 cm

8. Scale 1 in. : 50 mi; drawing 2.25 in.

9. Scale 1 cm : 10 km; drawing 6.3 cm

10. Scale 2 in. : 5 ft; drawing 6 in.

11. Scale $1/4$ in. : 1 ft; drawing 3 in.

PART D Exit Ticket

12. A model car uses 1 in. : 2 ft. The model is 7 in. long. How long is the real car?

WEEKLY CHECK
After Lesson 8.4

About 15 minutes

Weekly Check 22: Scale Factors

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each question carefully. Show your thinking.
- [ ] When you finish, check that every question has an answer.

Show what you know

Directions: Answer each item about scale. Show your work.

1. Is 6×15 a scaled copy of 2×5?

2. Original 8 cm, copy 20 cm. Scale factor?

3. Side 12 in., scale factor $\frac{2}{3}$

4. Area 7, scale factor 3. New area?

5. Scale 1 in. : 5 ft; drawing 6.4 in. Actual?

6. Scale 1 cm : 4 m; drawing 3.5 cm. Actual?

LESSON 8.5 · PRACTICE
Week 2 · Lesson 1 of 4

Unit 8: Scale Drawings and Geometric Figures

Lesson 8.5: Finding Drawing Lengths

Name _____ Date _____

TODAY'S GOAL

I can find the length on a scale drawing for a given actual length.

Standards: 7.G.1

WHAT TO DO

- [ ] Read Textbook Lesson 8.5 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Find the drawing length, Part D Plan a room, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Answer each question.

1. Scale 1 in. : 4 ft. Actual 22 ft. Drawing?

2. Scale 1 cm : 20 km. Actual 130 km. Drawing?

3. Scale 2 cm : 3 m. Actual 12 m. Drawing?

PART B Warm-Up

Directions: Solve each problem. Show your work.

4. $100 \div 5$

5. $10 \times \frac{1}{2}$

PART C Find the drawing length

6. Scale 1 in. : 10 ft; actual 45 ft

7. Scale 1 cm : 2 m; actual 15 m

8. Scale 1 in. : 25 mi; actual 175 mi

9. Scale 1 cm : 50 km; actual 325 km

10. Scale 3 cm : 1 m; actual 4 m

11. Scale $\frac{1}{4}$ in. : 1 ft; actual 18 ft

PART D Plan a room

12. Lillie's room is 12 ft by 10 ft. Draw it using the scale 1 square : 2 ft. What are the drawing dimensions?

PART E Exit Ticket

13. A 60-ft tree is drawn with 1 in. : 15 ft. How tall is the drawing?

LESSON 8.6 · PRACTICE
Week 2 · Lesson 2 of 4

Unit 8: Scale Drawings and Geometric Figures

Lesson 8.6: Area in Scale Drawings

Name _____ Date _____

TODAY'S GOAL

I can find actual areas from scale drawings.

Standards: 7.G.1

WHAT TO DO

- [ ] Read Textbook Lesson 8.6 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Find the actual area, Part D Solve, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Answer each question.

1. Drawing 3 in. × 5 in.; 1 in. : 4 ft. Actual area?

2. Drawing 2 cm × 2 cm; 1 cm : 10 m. Actual area?

3. Drawing area 6 in²; 1 in. : 3 ft. Actual area?

PART B Warm-Up

Directions: Multiply.

4 200 × 300

5 50 × 50

PART C Find the actual area

6 Drawing 2 × 3 in.; 1 in. : 5 ft

7 Drawing 5 × 8 cm; 1 cm : 2 m

8 Drawing 3/2 × 4 in.; 1 in. : 6 ft

9 Drawing 5/2 × 7/2 cm; 1 cm : 10 m

PART D Solve

10 A rectangular rug on a plan (1 in. : 2 ft) measures 2.5 in. by 4 in. Carpet costs $3 per ft². Cost of the rug?

PART E Exit Ticket

11 A plan uses 1 cm : 3 m. A room is 4 cm by 5 cm on the plan. Find the actual area.

LESSON 8.7 · PRACTICE
Week 2 · Lesson 3 of 4

Unit 8: Scale Drawings and Geometric Figures

Lesson 8.7: Maps and Scale

Name _____ Date _____

TODAY'S GOAL

I can use a map scale to find real distances and travel times.

Standards: 7.G.1, 7.RP.2

WHAT TO DO

- [ ] Read Textbook Lesson 8.7 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Find the actual distance, Part D Solve, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Answer each question.

1. Map 1 in. : 30 mi. Distance 4.5 in.?

2. Scale 1 : 100,000. Map 4 cm. Actual km?

3. 150 mi at 60 mph: hours?

PART B Warm-Up

Directions: Solve each problem. Show your work.

4. 7.2 × 25

5. 300,000 cm in km

PART C Find the actual distance

6. Scale 1 cm : 40 km; map 3.5 cm

7. Scale 1 in. : 15 mi; map 6 in.

8. Scale 1 : 25,000; map 8 cm

9. Scale 1 : 200,000; map 5 cm

PART D Solve

10. On a map (1 in. : 50 mi), Ellie's trip is 4.8 in. She drives 60 mph. How long is the drive?

11. A hiking map is 1 : 24,000. A trail is 15 cm on the map. How many kilometers is it?

PART E Exit Ticket

12. Two towns are 75 km apart. With 1 cm : 15 km, how far apart on the map?

LESSON 8.8 · PRACTICE
Week 2 · Lesson 4 of 4

Unit 8: Scale Drawings and Geometric Figures

Lesson 8.8: Changing the Scale

Name _____ Date _____

TODAY'S GOAL

I can reproduce a scale drawing at a different scale.

Standards: 7.G.1

WHAT TO DO

- [ ] Read Textbook Lesson 8.8 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Redraw at the new scale, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Answer each question.

1. Plan 8 in. at 1 in. : 5 ft; redraw at 1 in. : 10 ft

2. Plan 3 cm at 1 cm : 6 m; redraw at 1 cm : 2 m

3. Plan 12 cm at 1 cm : 1 m; redraw at 1 cm : 3 m

PART B Warm-Up

Directions: Solve each problem. Show your work.

4 10×2

5 $20 \div 4$

PART C Redraw at the new scale

Directions: Find the actual length, then the new drawing length at the new scale.

6 6 in. at 1 in. : 4 ft; new scale 1 in. : 8 ft

7 5 cm at 1 cm : 10 m; new scale 1 cm : 5 m

8 9 in. at 1 in. : 2 ft; new scale 1 in. : 3 ft

9 4 cm at 1 cm : 3 m; new scale 1 cm : 2 m

PART D Exit Ticket

10 A rectangle 8 cm × 4 cm uses 1 cm : 3 m. Redraw it at 1 cm : 6 m on the grid (1 square = 1 cm).

WEEKLY CHECK
After Lesson 8.8

About 15 minutes

Weekly Check 23: Scale Drawings and Maps

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each question carefully. Show your thinking.
- [ ] When you finish, check that every question has an answer.

Show what you know

1. Scale 1 in. : 8 ft; actual 36 ft. Drawing?

2. Drawing 3 cm × 4 cm; 1 cm : 5 m. Actual area?

3. Map 1 cm : 30 km; 4.5 cm. Actual?

4. Scale 1 : 50,000; map 4 cm. Actual km?

5. Drawing 10 cm at 1 cm : 2 m. Redraw at 1 cm : 5 m.

6. 180 mi at 45 mph: hours?

LESSON 8.9 · PRACTICE
Week 3 · Lesson 1 of 4

Unit 8: Scale Drawings and Geometric Figures

Lesson 8.9: Triangle Side Lengths

Name _____ Date _____

TODAY'S GOAL

I can decide whether three side lengths can form a triangle.

Standards: 7.G.2

WHAT TO DO

- [ ] Read Textbook Lesson 8.9 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Triangle? Write yes or no, Part D Find the range, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Decide whether each set of side lengths makes a triangle. Explain.

1. 3, 3, 5

2. 1, 2, 4

3. 8, 8, 8

PART B Warm-Up

Directions: Answer each item.

4. 2 + 3

5. Is 5 > 6?

PART C Triangle? Write yes or no.

6. 5, 6, 7

7. 2, 2, 5

8. 10, 4, 7

9. 3, 9, 6

10. 12, 5, 13

11. 1.5, 2.5, 3.5

PART D Find the range

12. Two sides are 5 cm and 8 cm. What lengths could the third side be?

13. Two sides are 10 in. and 4 in. Can the third side be 6 in.? 7 in.?

PART E Exit Ticket

14. Explain why 4, 5, 10 cannot form a triangle.

LESSON 8.10 · PRACTICE
Week 3 · Lesson 2 of 4

Unit 8: Scale Drawings and Geometric Figures

Lesson 8.10: Triangle Angles

Name _____ Date _____

TODAY'S GOAL

I can use the angle sum of a triangle to decide whether angles can form a triangle.

Standards: 7.G.2

WHAT TO DO

- [ ] Read Textbook Lesson 8.10 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Find the missing angle, Part D Can these be the angles of a triangle?, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Answer each question.

1. 30°, 60°, 90°: triangle?

2. Two angles 110° and 80°: triangle?

3. Missing angle: 35°, 35°, ?

PART B Warm-Up

Directions: Evaluate.

4. $180 - 48 - 75$

5. $90 + 60 + 40$

PART C Find the missing angle

Directions: The angles of a triangle add to 180°. Find the third angle.

6. 60° and 60°

7. 90° and 25°

8. 120° and 35°

9. 45° and 45°

10. 100° and 47°

11. 15° and 82°

PART D Can these be the angles of a triangle?

Directions: Write yes or no for each set. Explain.

12. 70°, 70°, 40°

13. 90°, 90°, 10°

14. 100°, 40°, 50°

15. 33°, 57°, 90°

PART E Exit Ticket

16. A right triangle has one 38° angle. Find the third angle.

LESSON 8.11 · PRACTICE
Week 3 · Lesson 3 of 4
Unit 8: Scale Drawings and Geometric Figures

Lesson 8.11: Drawing Triangles from Conditions

Name _____ Date _____

TODAY'S GOAL

I can draw triangles with a ruler and protractor and tell whether the conditions give one, many, or no triangles.

Standards: 7.G.2

WHAT TO DO

- [ ] Read Textbook Lesson 8.11 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C One, many, or none?, Part D Draw it, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Tell whether the conditions make one triangle, many triangles, or no triangle.

1. Sides 4, 4, 4 cm

2. Angles 45°, 45°, 90°

3. Angles 100°, 100°, ?

PART B Warm-Up

Directions: Answer each item.

4 Is 5 + 6 > 7?

5 50 + 60 + 70

PART C One, many, or none?

Directions: Write one, many, or none for each set of conditions.

6 sides 3, 4, 8 cm

7 sides 6, 8, 10 cm

8 angles 30°, 60°, 90°

9 sides 5 and 9 cm with a 70° angle between

10 angles 40° and 80° with a 6 cm side between

11 angles 95°, 95°, ?

PART D Draw it

12 Use a ruler and protractor to draw a triangle with sides 6 cm and 4 cm and a 60° angle between them. Measure the third side.

PART E Exit Ticket

13 Why do three angles not give a unique triangle?

LESSON 8.12 · PRACTICE
Week 3 · Lesson 4 of 4

Unit 8: Scale Drawings and Geometric Figures

Lesson 8.12: Drawing Quadrilaterals with Conditions

Name _____ Date _____

TODAY'S GOAL

I can draw quadrilaterals with given conditions and decide whether they are unique.

Standards: 7.G.2

WHAT TO DO

- [ ] Read Textbook Lesson 8.12 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C One or many?, Part D Draw it, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Answer each question.

1. Rectangle with sides 6 and 2: how many?

2. Parallelogram with sides 6 and 2: how many?

3. Square with sides 3 cm: how many?

PART B Warm-Up

Directions: Answer each item.

4 Angles in a rectangle

5 Angle sum of a quadrilateral

PART C One or many?

Directions: Write one or many: how many different shapes fit each description?

6 square, side 5 cm

7 rhombus, side 5 cm

8 rectangle, 4 cm by 7 cm

9 quadrilateral with sides 3, 4, 5, 6 cm

PART D Draw it

10 Draw a parallelogram with sides 5 cm and 3 cm and a 60° angle. Is it unique?

PART E Exit Ticket

11 A quadrilateral has angles 90°, 90°, 90°, and ? Find the fourth angle.

WEEKLY CHECK
After Lesson 8.12

About 15 minutes

Weekly Check 24: Triangles and Quadrilaterals

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each question carefully. Show your thinking.
- [ ] When you finish, check that every question has an answer.

Show what you know

Directions: Answer each item.

1. Triangle? 4, 6, 11

2. Triangle? 7, 8, 12

3. Missing angle: 64° and 71°

4. Angles 50°, 60°, 70°: one, many, or none?

5. Sides 5 cm, 7 cm with 45° between: one, many, or none?

6. Rhombus with side 4 cm: one or many?

LESSON 8.13 · PRACTICE
Week 4 · Lesson 1 of 4

Unit 8: Scale Drawings and Geometric Figures

Lesson 8.13: Cross Sections of Prisms

Name _____ Date _____

TODAY'S GOAL

I can describe the cross sections made by slicing a rectangular prism.

Standards: 7.G.3

WHAT TO DO

- [ ] Read Textbook Lesson 8.13 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Name the cross section, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Name the shape of each cross section.

1. Slice of a 4 × 6 × 10 box parallel to the 4 × 6 base

2. Vertical slice of a cube

3. Slice cutting off one corner of a cube

PART B Warm-Up

Directions: Answer each item.

4 How many faces does a cube have?

5 Shape of each face of a cube

PART C Name the cross section

6 A box 3 × 5 × 8, sliced parallel to the 3 × 5 face

7 A cube sliced diagonally from one top edge to the opposite bottom edge

8 A loaf of bread sliced like usual

9 A triangular prism sliced parallel to its triangle base

PART D Exit Ticket

10 Can a slice of a cube be a pentagon? Try to sketch one.

LESSON 8.14 · PRACTICE
Week 4 · Lesson 2 of 4

Unit 8: Scale Drawings and Geometric Figures

Lesson 8.14: Cross Sections of Pyramids

Name _____ Date _____

TODAY'S GOAL

I can describe the cross sections made by slicing a pyramid.

Standards: 7.G.3

WHAT TO DO

- [ ] Read Textbook Lesson 8.14 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Name the cross section, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Name the shape of each cross section.

1. Square pyramid, horizontal slice

2. Triangular pyramid, slice parallel to base

3. Square pyramid, vertical slice not through the top

PART B Warm-Up

Directions: Answer each item.

4 Shape of a square pyramid's base

5 Shape of its side faces

PART C Name the cross section

6 Square pyramid, slice parallel to the base near the top

7 Square pyramid, vertical slice through the apex

8 Rectangular pyramid, horizontal slice

9 Triangular pyramid, horizontal slice

PART D Exit Ticket

10 A square pyramid's base is 10 in. on a side. A slice halfway up and parallel to the base: what shape and size?

LESSON 8.15 · PRACTICE
Week 4 · Lesson 3 of 4

Unit 8: Scale Drawings and Geometric Figures

Lesson 8.15: Cross Sections of Cylinders and Cones

Name _____ Date _____

TODAY'S GOAL

I can describe the cross sections of cylinders and cones.

Standards: 7.G.3

WHAT TO DO

- [ ] Read Textbook Lesson 8.15 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Name the cross section, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Name the shape of each cross section.

1. Slice of a cylinder parallel to the base

2. Vertical slice of a cone through its tip

3. Vertical slice of a cylinder through its center

PART B Warm-Up

Directions: Answer each item.

4 Shape of a cylinder's bases

5 Shape of a cone's base

PART C Name the cross section

6 Cylinder, horizontal

7 Cylinder, vertical through center

8 Cone, horizontal

9 Cone, vertical through tip

10 Cylinder (r = 3, h = 10), vertical through center: dimensions

11 Sphere, any slice

PART D Exit Ticket

12 A cylinder has diameter 4 in. and height 9 in. Describe its vertical cross section through the center.

LESSON 8.16 · PRACTICE
Week 4 · Lesson 4 of 4

Unit 8: Scale Drawings and Geometric Figures

Lesson 8.16: Scale Drawing Project

Name _____ Date _____

TODAY'S GOAL

I can create a scale drawing of a real space and use it to solve problems.

Standards: 7.G.1

WHAT TO DO

- [ ] Read Textbook Lesson 8.16 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Project, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Use the scale 1 in. : 2 ft. Find each drawing length or area.

1. Rug 5 ft × 8 ft at 1 in. : 2 ft

2. Window 3 ft wide at 1 in. : 2 ft

3. Area of the room above

PART B Warm-Up

Directions: Solve each problem. Show your work.

4 $14 \div 2$

5 14×12

PART C Project

6 Measure your room. Length and width:

7 Choose a scale:

8 Draw your room and at least three objects on the grid.

9 Find the floor area of your room.

PART D Exit Ticket

10 Why did you choose your scale?

UNIT REVIEW
Study before the unit test

About 40 minutes

Unit 8 Review

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each question carefully. Show your thinking.
- [ ] When you finish, check that every question has an answer.

PART A Scale

Directions: Solve each problem. Show your work.

1. Scale factor from 15 to 6

2. Area 20, scale factor 1.5

3. Plan 1 in. : 3 ft; room 5 in. × 4.5 in. Actual size?

4. Same room: actual area?

5. Map 1 : 100,000; 7.5 cm. Actual km?

6. Redraw 9 cm at 1 cm : 4 m using 1 cm : 6 m

PART B Geometry

Directions: Answer each item.

7. Two sides 6 and 9: possible third sides

8. Missing angle: 90° and 27°

9. ASA: one, many, or none?

10. Cross section of a cylinder, horizontal

11 Cross section of a square pyramid through the apex, vertical

12 Cross section of a 2 × 3 × 7 box parallel to the 2 × 3 face

UNIT TEST
At the end of the unit

About 45 minutes

Unit 8 Test

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] This test has 21 points.
- [ ] Read each question carefully. Show your thinking.
- [ ] When you finish, check that every question has an answer.

PART A · Part A · Scale (1 point each)

Directions: Answer each item about scale.

1. Is 9 × 12 a scaled copy of 3 × 4?

2. Scale factor from 10 to 25

3. Side 18 cm, scale factor $\frac{1}{3}$

4. Area 12, scale factor 2

5. Scale 1 in. : 6 ft; drawing 4.5 in. Actual?

6. Scale 1 cm : 25 km; actual 200 km. Drawing?

PART B · Part B · Triangles and figures (1 point each)

Directions: Answer each item.

7. Triangle? 3, 5, 9

8. Missing angle: 38° and 106°

9. Sides 6, 8, 10: one, many, or none?

10. Angles 45°, 45°, 90°: one, many, or none?

11 Cone, horizontal slice

12 Cube, slice parallel to a face

PART C Part C · Problem solving (3 points each)

13 A floor plan uses 1 in. : 4 ft. A living room is 5 in. by 3.5 in. on the plan. Find the actual dimensions and area. Carpet costs $2.50 per square foot: how much to carpet the room?

14 A map uses 1 cm : 20 km. Two towns are 9 cm apart. Ellie drives 60 km/h. How long is the drive? Show your work.

15 Use a ruler and protractor to draw a triangle with a 6 cm side between angles of 50° and 70°. Find the third angle. Is the triangle unique?

UNIT 9

Angles, Circles, Area, and Volume

Angle relationships, circles, composite area, surface area and volume of prisms, and a review of the whole year.

Your practice pages for Unit 9. Each lesson matches the same lesson number in your textbook.

UNIT VOCABULARY

supplementary	Sum to 180°.
complementary	Sum to 90°.
circumference	$C = \pi d$.
area of a circle	$A = \pi r^2$.
volume of a prism	$V = Bh$.

LESSON 9.1 · PRACTICE
Week 1 · Lesson 1 of 4

Unit 9: Angles, Circles, Area, and Volume

Lesson 9.1: Angle Relationships

Name _____ Date _____

TODAY'S GOAL

I can identify complementary, supplementary, vertical, and adjacent angles.

Standards: 7.G.5

WHAT TO DO

- [ ] Read Textbook Lesson 9.1 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Find the angle, Part D Name the relationship, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Find each angle measure.

1. Complement of 28°

2. Supplement of 28°

3. Two lines cross; one angle is 75°. Its vertical angle?

PART B Warm-Up

Directions: Subtract.

4 90 − 35

5 180 − 130

PART C Find the angle

Directions: Find each complement or supplement.

6 Complement of 40°

7 Supplement of 40°

8 Complement of 72°

9 Supplement of 115°

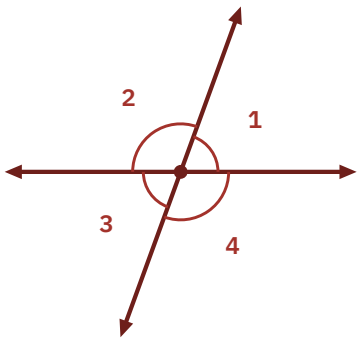
10 Vertical angle to 33°

11 Neighbor (adjacent, on a line) of 33°

PART D Name the relationship

Directions: Name the angle relationship, or answer the question.

12



13 Can two angles be both complementary and vertical? Explain.

PART E Exit Ticket

Directions: Write an equation and find both angles.

14 Two angles are supplementary. One is 4 times the other. Find both.

LESSON 9.2 · PRACTICE
Week 1 · Lesson 2 of 4

Unit 9: Angles, Circles, Area, and Volume

Lesson 9.2: Solving for Unknown Angles

Name _____ Date _____

TODAY'S GOAL

I can write and solve equations to find unknown angles.

Standards: 7.G.5, 7.EE.4a

WHAT TO DO

- [ ] Read Textbook Lesson 9.2 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Write an equation and solve, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Write an equation and solve for x.

1. x + 48 = 90

2. Vertical angles (4x − 20)° and 60°

3. Supplementary: 5x and 4x

PART B Warm-Up

Directions: Solve.

4 $2x + 80 = 180$

5 $3x = 105$

PART C Write an equation and solve

6 Complementary: $x°$ and $64°$

7 Supplementary: $(x + 30)°$ and $2x°$

8 Vertical: $(5x - 10)°$ and $80°$

9 Complementary: $(3x)°$ and $(x + 10)°$

10 Three angles on a line: $40°$, $x°$, and $2x°$

11 Angles around a point: $90°$, $150°$, and $x°$

PART D Exit Ticket

Directions: Write an equation, solve for x, and find the angle measure.

12 Two vertical angles measure $(6x + 4)°$ and $(4x + 20)°$. Find x and the angle.

LESSON 9.3 · PRACTICE
Week 1 · Lesson 3 of 4

Unit 9: Angles, Circles, Area, and Volume

Lesson 9.3: Circles and Circumference

Name _____ Date _____

TODAY'S GOAL

I can find the circumference of a circle from its radius or diameter.

Standards: 7.G.4

WHAT TO DO

- [ ] Read Textbook Lesson 9.3 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Find the circumference, Part D Solve, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Use $C = \pi d$ or $C = 2\pi r$. Use 3.14 for π.

1. C when d = 6 ft

2. C when r = 3 m

3. d when C = 31.4 cm

PART B Warm-Up

Directions: Multiply.

4. 3.14 × 10

5. 2 × 7

PART C Find the circumference

Directions: Use C = πd or C = 2πr. Use 3.14 for π. Write the units.

6. d = 4 cm

7. r = 5 in.

8. d = 20 ft

9. r = 12 m

10. d = 5/2 cm

11. r = 50 mm

PART D Solve

12. A bike wheel has diameter 26 in. How far does it roll in one turn? In 100 turns?

13. A circular track is 314 m around. What is its diameter?

PART E Exit Ticket

14. A circle has radius 9 cm. Find its circumference.

LESSON 9.4 · PRACTICE
Week 1 · Lesson 4 of 4

Unit 9: Angles, Circles, Area, and Volume

Lesson 9.4: Area of a Circle

Name _____ Date _____

TODAY'S GOAL

I can find the area of a circle.

Standards: 7.G.4

WHAT TO DO

- [ ] Read Textbook Lesson 9.4 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Find the area, Part D Solve, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Answer each question.

1. A when r = 10 m

2. A when d = 6 ft

3. A when r = 1.5 cm

PART B Warm-Up

Directions: Evaluate.

4. 5^2

5. 3.14×25

PART C Find the area

Directions: Use $A = \pi r^2$. Use 3.14 for π. Write square units.

6. r = 2 cm

7. r = 7 in.

8. d = 10 ft

9. d = 12 m

10. r = 20 yd

11. d = 3 cm

PART D Solve

12. A sprinkler waters a circle of radius 15 ft. What area does it water?

13. A 12-in. pizza (diameter) costs $12. A 16-in. pizza costs $18. Which is the better deal per square inch?

PART E Exit Ticket

14. Find the area of a circle with diameter 14 cm.

WEEKLY CHECK
After Lesson 9.4

About 15 minutes

Weekly Check 25: Angles and Circles

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each question carefully. Show your thinking.
- [ ] When you finish, check that every question has an answer.

Show what you know

Directions: Find each angle, or solve for x.

1. Complement of 37°

2. Vertical angles (3x + 6)° and 51°: x?

3. Supplementary: x° and (2x + 30)°

4. Circumference, r = 5 in.

5. Area, d = 10 cm

6. Diameter if C = 47.1 m

LESSON 9.5 · PRACTICE
Week 2 · Lesson 1 of 4

Unit 9: Angles, Circles, Area, and Volume

Lesson 9.5: Problems with Circles

Name _____ Date _____

TODAY'S GOAL

I can find areas and perimeters of figures made from parts of circles.

Standards: 7.G.4

WHAT TO DO

- [ ] Read Textbook Lesson 9.5 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Find the area, Part D Solve, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Find each area or perimeter. Use 3.14 for π.

1. Area of a semicircle, r = 4 cm

2. Area of a quarter circle, r = 10 ft

3. Perimeter of a semicircle, d = 10 in. (curve + diameter)

PART B Warm-Up

Directions: Evaluate.

4. $3.14 \times 25 \div 2$

5. $3.14 \times 10 \div 2$

PART C Find the area

Directions: Find each area. Use 3.14 for π. Write square units.

6. Semicircle, r = 6 m

7. Quarter circle, r = 2 cm

8. Square 8 × 8 with a circle (d = 8) cut out

9. Rectangle 12 × 4 with semicircles on both short ends

PART D Solve

10. A running track is a 100 m by 60 m rectangle with semicircles on both 60 m ends. Find the distance around the track.

PART E Exit Ticket

11. Find the area of the window above if the rectangle is 4 ft by 3 ft (semicircle on the 4 ft side).

LESSON 9.6 · PRACTICE
Week 2 · Lesson 2 of 4

Unit 9: Angles, Circles, Area, and Volume

Lesson 9.6: Area of Composite Figures

Name _____ Date _____

TODAY'S GOAL

I can find the area of composite figures made of triangles, rectangles, and trapezoids.

Standards: 7.G.6

WHAT TO DO

- [ ] Read Textbook Lesson 9.6 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Find the area, Part D Solve, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Find each area. Write square units.

1. Triangle, b = 9, h = 6

2. Trapezoid, bases 7 and 11, h = 5

3. Rectangle 10 × 8 with a 3 × 3 square cut out

PART B Warm-Up

Directions: Evaluate.

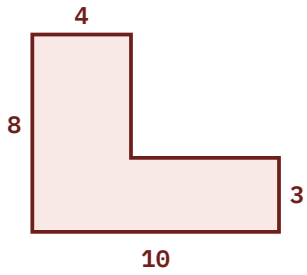
4 $12 \times 4 + 5 \times 5$

5 $(10 + 5) \div 2 \times 4$

PART C Find the area

Directions: Split each figure into simpler shapes. Find each area, then add or subtract.

6 L-shape: full width 10, full height 8, top width 4, bottom height 3



7 Trapezoid: bases 6 and 14, height 5

8 A rectangle 8 × 5 with a triangle (base 8, height 3) on top

9 Parallelogram b = 12, h = 7

PART D Solve

10 A deck is a 16 ft by 12 ft rectangle with a 4 ft by 4 ft square cut out for a tree. Stain covers 50 ft² per can. How many cans?

PART E Exit Ticket

11 Find the area: a 10 × 6 rectangle with a right triangle (legs 6 and 4) attached to the right side.

LESSON 9.7 · PRACTICE
Week 2 · Lesson 3 of 4

Unit 9: Angles, Circles, Area, and Volume

Lesson 9.7: Surface Area of Prisms

Name _____ Date _____

TODAY'S GOAL

I can find the surface area of rectangular and triangular prisms.

Standards: 7.G.6

WHAT TO DO

- [ ] Read Textbook Lesson 9.7 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Find the surface area, Part D Solve, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Find the surface area. Add the areas of all the faces.

1. Cube with edge 3 cm

2. Box 10 × 2 × 4

3. Box 6 × 6 × 1

PART B Warm-Up

Directions: Evaluate.

4. $2 \times (15 + 20 + 12)$

5. $\frac{1}{2} \times 6 \times 8$

PART C Find the surface area

Directions: Find the surface area of each box. Write square units.

6. Box $4 \times 4 \times 4$

7. Box $8 \times 3 \times 2$

8. Box $10 \times 5 \times 6$

9. Box $3/2 \times 2 \times 4$

10. Triangular prism: legs 3 and 4, hypotenuse 5, length 10

11. Triangular prism: legs 5 and 12, hypotenuse 13, length 8

PART D Solve

12. Lillie wraps a gift box 12 in. × 8 in. × 3 in. How much wrapping paper covers it exactly?

PART E Exit Ticket

13. Find the surface area of a box 7 cm × 2 cm × 5 cm.

LESSON 9.8 · PRACTICE
Week 2 · Lesson 4 of 4
Unit 9: Angles, Circles, Area, and Volume

Lesson 9.8: Volume of Prisms

Name _____ Date _____

TODAY'S GOAL

I can find the volume of any right prism using V = Bh.

Standards: 7.G.6

WHAT TO DO

- [ ] Read Textbook Lesson 9.8 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Find the volume, Part D Solve, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Find the volume. Write cubic units.

1. Box 6 × 2 × 5

2. Cube edge 4

3. Prism with B = 18 cm², h = 7 cm

PART B Warm-Up

Directions: Multiply.

4. $5 \times 3 \times 4$

5. 24×12

PART C Find the volume

Directions: Find the volume of each box. Write cubic units.

6. Box $9 \times 2 \times 3$

7. Box $5/2 \times 4 \times 6$

8. Box $10 \times 10 \times 1/2$

9. Triangular prism: legs 3 and 4, length 10

10. Trapezoid base (bases 5, 9; height 2), length 6

11. $B = 32.5$ in², $h = 4$ in.

PART D Solve

12. A fish tank is 30 in. × 12 in. × 16 in. One gallon is about 231 in³. About how many gallons does it hold?

PART E Exit Ticket

13. A prism's base is a triangle with base 10 cm and height 6 cm. The prism is 9 cm long. Find its volume.

WEEKLY CHECK
After Lesson 9.8

About 15 minutes

Weekly Check 26: Area, Surface Area, and Volume

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each question carefully. Show your thinking.
- [ ] When you finish, check that every question has an answer.

Show what you know

Directions: Find each area, surface area, or volume. Write the units.

1. Semicircle area, r = 10

2. Trapezoid: bases 8 and 12, height 5

3. Surface area, box 4 × 3 × 2

4. Volume, box 4 × 3 × 2

5. Volume, triangular prism: legs 5, 6; length 10

6. SA, cube edge 5

LESSON 9.9 · PRACTICE
Week 3 · Lesson 1 of 4

Unit 9: Angles, Circles, Area, and Volume

Lesson 9.9: Volume and Surface Area in Context

Name _____ Date _____

TODAY'S GOAL

I can decide whether a problem needs volume or surface area and solve it.

Standards: 7.G.6

WHAT TO DO

- [ ] Read Textbook Lesson 9.9 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Volume or surface area? Then solve, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Write whether each job needs volume or surface area.

1. Water to fill a pool

2. Tiles to line the pool's walls and floor

3. Label around a soup can

PART B Warm-Up

Directions: Evaluate.

4 $6 \times 4 \times 1$

5 $2 \times 6 + 2 \times 4$

PART C Volume or surface area? Then solve.

6 How much soil fills a planter 3 ft × 2 ft × 1.5 ft?

7 How much fabric covers a cube-shaped ottoman, 2 ft edges, all faces except the bottom?

8 An open-top box (no lid) is 10 cm × 8 cm × 5 cm. How much cardboard?

9 Ice cubes are 2 cm on each edge. How many fit in a 10 cm × 8 cm × 6 cm tray (exactly packed)?

PART D Exit Ticket

10 Paint covers 350 ft² per gallon. Walls of a room total 1,050 ft². How many gallons?

LESSON 9.10 · PRACTICE
Week 3 · Lesson 2 of 4

Unit 9: Angles, Circles, Area, and Volume

Lesson 9.10: Composite Solids

Name _____ Date _____

TODAY'S GOAL

I can find the volume and surface area of solids made from prisms.

Standards: 7.G.6

WHAT TO DO

- [ ] Read Textbook Lesson 9.10 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Find the volume, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Find the volume of each composite solid.

1. Two cubes, edges 3 and 2, stacked

2. Box 8 × 5 × 4 with a 2 × 2 × 4 hole

3. Box 5 × 5 × 5 plus a 5 × 5 × 2 box on top

PART B Warm-Up

Directions: Evaluate.

4. $6 \times 3 \times 1 + 3 \times 3 \times 1$

5. $1{,}000 - 40$

PART C Find the volume

Directions: Split each solid into simpler prisms. Find each volume, then add or subtract.

6. L-shaped block: $10 \times 4 \times 2$ plus $4 \times 4 \times 2$

7. Box $12 \times 6 \times 6$ minus a cube of edge 3

8. Triangular prism (legs 4, 6; length 10) on a box $6 \times 10 \times 2$

9. Three stacked boxes: $6 \times 6 \times 1$, $4 \times 4 \times 1$, $2 \times 2 \times 1$

PART D Exit Ticket

10. A planter is a 4 ft × 2 ft × 2 ft box with a 4 ft × 1 ft × 1 ft box attached on the side. How much soil does it hold?

LESSON 9.11 · PRACTICE
Week 3 · Lesson 3 of 4
Unit 9: Angles, Circles, Area, and Volume

Lesson 9.11: Comparing Volume and Surface Area

Name _____ Date _____

TODAY'S GOAL

I can compare solids with the same volume but different surface areas.

Standards: 7.G.6

WHAT TO DO

- [ ] Read Textbook Lesson 9.11 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Volume 24: find the surface area, Part D Explain, Part E Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Find each surface area, or answer the question.

1. SA of 12 × 1 × 1

2. SA of 3 × 2 × 2

3. Both have volume 12. Which uses less material?

PART B Warm-Up

Directions: Multiply.

4. $8 \times 1 \times 1$

5. $2 \times 2 \times 2$

PART C Volume 24: find the surface area

Directions: Each box has volume 24. Find its surface area.

6. $24 \times 1 \times 1$

7. $12 \times 2 \times 1$

8. $6 \times 4 \times 1$

9. $4 \times 3 \times 2$

PART D Explain

10. A company wants a box that holds 24 cubes and uses the least cardboard. Which box above should it choose? Why?

PART E Exit Ticket

11. Can two boxes have the same surface area but different volumes? Give an example.

LESSON 9.12 · PRACTICE
Week 3 · Lesson 4 of 4

Unit 9: Angles, Circles, Area, and Volume

Lesson 9.12: Geometry Problem Solving

Name _____ Date _____

TODAY'S GOAL

I can choose and use the right geometry formulas to solve multi-step problems.

Standards: 7.G.4, 7.G.5, 7.G.6

WHAT TO DO

- [ ] Read Textbook Lesson 9.12 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Solve, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Use the pool and path in Lesson 9.12 to answer each question.

1. Circumference of the pool

2. Outer edge of the path

3. Volume of a 1 ft deep layer of water in the pool

PART B Warm-Up

Directions: Evaluate.

4 13^2

5 $169 - 100$

PART C Solve

Directions: Solve each problem. Show your work.

6 A circular table has diameter 4 ft. A cloth hangs 1 ft over the edge all around. What is the cloth's area?

7 Two angles are supplementary. One is 24° more than the other. Find both.

8 A tent is a triangular prism: triangle base 6 ft, height 4 ft, length 8 ft. Find the space inside.

9 A cylindrical-looking hay bale is wrapped with a band around its 5 ft diameter. How long is the band?

PART D Exit Ticket

10 Write your own two-step geometry problem and solve it.

WEEKLY CHECK
After Lesson 9.12

About 15 minutes

Weekly Check 27: Geometry Problem Solving

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each question carefully. Show your thinking.
- [ ] When you finish, check that every question has an answer.

Show what you know

Directions: Solve each problem. Show your work.

1. Filling a tank: volume or SA?

2. Open box 5 × 4 × 3: cardboard

3. Box 6 × 6 × 6 minus a 2 × 2 × 6 hole

4. Which has less SA: 6 × 1 × 1 or 3 × 2 × 1?

5. Ring: outer r = 5, inner r = 3

6. Two complementary angles; one is twice the other

LESSON 9.13 · PRACTICE
Week 4 · Lesson 1 of 4

Unit 9: Angles, Circles, Area, and Volume

Lesson 9.13: Year Review: Ratios, Proportions, and Percents

Name _____ Date _____

TODAY'S GOAL

I can solve problems with unit rates, proportional relationships, and percents.

Standards: 7.RP.1–3

WHAT TO DO

- [ ] Read Textbook Lesson 9.13 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Review: Units 1–2, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Solve each problem. Show your work.

1. Unit rate: $\frac{3}{4}$ mi in $\frac{1}{4}$ hr

2. y is proportional to x; y = 10 when x = 4. Find k.

3. 20% off $65

PART B Warm-Up

Directions: Divide.

4. 4.47 ÷ 3

5. 7.25 ÷ 5

PART C Review: Units 1–2

Directions: Solve each problem. Show your work.

6. Ellie reads 45 pages in 1.5 hours. Pages per hour?

7. Solve: $\frac{6}{15} = \frac{x}{40}$

8. Is y = 4x + 1 proportional?

9. 15% tip on $48

10. Price $120 → $90: percent change

11. Simple interest: $500 at 4% for 3 years

12. 6% tax on $35

13. Map: 2 cm : 15 km. 7 cm?

PART D Exit Ticket

14. A recipe uses 3 cups flour for 2 dozen cookies. How much for 5 dozen?

LESSON 9.14 · PRACTICE
Week 4 · Lesson 2 of 4

Unit 9: Angles, Circles, Area, and Volume

Lesson 9.14: Year Review: Rational Numbers

Name _____ Date _____

TODAY'S GOAL

I can add, subtract, multiply, and divide rational numbers.

Standards: 7.NS.1–3

WHAT TO DO

- [ ] Read Textbook Lesson 9.14 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Review: Units 3–5, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Add, subtract, or multiply.

1. $-7 + 12$

2. $-3 - (-8)$

3. -2.5×4

PART B Warm-Up

Directions: Add or subtract.

4. $-6 + 14$

5. $8 - 9$

PART C Review: Units 3–5

Directions: Add or subtract.

6. $-15 + (-9)$

7. $4 - 11$

8. $-1.2 + 3.5$

9. $(-6)(-7)$

10. $-48 \div 6$

11. $\frac{2}{3} \times (-\frac{9}{10})$

12. Write $\frac{5}{8}$ as a decimal

13. Write $\frac{2}{3}$ as a decimal

14. $(-2)^3$

15. $|-9| - |4|$

PART D Exit Ticket

16. A diver at −40 ft rises 15 ft, then descends 28 ft. Where is she?

LESSON 9.15 · PRACTICE
Week 4 · Lesson 3 of 4

Unit 9: Angles, Circles, Area, and Volume

Lesson 9.15: Year Review: Expressions, Equations, and Inequalities

Name _____ Date _____

TODAY'S GOAL

I can simplify expressions and solve equations and inequalities.

Standards: 7.EE.1–4

WHAT TO DO

- [ ] Read Textbook Lesson 9.15 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Review: Unit 6, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Factor, solve the equation, or solve the inequality.

1. Factor $18y + 12$

2. $4x - 7 = 21$

3. $-3x + 5 > 17$

PART B Warm-Up

Directions: Evaluate.

4 110 − 20

5 90 ÷ 15

PART C Review: Unit 6

Directions: Simplify or evaluate.

6 Simplify $5a - 2(a + 3)$

7 Simplify $\frac{1}{2}(8x - 6) + x$

8 Evaluate $2x^2 - x$ when $x = -3$

9 $-2(x + 3) = 10$

10 $\frac{1}{2}x - 2 = 4$

11 $0.4n + 1.2 = 3.6$

12 $-4y \geq 20$

13 $2x - 9 < 7$

PART D Exit Ticket

14 Write and solve: three times a number, minus 8, is 25.

LESSON 9.16 · PRACTICE
Week 4 · Lesson 4 of 4

Unit 9: Angles, Circles, Area, and Volume

Lesson 9.16: Year Review: Statistics, Probability, and Geometry

Name _____ Date _____

TODAY'S GOAL

I can solve problems using statistics, probability, and geometry.

Standards: 7.SP, 7.G

WHAT TO DO

- [ ] Read Textbook Lesson 9.16 first.
- [ ] Do every part on these pages: Part A Try It, Part B Warm-Up, Part C Review: Units 7–9, Part D Exit Ticket.
- [ ] Show your thinking. Write final answers on the lines.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Try It

Directions: Solve each problem. Show your work.

1. P(heads, then heads)

2. Area of a circle, r = 6

3. Missing angle in a triangle: 50°, 75°

PART B Warm-Up

Directions: Multiply.

4 0.35 × 600

5 6 × 6

PART C Review: Units 7–9

Directions: Solve each problem. Show your work.

6 Mean and MAD of 4, 6, 8, 10, 12

7 Bag: 3 red, 5 blue, 2 green. P(not blue)

8 Outfits: 4 shirts, 3 pants

9 Scale 1 in. : 6 ft. Drawing 3.5 in.?

10 Triangle? 3, 4, 8

11 Supplement of 47°

12 Circumference, d = 15 cm

13 Volume, box 6 × 4 × 2.5

14 Surface area, box 6 × 4 × 2

15 Cross section of a cylinder, horizontal

PART D Exit Ticket

16 Which topic from this year do you feel strongest in? Which one will you review before the test?

YEAR-END REVIEW
Study before the unit test

About 50 minutes

Year-End Review

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each question carefully. Show your thinking.
- [ ] When you finish, check that every question has an answer.

PART A Ratios, percents, and rational numbers

Directions: Solve each problem. Show your work.

1. Unit price: 6 for $8.70

2. $y = kx$; $y = 21$ when $x = 3$. Find y when $x = 8$.

3. 35% of 240

4. Price $60 → $75: percent change

5. $-8 - (-3) + 12$

6. $-\frac{5}{6} \times \frac{3}{10}$

7. $-3.6 \div (-0.4)$

8. $\frac{7}{20}$ as a decimal

PART B Algebra

Directions: Simplify, or solve each equation.

9. Simplify $4(2x - 1) - 3x$

10. $-5x + 8 = -17$

11. $3(n - 2) = 15$

12. $6 - 2x \geq 14$

PART C Statistics, probability, and geometry

Directions: Solve each problem. Show your work.

13. 18 of 60 sampled like a show; estimate for 2,500

14. Coin + cube: P(T and even)

15. Area, circle r = 4

16. Volume, triangular prism: legs 4 and 9, length 5

17. Scale 1 cm : 8 m; 6.5 cm

18. Missing angle: 90° and 34° in a triangle

YEAR-END TEST
At the end of the unit
About 60 minutes

Grade 7 Year-End Test

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] This test has 32 points.
- [ ] Read each question carefully. Show your thinking.
- [ ] When you finish, check that every question has an answer.

PART A Part A · Ratios and percents (1 point each)

Directions: Solve each problem. Show your work.

1. Unit rate: 150 miles in 2.5 hours

2. $y = kx$ passes through $(5, 35)$. Find k.

3. 30% of 90

4. A $40 shirt is 15% off. Sale price?

5. Price $50 → $65: percent change

PART B Part B · Rational numbers (1 point each)

Directions: Add, subtract, multiply, or divide.

6. $-14 + 9$

7. $6 - (-11)$

8. $(-4)(-2.5)$

9. $-\frac{2}{3} \div \frac{4}{9}$

10 Write $^{3}/_{8}$ as a decimal

PART C Part C · Expressions, equations, inequalities (1 point each)

Directions: Simplify, factor, or solve.

11 Simplify $2(3x - 5) + 4x$

12 Factor $24n - 16$

13 $-6x + 4 = 40$

14 $5(y + 2) = -15$

15 $-2k + 7 \leq 15$

PART D Part D · Statistics and probability (1 point each)

Directions: Solve each problem. Show your work.

16 A random sample of 50 has 12 left-handed people. Estimate for 1,500.

17 Mean of 3, 7, 8, 10, 12

18 Bag: 4 red, 6 blue. P(blue)

19 Two coins: P(at least one head)

20 Two number cubes: P(sum is 10)

PART E Part E · Geometry (1 point each)

Directions: Solve each problem. Show your work.

21 Scale 1 in. : 12 ft; 3.5 in. Actual?

22 Supplement of 68°

23 Circumference, d = 12 m

24 Area, r = 9 cm

25 Volume, box 7 × 3 × 4

26 Surface area, box 7 × 3 × 4

PART F Part F · Problem solving (3 points each)

27 A phone plan costs $25 per month plus $0.10 per text. Ellie can spend at most $40 per month. Write and solve an inequality for the number of texts. Interpret.

28 A circular garden has diameter 14 ft. Find its area. Edging costs $2 per foot: how much to edge it? Show your work.

Grade 7 Mathematics

Student Workbook

A full year of Grade 7 mathematics, taught one clear idea at a time. Every lesson opens with a question worth thinking about, walks through worked examples step by step, and builds from models and pictures to fluent, confident problem solving.

This Student Workbook holds the daily practice pages for every lesson, plus a Weekly Check at the end of each week, the unit reviews, and the unit tests, with room to show thinking and write complete answers.

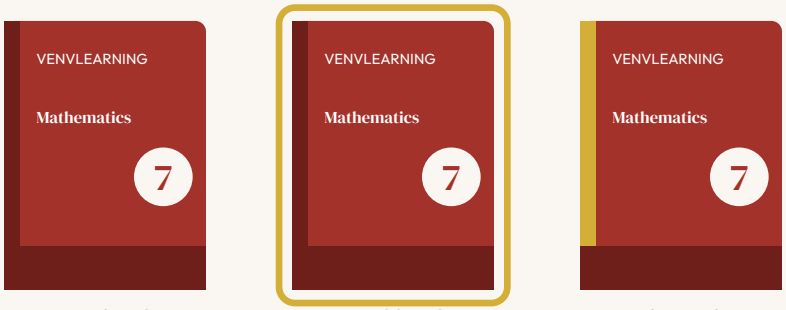
INSIDE THIS YEAR

01 Proportional Relationships
02 Percents
03 Adding and Subtracting Integers
04 Adding and Subtracting Rational Numbers
05 Multiplying and Dividing Rational Numbers
06 Expressions, Equations, and Inequalities
07 Statistics and Probability
08 Scale Drawings and Geometric Figures
09 Angles, Circles, Area, and Volume

IN THIS BOOK

- Warm-ups, practice, and exit tickets for every lesson
- A Weekly Check for each week of lessons
- Unit reviews and unit tests
- Space to show work and explain thinking

THE COMPLETE SET



Textbook

Workbook

You are holding this book

Teacher Edition

Each lesson has the same number in all three books, so the textbook, workbook, and teacher pages always match.